

# Deep Waters

Costas Arkolakis\*

Yang Pei<sup>†</sup>

Anton Yang<sup>‡</sup>

## Abstract

We provide a spatial theory of water allocation to assess the global distribution of water resources across space, states, and time. We model the details of the hydrological cycle—precipitation, infiltration, groundwater recharge, evapotranspiration, and surface runoff—that determine the supply and movement of water across a grid of locations. The local evolution of water stocks depends on climatic forcings, subsurface storage capacity, and a flow-direction network that routes surface runoff downstream. We embed this hydrological system in a dynamic multi-region, multi-sector spatial economy where water is a key input to production. We use the model to measure the aggregate and spatial implications of water scarcity, groundwater depletion, and changes in precipitation patterns. We find substantial heterogeneity across regions in the welfare consequences of water reallocation, with arid regions facing large losses from groundwater depletion and downstream regions benefiting from upstream runoff. Incorporating policy, we evaluate the impact of water pricing, infrastructure investments, and inter-basin transfers on the spatial distribution of economic activity.

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\*Yale University. Email: [costas.arkolakis@yale.edu](mailto:costas.arkolakis@yale.edu).

<sup>†</sup>Tsinghua University. Email: [ypei1.work@gmail.com](mailto:ypei1.work@gmail.com).

<sup>‡</sup>Yale University. Email: [anton.yang@yale.edu](mailto:anton.yang@yale.edu)

# 1 Introduction

Water is the most fundamental natural resource for human life and economic activity, yet its spatial distribution is profoundly unequal. Some regions receive abundant rainfall and sit atop vast aquifers, while others face chronic scarcity. Unlike most tradable commodities, water is heavy, difficult to transport over long distances, and governed by the laws of gravity and topography. It flows downhill, seeps into the ground, evaporates into the atmosphere, and accumulates in underground reservoirs over millennia. Understanding how water moves through space—and how economic agents interact with these hydrological flows—is essential for evaluating the welfare consequences of water scarcity and the design of water policy.

The challenge of modeling water allocation across space has three distinctive features. First, water exists in multiple *states*: it falls as precipitation, infiltrates into the root zone, percolates to shallow and deep aquifers, runs off into rivers and lakes, evaporates back into the atmosphere, and freezes as snow. The transitions between these states are governed by physical processes that depend on local soil properties, land cover, temperature, and antecedent moisture conditions. Second, water moves across *locations* through a fixed drainage network determined by topography. Surface runoff generated at one location becomes inflow at downstream locations, creating spatial linkages that propagate local shocks through the hydrological network. Third, the system is inherently *dynamic*: groundwater stocks accumulate over decades and deplete with extraction, snowpack builds in winter and releases meltwater in spring, and the entire system responds to seasonal and long-run changes in climatic forcing.

In order to answer questions about the welfare consequences of water scarcity and the spatial design of water policy, we build a theory of water allocation across space that integrates these three dimensions. We start in Section 2 by embedding the details of the hydrological cycle in a multi-region, multi-sector spatial economy. We model preferences, production technology, water extraction, and trade, showing how water availability shapes comparative advantage and the spatial distribution of economic activity. The demand for water is determined by a multi-sector Armington trade model. The cost of water extraction depends on the local state of the hydrological system, creating a link between physical hydrology and economic outcomes.

In Section 3, we present the water balance model that constitutes the physical core of the paper. The model tracks water quantities at a grid of discrete locations across multiple states—root-zone moisture, shallow groundwater, deep groundwater, snowpack, and surface water—over time. We detail the hydrological processes that govern the evolution of these stocks: a snow module with degree-day melt, a storage-limited runoff rule, recharge partitioning between shallow and deep aquifers, and a flow-direction routing operator that transports surface runoff downstream. The key equations are presented in the main text; detailed derivations, proofs of mass conservation, and the alternative curve-number specification are deferred to the appendix.

To create a quantitative portrait of water allocation across space, in Section 4 we complement the theory with high-resolution hydrological, climatic, and geospatial datasets harmonized at the grid-cell level. We use NASA GLDAS land-surface model outputs for climatic forcing,

GRACE satellite gravity data for groundwater validation, HydroSHEDS flow-direction grids for the routing network, and hydrogeology databases for subsurface storage parameters. We provide a summary of each dataset’s role in the main text; detailed processing steps, unit conversions, and variable definitions are in Appendix A9.

In Section 5, we calibrate the model and validate it against independent observations. We then use the calibrated model to trace out the implications of the current water distribution for welfare across the globe, finding substantial heterogeneity driven by differences in renewable versus non-renewable water endowments and by the spatial structure of the routing network.

Section 6 presents the deep water scenarios: forward-looking projections of the hydrological-economic system under baseline dynamics and alternative assumptions about climate, extraction, and infrastructure. We find that the world’s water system faces a growing divergence between regions with sustainable surface-water supplies and those mining non-renewable groundwater.

In Section 7, we consider sustainable water policies. We evaluate the welfare effects of water pricing reform, inter-basin transfer infrastructure, and adaptation to changing precipitation patterns. We find that pricing reform generates aggregate welfare gains by reallocating water to higher-value uses, while inter-basin transfers can yield large gains in recipient regions at the cost of reduced downstream flows in donor basins.

Section 8 concludes.

**Related literature.** This paper contributes to several strands of the economics literature. First, we build on the quantitative spatial economics framework developed by [Eaton and Kortum \(2002\)](#), [Allen and Arkolakis \(2014\)](#), [Redding and Rossi-Hansberg \(2017\)](#), and [Arkolakis and Walsh \(2023\)](#), among others. Our model shares the multi-region, multi-sector structure of these papers but adds a physical resource layer that governs the spatial distribution of water. The closest paper in structure is [Arkolakis and Walsh \(2023\)](#), who embed the physics of the electrical grid in a spatial growth model to study the renewable energy transition. We follow an analogous approach for the hydrological cycle: just as they model electricity transmission losses and grid constraints, we model infiltration losses, storage constraints, and downstream routing.

Second, we contribute to the literature on water economics and groundwater management. The economics of groundwater extraction has been studied extensively in the context of common-pool resource problems ([Gisser and Sánchez, 1980](#); [Provencher and Burt, 1993](#); [Burness and Quirk, 1979](#)). [Smith \(2016\)](#) provide a comprehensive review. Our contribution is to embed groundwater dynamics in a general-equilibrium spatial model where water availability affects production costs, trade patterns, and the spatial distribution of economic activity.

Third, we relate to the growing literature on natural resources in spatial economics. [Desmet and Rossi-Hansberg \(2024\)](#) and [Conte et al. \(2021\)](#) study the spatial implications of climate change. [Costinot et al. \(2016\)](#) examine agricultural comparative advantage under climate change. Our paper complements this work by providing a micro-founded model of the physical resource that mediates much of the impact of climate on the economy.

Fourth, our hydrological model draws on a large engineering and geoscience literature on land-surface modeling and runoff generation. The storage-limited runoff approach is related to variable infiltration capacity models (Liang et al., 1994), while the curve-number method originates with the USDA Soil Conservation Service (USDA Soil Conservation Service, 1985). We adapt these approaches to a spatial economics setting where computational tractability and transparent comparative statics are essential.

## 2 Setup

This section describes the economic environment. There are  $I$  locations on a spatial grid,  $J$  sectors, and discrete time periods  $t = 0, 1, \dots, T$ . Each location can hold water in multiple hydrological states; the physical evolution of these states is governed by the water balance model described in Section 3. Here we take the current hydrological state as given and describe how it shapes the within-period economic equilibrium.

### 2.1 Water System: States, Dynamics, and Extraction Constraints

Before introducing the economic model, we summarize the complete water system. The purpose of this subsection is to distinguish clearly (i) the physical water states, (ii) exogenous hydrological forcings, (iii) physical flows generated by the hydrological system, (iv) spatial movements across grid cells, and (v) the three endogenous economic extraction decisions.

The current economic model has exactly three endogenous water sources:

$$\boxed{\text{Surface water,} \quad \text{Shallow groundwater,} \quad \text{Deep groundwater.}} \quad (1)$$

Root-zone moisture remains an important physical state and can affect land productivity, but it is not treated as a fourth endogenous extraction source in the current specification.

#### 1. Six-state water accounting system

At location  $i$  and time  $t$ , water is represented by the six-dimensional accounting vector

$$\boxed{W_{ti} \equiv \begin{pmatrix} W_{ti}^1 \\ W_{ti}^2 \\ W_{ti}^3 \\ W_{ti}^4 \\ W_{ti}^5 \\ W_{ti}^6 \end{pmatrix}} \in \mathbb{R}^6, \quad (2)$$

where

- $W_{ti}^1$ : Out — water in the atmosphere and oceans,
- $W_{ti}^2$ : Surface — rivers, lakes, reservoirs, and related surface water,
- $W_{ti}^3$ : Root Zones — soil moisture,
- $W_{ti}^4$ : Shallow Groundwater Storage Pool,
- $W_{ti}^5$ : Unsustainable (deep) Groundwater,
- $W_{ti}^6$ : Econ — water transferred into economic use.

The operational hydrological variables corresponding to the principal physical states are

$$\boxed{W_{ti}^2 \leftrightarrow X_{2,i,t}, \quad W_{ti}^3 \leftrightarrow M_{i,t}, \quad W_{ti}^4 \leftrightarrow S_{i,t}, \quad W_{ti}^5 \leftrightarrow D_{i,t}.} \quad (3)$$

Here,

- $X_{2,i,t}$ : surface-water stock,
- $M_{i,t}$ : root-zone moisture,
- $S_{i,t}$ : shallow-groundwater stock,
- $D_{i,t}$ : deep-groundwater stock.

The physical model additionally tracks snow water equivalent  $SWE_{i,t}$ .

Stacking all  $I$  locations gives

$$\boxed{\mathbf{W}_t = \begin{bmatrix} W_{t1} & W_{t2} & \cdots & W_{tI} \end{bmatrix}} \in \mathbb{R}^{6 \times I}. \quad (4)$$

## 2. Exogenous forcings, hydrological variables, and economic choices

It is useful to distinguish three groups of objects.

**Exogenous physical forcings and fixed characteristics.** These include rainfall  $r_{i,t}^{\text{rain}}$ , snowfall  $r_{i,t}^{\text{snow}}$ , temperature  $T_{i,t}$ , externally measured evapotranspiration where used, initial water conditions, grid characteristics, groundwater capacities  $\bar{S}_i$  and  $\bar{D}_i$ , routing weights  $w_{j \rightarrow i}$ , open-water share  $\omega_i$ , and fixed hydrological parameters such as  $\text{ddf}$ ,  $T^{\text{melt}}$ ,  $\zeta_i$ ,  $\kappa_i^{\text{rout}}$ , and  $\varsigma_i^{\text{bed}}$ .

**Hydrologically determined stocks and flows.** Conditional on inherited water stocks, exogenous forcings, and economic withdrawals, the physical block determines objects such as snowmelt  $m_{i,t}$ , effective precipitation  $\tilde{r}_{i,t}$ , recharge  $f_{i,t}^{\text{Sh}}$  and  $f_{i,t}^{\text{Dp}}$ , runoff  $q_{i,t}$ , routed inflow, baseflow  $b_{i,t}$ , seepage  $s_{i,t}$ , and the updated stocks  $M_{i,t}$ ,  $X_{2,i,t}$ ,  $S_{i,t}$ , and  $D_{i,t}$ .

**Endogenous economic extraction choices.** The economic model chooses

$$\boxed{\text{Div}_{i,t}^{\text{sur}}, \quad \text{Div}_{i,t}^{\text{Sh}}, \quad \text{Div}_{i,t}^{\text{Dp}}.} \quad (5)$$

These three decisions feed back into the physical laws of motion.

Return flows also become endogenous because they depend on economic withdrawal.

### 3. Global law of motion

Let  $\mathbf{E}_0$  denote the initial distribution of water. Then

$$\mathbf{W}_0 = \mathbf{E}_0. \quad (6)$$

For subsequent periods,

$$\boxed{\mathbf{W}_t = \mathbf{W}_{t-1} + \mathbf{\Omega}_t, \quad \mathbf{\Omega}_t = \mathbf{T}_t + \mathbf{S}_t.} \quad (7)$$

The two components of  $\mathbf{\Omega}_t$  have different meanings:

- $\mathbf{T}_t$  : net movements across water states within grid cells,
- $\mathbf{S}_t$  : net movements of water across grid cells.

Both matrices have dimension  $6 \times I$ .

The distinction is important. Economic extraction is a *within-location state transition* and therefore enters  $\mathbf{T}_t$ . Spatial routing is represented by  $\mathbf{S}_t$  and cannot in general be omitted from the law of motion.

*Notation.* The bold matrix  $\mathbf{S}_t$  denotes spatial water movement. It is different from the scalar  $S_{i,t}$  denoting shallow-groundwater storage.

### 4. Within-location transitions: $\mathbf{B}_{it}$

At each location  $i$ , gross flows between the six water states are recorded in

$$\mathbf{B}_{it} \in \mathbb{R}^{6 \times 6}.$$

For state indices  $r, s \in \{1, \dots, 6\}$ ,

$$\boxed{B_{it}^{rs} = \text{gross water flow from state } r \text{ to state } s \text{ during period } t.} \quad (8)$$

Thus,

$$\boxed{\text{row} = \text{origin state}, \quad \text{column} = \text{destination state}.}$$

Using the six-state ordering above, the current schematic transition matrix is

$$\mathbf{B}_{it} = \begin{pmatrix} 0 & \text{Precip. Surf. Run. Glacier Run.} & \text{Precip. and Snow} & 0 & 0 & \text{Precipitation and Snow} \\ E & 0 & 0 & 0 & 0 & \text{Div}_{i,t}^{\text{sur}} \\ ET & \text{Surface Runoff} & 0 & \text{Recharge} & 0 & 0 \\ 0 & ET & \text{Base-flow} & 0 & 0 & \text{Div}_{i,t}^{\text{Sh}} \\ 0 & 0 & 0 & 0 & 0 & \text{Div}_{i,t}^{\text{Dp}} \\ ET & \text{Surface Runoff} & \text{Infiltr.} & 0 & 0 & 0 \end{pmatrix}. \quad (9)$$

Here  $E$  denotes evaporation and  $ET$  evapotranspiration. Runoff, recharge, baseflow, and infiltration denote the corresponding physical water transfers.

The “Glacier Run.” entry is retained from the project transition matrix; its separate operational equation has not yet been fixed in the physical block.

For the economic part, the important correction is the last column:

$$\boxed{B_{it}^{2,6} = \text{Div}_{i,t}^{\text{sur}}, \quad B_{it}^{3,6} = 0, \quad B_{it}^{4,6} = \text{Div}_{i,t}^{\text{Sh}}, \quad B_{it}^{5,6} = \text{Div}_{i,t}^{\text{Dp}}.} \quad (10)$$

Hence,

$$\boxed{W^2 \xrightarrow{\text{Div}_{i,t}^{\text{sur}}} W^6, \quad W^4 \xrightarrow{\text{Div}_{i,t}^{\text{Sh}}} W^6, \quad W^5 \xrightarrow{\text{Div}_{i,t}^{\text{Dp}}} W^6.} \quad (11)$$

Root-zone moisture  $W^3$  is therefore not an endogenous extraction source.

## 5. From $\mathbf{B}_{it}$ to $\mathbf{T}_t$

Because  $\mathbf{B}_{it}$  records gross flows, it must be converted into net state changes.

Let

$$\mathbf{1} = (1, 1, 1, 1, 1, 1)^\top.$$

The net within-location change is

$$\boxed{\boldsymbol{\tau}_{it} = \mathbf{B}_{it}^\top \mathbf{1} - \mathbf{B}_{it} \mathbf{1}} \in \mathbb{R}^6. \quad (12)$$

For state  $s$ ,

$$\tau_{it}^s = \underbrace{\sum_{r=1}^6 B_{it}^{rs}}_{\text{all inflows into state } s} - \underbrace{\sum_{k=1}^6 B_{it}^{sk}}_{\text{all outflows from state } s}. \quad (13)$$

The location-specific vectors are stacked into

$$\mathbf{T}_t = \begin{bmatrix} \tau_{1t} & \cdots & \tau_{It} \end{bmatrix} \in \mathbb{R}^{6 \times I}. \quad (14)$$

Thus,

$$\mathbf{B}_{it} \longrightarrow \tau_{it} \longrightarrow \mathbf{T}_t$$

describes within-cell movements only.

## 6. Across-location transitions: $\mathbf{S}_t$

The second component of the global law of motion,  $\mathbf{S}_t$ , records movements across locations.

In the current hydrological specification, direct lateral movement is confined to Surface water. Therefore,

$$(\mathbf{S}_t)_{ri} = 0 \quad \text{for } r \in \{1, 3, 4, 5, 6\}. \quad (15)$$

Only the Surface row,  $r = 2$ , has a direct spatial component.

This does *not* imply that spatial water movements are irrelevant for groundwater. Rather, Surface water can arrive from upstream and subsequently affect Shallow and Deep groundwater through seepage, infiltration, and recharge.

There are two equivalent implementations of surface routing used in the project materials.

**Baseline zero-residence routing.** Let  $q_{j,t}$  denote runoff generated at cell  $j$ . With routing weights  $w_{j \rightarrow i} \geq 0$ ,

$$l_{i,t} = \sum_{j=1}^I w_{j \rightarrow i} q_{j,t}. \quad (16)$$

The routed flow generated in period  $t$  arrives downstream in the next period:

$$u_{i,t+1} = \tilde{r}_{i,t+1} + l_{i,t}. \quad (17)$$

Here  $l_{i,t}$  is a flow, not an additional water stock.

**Explicit Surface-stock routing.** When Surface water has positive residence time and is represented by the stock  $X_{2,i,t}$ , the same spatial mechanism is represented directly in the Surface-water law of motion.

The direct spatial contribution to the Surface row is

$$(\mathbf{S}_t)_{2i} = \underbrace{\sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1}}_{\text{surface water arriving from other cells}} - \underbrace{\kappa_i^{\text{rout}} X_{2,i,t-1}}_{\text{surface water released from cell } i}. \quad (18)$$

Thus,  $\mathbf{S}_t$  is explicitly present in the surface-water dynamics.

The baseline  $\iota$  formulation and the explicit  $X_2$  routing formulation are two representations of the same spatial mechanism. When the explicit  $X_2$  balance is used, routed inflow must not be added a second time through an additional  $\mathbf{S}_t$  term in that same scalar equation.

## 7. Snow and effective liquid water

Let  $r_{i,t}^{\text{rain}}$  and  $r_{i,t}^{\text{snow}}$  denote rainfall and snowfall water equivalent, and let  $T_{i,t}$  denote temperature. Snowmelt is

$$m_{i,t} = \min \left\{ \text{SWE}_{i,t-1} + r_{i,t}^{\text{snow}}, \text{ddf} \max(T_{i,t} - T^{\text{melt}}, 0) \Delta_t \right\}. \quad (19)$$

Snowpack evolves according to

$$\text{SWE}_{i,t} = \max \left\{ \text{SWE}_{i,t-1} + r_{i,t}^{\text{snow}} - m_{i,t}, 0 \right\}. \quad (20)$$

Effective liquid precipitation is

$$\tilde{r}_{i,t} = r_{i,t}^{\text{rain}} + m_{i,t}. \quad (21)$$

## 8. Local water partition: abstraction, recharge, and runoff

Under the baseline zero-residence routing specification, total liquid input is

$$u_{i,t} = \tilde{r}_{i,t} + l_{i,t-1}. \quad (22)$$

Let  $\bar{a}_i$  denote initial-abstraction capacity. Then

$$a_{i,t} = \min \{ u_{i,t}, \bar{a}_i \}, \quad u_{i,t}^{\text{net}} = u_{i,t} - a_{i,t}. \quad (23)$$

Let  $\xi_i \in [0, 1]$  denote the potential deep-percolation share. Potential recharge is

$$f_{i,t}^{\text{Sh,raw}} = (1 - \xi_i) u_{i,t}^{\text{net}}, \quad f_{i,t}^{\text{Dp,raw}} = \xi_i u_{i,t}^{\text{net}}. \quad (24)$$

With storage capacities  $\bar{S}_i$  and  $\bar{D}_i$ ,

$$\begin{aligned} f_{i,t}^{\text{Sh}} &= \min \left\{ (1 - \xi_i) u_{i,t}^{\text{net}}, (\bar{S}_i - S_{i,t-1})_+ \right\}, \\ f_{i,t}^{\text{Dp}} &= \min \left\{ \xi_i u_{i,t}^{\text{net}}, (\bar{D}_i - D_{i,t-1})_+ \right\}. \end{aligned} \quad (25)$$

Surface runoff is the residual:

$$q_{i,t} = u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}} \geq 0. \quad (26)$$

Therefore,

$$u_{i,t} = a_{i,t} + f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}} + q_{i,t}. \quad (27)$$

## 9. Root-zone moisture

Root-zone moisture follows

$$M_{i,t} = \max \{M_{i,t-1} + a_{i,t} - e_{i,t}, 0\}. \quad (28)$$

In the current economic extraction problem,

$$B_{it}^{3,6} = 0. \quad (29)$$

Hence no endogenous Div variable is associated with  $M_{i,t}$ .

Moisture remains economically relevant through land productivity. For example, the land-productivity component can depend on

$$\Phi_{i,j,t} = \bar{\Phi}_{i,j} g_j(M_{i,t}, S_{i,t}). \quad (30)$$

Thus,

$M_{i,t}$  is a hydrological/productivity state,

not one of the three extraction states.

## 10. Explicit Surface-water dynamics

For the integrated hydro-economic model, Surface water is represented explicitly by  $X_{2,i,t}$ .

Let  $\omega_i$  denote the open-water share of grid cell  $i$ . Then

$$r_{i,t}^{\text{land}} = (1 - \omega_i)r_{i,t}, \quad r_{i,t}^{\text{water}} = \omega_i r_{i,t}. \quad (31)$$

Let  $\kappa_i^{\text{rout}} \in (0, 1]$  denote the fraction of surface-water storage released from cell  $i$  during period  $t$ .

The Surface-water balance is

$$\begin{aligned} X_{2,i,t} = & (1 - \kappa_i^{\text{rout}})X_{2,i,t-1} + q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} \\ & + \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} + v_i^{\text{surf}} \mathcal{R}_{i,t} \\ & - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}}. \end{aligned} \quad (32)$$

Every term has a distinct physical meaning:

$$\begin{aligned}
(1 - \kappa_i^{\text{rout}})X_{2,i,t-1} &: \text{retained Surface stock,} \\
q_{i,t} &: \text{local runoff entering Surface water,} \\
b_{i,t} &: \text{baseflow from Shallow groundwater,} \\
r_{i,t}^{\text{water}} &: \text{precipitation directly on open water,} \\
\sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} &: \text{Surface inflow from upstream cells,} \\
v_i^{\text{surf}} \mathcal{R}_{i,t} &: \text{return flow entering Surface water,} \\
\text{Div}_{i,t}^{\text{sur}} &: \text{economic Surface-water extraction,} \\
e_{i,t}^{\text{open}} &: \text{open-water evaporation,} \\
s_{i,t} &: \text{seepage from Surface to Shallow groundwater,} \\
O_{i,t}^{\text{ocean}} &: \text{terminal water export.}
\end{aligned}$$

The same equation can be written to display explicitly the  $\mathbf{T}_t$ – $\mathbf{S}_t$  decomposition:

$$\begin{aligned}
X_{2,i,t} = & X_{2,i,t-1} \\
& + \underbrace{\left[ q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} + v_i^{\text{surf}} \mathcal{R}_{i,t} - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}} \right]}_{\text{within-location / source-state changes}} \\
& + \underbrace{\left[ \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} - \kappa_i^{\text{rout}} X_{2,i,t-1} \right]}_{\text{direct Surface component of } \mathbf{S}_t}.
\end{aligned} \tag{33}$$

Equation (33) is the scalar Surface-row version of

$$\mathbf{W}_t = \mathbf{W}_{t-1} + \mathbf{T}_t + \mathbf{S}_t.$$

Thus,  $\mathbf{S}_t$  has not disappeared: it is the last bracket in the Surface-water equation.

## 11. Baseflow and Surface-to-Shallow seepage

Surface and Shallow water are physically connected in both directions.

Baseflow transfers water from Shallow groundwater to Surface water. A simple representation is

$$b_{i,t} = \beta_i^{\text{base}} S_{i,t-1}, \tag{34}$$

where  $\beta_i^{\text{base}}$  is a recession parameter.

Conversely, Surface-water seepage transfers water into the Shallow aquifer. In the baseline

explicit-Surface specification,

$$s_{i,t} = \varsigma_i^{\text{bed}}(1 - \kappa_i^{\text{rout}})X_{2,i,t-1}. \quad (35)$$

These two flows must enter the Surface and Shallow equations with opposite signs in order to preserve mass accounting.

## 12. Shallow-groundwater dynamics

Combining recharge, Surface-water seepage, baseflow, and economic pumping gives

$$S_{i,t} = S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t} - \text{Div}_{i,t}^{\text{Sh}}. \quad (36)$$

This equation is especially important for the economic model.

Its components are

- $S_{i,t-1}$  : inherited Shallow stock,
- $f_{i,t}^{\text{Sh}}$  : new recharge into the Shallow aquifer,
- $s_{i,t}$  : seepage from Surface water,
- $b_{i,t}$  : baseflow leaving Shallow groundwater,
- $\text{Div}_{i,t}^{\text{Sh}}$  : economic Shallow-water pumping.

There is no direct  $\mathbf{S}_t$  term in (36), because Shallow groundwater does not move directly across grid cells in the current routing specification:

$$(\mathbf{S}_t)_{4i} = 0. \quad (37)$$

Spatial water movements still affect  $S_{i,t}$  indirectly because Surface water arriving from upstream can affect subsequent seepage and recharge.

## 13. Deep-groundwater dynamics

Deep groundwater evolves according to

$$D_{i,t} = D_{i,t-1} + f_{i,t}^{\text{Dp}} - \text{Div}_{i,t}^{\text{Dp}}. \quad (38)$$

Its components are

- $D_{i,t-1}$  : inherited Deep stock,
- $f_{i,t}^{\text{Dp}}$  : Deep recharge,
- $\text{Div}_{i,t}^{\text{Dp}}$  : economic Deep-water pumping.

Again,

$$(\mathbf{S}_t)_{5i} = 0, \quad (39)$$

so there is no direct across-grid movement of Deep groundwater.

When deep recharge is negligible,

$$f_{i,t}^{\text{Dp}} \simeq 0,$$

the law of motion becomes

$$D_{i,t} \simeq D_{i,t-1} - \text{Div}_{i,t}^{\text{Dp}}, \quad (40)$$

which makes Deep groundwater close to a standard exhaustible resource.

#### 14. Three dynamic constraints for extraction

The three state equations relevant for the economic extraction problem can now be displayed together.

$$\begin{aligned} X_{2,i,t} &= (1 - \kappa_i^{\text{rout}}) X_{2,i,t-1} + q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} \\ &\quad + \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} + v_i^{\text{surf}} \mathcal{R}_{i,t} \\ &\quad - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}}, \\ S_{i,t} &= S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t} - \text{Div}_{i,t}^{\text{Sh}}, \\ D_{i,t} &= D_{i,t-1} + f_{i,t}^{\text{Dp}} - \text{Div}_{i,t}^{\text{Dp}}. \end{aligned} \quad (41)$$

These equations are not additional laws unrelated to the global water-motion equation. They are the expanded scalar laws for the three extractable states contained in

$$\mathbf{W}_t = \mathbf{W}_{t-1} + \mathbf{T}_t + \mathbf{S}_t.$$

The Surface equation contains both the within-location component and the direct spatial-routing component. The Shallow and Deep equations have no direct  $\mathbf{S}_t$  term under the current specification.

#### 15. Physical feasibility

Economic withdrawals must satisfy

$$\text{Div}_{i,t}^{\text{sur}} \geq 0, \quad \text{Div}_{i,t}^{\text{Sh}} \geq 0, \quad \text{Div}_{i,t}^{\text{Dp}} \geq 0. \quad (42)$$

The extractable stocks must remain non-negative:

$$X_{2,i,t} \geq 0, \quad S_{i,t} \geq 0, \quad D_{i,t} \geq 0. \quad (43)$$

The Shallow law therefore implies

$$0 \leq \text{Div}_{i,t}^{\text{Sh}} \leq S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t}. \quad (44)$$

The Deep law implies

$$0 \leq \text{Div}_{i,t}^{\text{Dp}} \leq D_{i,t-1} + f_{i,t}^{\text{Dp}}. \quad (45)$$

For Surface water, feasibility is imposed through equation (32) together with  $X_{2,i,t} \geq 0$ .

We retain the Surface equality constraint rather than replacing it with a simple exogenous upper bound because upstream routing and return flows are themselves endogenous components of the Surface-water balance.

## 16. Return flows

Let  $v_i$  denote the consumptive-use fraction of extracted water. For total extraction  $\text{Div}_{i,t}$ , total return flow is

$$\mathcal{R}_{i,t} = (1 - v_i)\text{Div}_{i,t}. \quad (46)$$

The currently specified Surface-return component is

$$v_i^{\text{surf}} \mathcal{R}_{i,t}.$$

The current project materials do not yet provide a complete formal allocation rule for the remaining return flow across Shallow and Deep groundwater. Therefore no additional subsurface return-flow term is inserted into (36) or (38) until that physical allocation is fixed.

## 17. Alternative runoff algorithm: Curve Number

The storage-limited recharge specification above is the baseline. The project materials also contain a Curve-Number alternative.

Let  $v_i \in (0, 100]$  denote the curve number and define

$$\zeta_i = 10 \left( \frac{100}{v_i} - 1 \right) = \frac{1000}{v_i} - 10, \quad I_i = 0.2\zeta_i. \quad (47)$$

Given liquid input  $u_{i,t}$ ,

$$a_{i,t} = \min\{u_{i,t}, I_i\}, \quad (48)$$

and runoff is

$$q_{i,t} = \frac{(u_{i,t} - a_{i,t})^2}{u_{i,t} - a_{i,t} + \zeta_i}. \quad (49)$$

Infiltration is the residual:

$$f_{i,t} = u_{i,t} - a_{i,t} - q_{i,t}, \quad (50)$$

with

$$f_{i,t}^{\text{Sh}} = (1 - \zeta_i) f_{i,t}, \quad f_{i,t}^{\text{Dp}} = \zeta_i f_{i,t}. \quad (51)$$

Hence,

Storage-limited baseline: recharge is determined first and runoff is residual,

Curve-Number alternative: runoff is determined first and infiltration is residual.

The three extraction-state laws in (41) remain the same; only the physical determination of runoff and recharge changes.

## 18. Intertemporal structure passed to economics

The three economic controls and their corresponding states are

$$(X_{2,i,t}, \text{Div}_{i,t}^{\text{sur}}), \quad (S_{i,t}, \text{Div}_{i,t}^{\text{Sh}}), \quad (D_{i,t}, \text{Div}_{i,t}^{\text{Dp}}). \quad (52)$$

The timing is

inherited stocks  $\longrightarrow$  current extraction  $\longrightarrow$  current ending stocks  $\longrightarrow$  next-period extraction possibilities.

There is therefore no need for a separate mechanical equation linking  $\text{Div}_{i,t}$  directly to  $\text{Div}_{i,t+1}$ .

The intertemporal connection operates through the state equations:

$$\begin{aligned} \text{Div}_{i,t}^{\text{sur}} \uparrow &\implies X_{2,i,t} \downarrow, \\ \text{Div}_{i,t}^{\text{Sh}} \uparrow &\implies S_{i,t} \downarrow, \\ \text{Div}_{i,t}^{\text{Dp}} \uparrow &\implies D_{i,t} \downarrow. \end{aligned}$$

These three dynamic constraints provide the physical basis for the shadow values and intertemporal extraction conditions derived in the economic model.

## 2.2 Water Extraction

The hydrological system above determines the states and physical constraints faced by the economic water-extraction problem. We now collect the three extraction decisions and their dynamic constraints in the form used by the economic model.

Water can be supplied to the economy from exactly three sources:

$$\text{Div}_{i,t}^{\text{sur}}, \quad \text{Div}_{i,t}^{\text{Sh}}, \quad \text{Div}_{i,t}^{\text{Dp}}. \quad (53)$$

Root-zone moisture  $M_{i,t}$  is not an endogenous extraction choice in the current model. It continues to affect the economy through the hydrological and productivity channels.

### Three dynamic extraction constraints

For emphasis, the three state equations faced by the economic extraction problem are repeated here.

#### Surface water.

$$\begin{aligned} X_{2,i,t} = & (1 - \kappa_i^{\text{rout}})X_{2,i,t-1} + q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} \\ & + \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} + v_i^{\text{surf}} \mathcal{R}_{i,t} \\ & - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}}. \end{aligned} \quad (54)$$

This constraint contains both local Surface-water dynamics and the direct spatial-routing component inherited from  $\mathbf{S}_t$ .

#### Shallow groundwater.

$$S_{i,t} = S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t} - \text{Div}_{i,t}^{\text{Sh}}. \quad (55)$$

#### Deep groundwater.

$$D_{i,t} = D_{i,t-1} + f_{i,t}^{\text{Dp}} - \text{Div}_{i,t}^{\text{Dp}}. \quad (56)$$

These three equalities are the source-specific dynamic constraints that will enter the subsequent constrained maximization problem.

They are the scalar expansions of the relevant rows of the global hydrological law

$$\mathbf{W}_t = \mathbf{W}_{t-1} + \mathbf{T}_t + \mathbf{S}_t. \quad (57)$$

They should therefore not be imposed twice as independent restrictions: one may work with the global transition system or with its equivalent expanded source equations.

### Total economic extraction

Total extraction at grid cell  $i$  is

$$\text{Div}_{i,t} = \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}}. \quad (58)$$

Let  $\mathcal{A}_i$  denote the area of grid cell  $i$ . If the Div variables are expressed as water-equivalent depths, the corresponding volume of water supplied to the within-period economy is

$$\bar{F}_{i,t}^{\text{Water}} = \mathcal{A}_i \text{Div}_{i,t} = \mathcal{A}_i \left( \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right). \quad (59)$$

Thus the dynamic hydrological problem determines the local water endowment available to the static production economy.

### Return flows

Let  $v_i$  denote the consumptive-use fraction. Total return flow generated by economic water use is

$$\mathcal{R}_{i,t} = (1 - v_i)\text{Div}_{i,t}. \quad (60)$$

The currently specified Surface component is

$$v_i^{\text{surf}} \mathcal{R}_{i,t}, \quad (61)$$

which re-enters the Surface-water dynamic constraint.

Hence extraction has two effects on future water states: it directly removes water from the source stock, while the non-consumed part can partially return to the hydrological system.

### Source-specific extraction costs

For

$$a \in \{\text{sur}, \text{Sh}, \text{Dp}\},$$

let

$$\text{ExtractCost}_i^a(\mathcal{A}_i \text{Div}_{i,t}^a, \mathbf{W}_{ti}; GF_i^a) \quad (62)$$

denote the cost of extracting water from source  $a$ .

Here,

$$\begin{aligned} \mathcal{A}_i \text{Div}_{i,t}^a &: \text{physical extraction volume,} \\ \mathbf{W}_{ti} &: \text{current local hydrological state,} \\ GF_i^a &: \text{source-specific extraction technology or cost shifter.} \end{aligned}$$

This formulation permits pumping or diversion costs to depend on both the amount extracted and the hydrological state. For example, pumping cost can rise as groundwater becomes deeper or more depleted.

The exact functional form of  $\text{ExtractCost}_i^a(\cdot)$  is specified in the extraction-cost block below.

### Physical feasibility of economic extraction

The economic controls satisfy

$$\text{Div}_{i,t}^{\text{sur}} \geq 0, \quad \text{Div}_{i,t}^{\text{Sh}} \geq 0, \quad \text{Div}_{i,t}^{\text{Dp}} \geq 0. \quad (63)$$

The corresponding physical states satisfy

$$X_{2,i,t} \geq 0, \quad S_{i,t} \geq 0, \quad D_{i,t} \geq 0. \quad (64)$$

For Shallow groundwater this implies

$$0 \leq \text{Div}_{i,t}^{\text{Sh}} \leq S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t}. \quad (65)$$

For Deep groundwater,

$$0 \leq \text{Div}_{i,t}^{\text{Dp}} \leq D_{i,t-1} + f_{i,t}^{\text{Dp}}. \quad (66)$$

Surface feasibility is represented directly by equation (54) and the non-negativity condition  $X_{2,i,t} \geq 0$ .

### Bridge to the dynamic economic problem

The economic extraction problem therefore has three controls,

$$\text{Div}_{i,t}^{\text{sur}}, \quad \text{Div}_{i,t}^{\text{Sh}}, \quad \text{Div}_{i,t}^{\text{Dp}},$$

and three corresponding physical state constraints,

$$X_{2,i,t}, \quad S_{i,t}, \quad D_{i,t}.$$

Current extraction generates current water supply  $\bar{F}_{i,t}^{\text{Water}}$  and therefore current economic revenue, but it also changes the water stocks available in future periods.

The subsequent dynamic problem can therefore be written as a constrained maximization of discounted water revenue net of source-specific extraction costs, subject directly to

$$(54), \quad (55), \quad (56). \quad (67)$$

The Lagrange multipliers on these three dynamic constraints have a direct economic interpretation as the marginal value of preserving Surface, Shallow, and Deep water for future use. These shadow values are the objects that enter the source-choice first-order conditions and the corresponding Hotelling-type intertemporal equations derived below.

## 2.3 Technology, Production, and Land Allocation

Countries are indexed by  $n, m \in \mathcal{N} \equiv \{1, \dots, N\}$ , physical water-land grid cells by  $i \in \mathcal{I} \equiv \{1, \dots, I\}$ , sectors by  $j, k \in \mathcal{J} \equiv \{1, \dots, J\}$ , and time by  $t$ . The physical-grid index  $i$  and the country index  $n$  refer to distinct spatial dimensions. Each physical grid cell belongs to one and only one country, while each country generally contains many physical grid cells. International trade uses  $n$  for the destination country and  $m$  for the origin country throughout.

**Country assignment of physical grids.** Let  $n(i) \in \mathcal{N}$  denote the unique country containing physical grid cell  $i$ . For each country  $n$ , define the set of physical grids located in that country as

$$\mathcal{I}_n \equiv \{i \in \mathcal{I} : n(i) = n\}. \quad (68)$$

Because every physical grid belongs to exactly one country,

$$\mathcal{I} = \bigsqcup_{n \in \mathcal{N}} \mathcal{I}_n, \quad (69)$$

where  $\bigsqcup$  denotes a disjoint union. Thus  $\mathcal{I}_n \cap \mathcal{I}_m = \emptyset$  for  $n \neq m$ , and every  $i \in \mathcal{I}$  appears in exactly one country-specific grid set.

For  $i \in \mathcal{I}_n$ , we retain the compact notation  $i_n$  whenever it is useful to make country membership explicit. The notation  $i_n$  does not represent a separate fragment of a shared grid: it is simply physical grid  $i$ , which is uniquely located in country  $n$ . Accordingly, all grid-level economic variables below are defined only for  $i \in \mathcal{I}_n$ .

Let  $\mathcal{A}_i$  denote the physical area of grid  $i$ . Since the entire grid lies in one country,

$$\mathcal{A}_{i_n} \equiv \mathcal{A}_i, \quad i \in \mathcal{I}_n, \quad (70)$$

and total country area is

$$\mathcal{A}_n \equiv \sum_{i \in \mathcal{I}_n} \mathcal{A}_i. \quad (71)$$

For country-level averages of intensive grid variables, define

$$\omega_{i_n}^A \equiv \frac{\mathcal{A}_i}{\mathcal{A}_n}, \quad \sum_{i \in \mathcal{I}_n} \omega_{i_n}^A = 1. \quad (72)$$

Let  $\bar{F}_i^{\text{Land}}$  denote the exogenous physical land endowment of grid  $i$ . Land endowments are primitive at the physical-grid level, so the endowment carries the grid index  $i$  but no separate country index. Country-level physical land is obtained only by aggregation,  $\sum_{i \in \mathcal{I}_n} \bar{F}_i^{\text{Land}}$ .

**Current-period local water endowment before production.** Water available to production is determined by the extraction problem. Let  $\text{Div}_{i,t}^{\text{sur}}$ ,  $\text{Div}_{i,t}^{\text{Sh}}$ , and  $\text{Div}_{i,t}^{\text{Dp}}$  denote current extraction from surface water, shallow groundwater, and deep groundwater. These quantities are endogenous in the intertemporal extraction problem but are taken as given by the within-period production equilibrium. Total extraction depth is

$$\text{Div}_{i,t} = \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}}. \quad (73)$$

The three components distinguish physical extraction sources in the water dynamics; in the static production economy they enter as a single water input.

After converting depth and area into compatible units, the current-period water volume available in grid  $i$  is

$$\bar{F}_{i,t}^{\text{Water}} \equiv \mathcal{A}_i \text{Div}_{i,t}, \quad i \in \mathcal{I}_n. \quad (74)$$

Country-level water availability is obtained only by aggregation,  $\sum_{i \in \mathcal{I}_n} \bar{F}_{i,t}^{\text{Water}}$ . This is an accounting sum only: it does not create a national or global water market and does not permit water to move across physical grids.

Conditional on the period- $t$  extraction vector inherited from the water dynamics,  $\bar{F}_{i,t}^{\text{Water}}$  is the local water endowment of the static production problem. Let  $F_{i,n,j,t}^{\text{Water}}$  denote sector- $j$  water use in grid  $i$  of country  $n$ , and let  $r_{i,n,t}^{\text{Water}}$  denote the corresponding local water price. Water is non-tradable across physical grids, so each grid has its own water-market-clearing condition:

$$\sum_{j \in \mathcal{J}} F_{i,n,j,t}^{\text{Water}} = \bar{F}_{i,t}^{\text{Water}}, \quad \forall i \in \mathcal{I}_n, n \in \mathcal{N}. \quad (75)$$

Thus there is one local water price  $r_{i,n,t}^{\text{Water}}$  for each physical grid. Country totals are useful for accounting but do not determine the local water price.

The exogenous labor endowment is national rather than grid-specific. Let  $\bar{F}_{n,t}^{\text{LAB}}$  denote the total labor endowment of country  $n$  in period  $t$ . Labor is mobile across grids and sectors within the country, while land and water remain tied to physical grid  $i$ .

**Primary-endowment CDE structure.** Within grid  $i$  of country  $n$ , sector  $j$  uses three primary endowments: units of direct production labor, units of local water, and efficiency units of land. We use  $F$  for factor quantities and reserve  $E$  for land-efficiency and setup-requirement means. Labor used directly in production is denoted by  $F_{i,n,j,t}^{\text{LAB}}$  and water use by  $F_{i,n,j,t}^{\text{Water}}$ . Physical land is heterogeneous. If physical plot  $\omega$  is assigned to sector  $j$ , one unit of that plot supplies  $e_{i,n,j,t}(\omega)$  efficiency units of land. Hence the price of one efficiency unit of land for that plot is the physical-plot return divided by plot productivity,

$$\frac{r_{i,n,j,t}^{\text{Land}}(\omega)}{e_{i,n,j,t}(\omega)}. \quad (76)$$

The plot index in the production-cost system enters only through this land term. The national wage  $w_{n,t}^{\text{LAB}}$  and the local water price  $r_{i,n,t}^{\text{Water}}$  do not receive plot-specific productivity shocks.

Let  $V_{i,n,j,t}$  denote the primary-endowment composite. On the quantity side, the CDE technology is represented by the implicit production frontier

$$\mathcal{F}_{n,j}^V \left( V_{i,n,j,t}, F_{i,n,j,t}^{\text{LAB}}, e_{i,n,j,t}(\omega) F_{i,n,j,t}^{\text{Land}}, F_{i,n,j,t}^{\text{Water}} \right) = 0. \quad (77)$$

The three quantity arguments are direct production labor, efficiency units of land, and units of local water. The cost relation below is the dual of this production frontier. Direct production labor is valued at the national wage  $w_{n,t}^{\text{LAB}}$ ; one efficiency unit of land on plot  $\omega$  is valued at

$r_{i_n,j,t}^{\text{Land}}(\omega)/e_{i_n,j,t}(\omega)$ ; and local water is valued at  $r_{i_n,t}^{\text{Water}}$ . Thus plot heterogeneity enters the production system only through land efficiency  $e_{i_n,j,t}(\omega)$ ; direct production labor and water carry no plot-specific productivity shock.

Let  $P_{i_n,j,t}^V$  denote the price of the primary-endowment composite. This price is a grid-sector equilibrium object and does not carry a plot index. The CDE cost relation for a candidate plot  $\omega$  is

$$\begin{aligned} 1 = & \beta_{n,j}^{\text{LAB}} V_{i_n,j,t} \epsilon_j^{\text{LAB}(1-\sigma_j^{\text{LAB}})} \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_n,j,t}^V V_{i_n,j,t}} \right)^{1-\sigma_j^{\text{LAB}}} \\ & + \beta_{n,j}^L V_{i_n,j,t} \epsilon_j^{L(1-\sigma_j^L)} \left( \frac{r_{i_n,j,t}^{\text{Land}}(\omega)/e_{i_n,j,t}(\omega)}{P_{i_n,j,t}^V V_{i_n,j,t}} \right)^{1-\sigma_j^L} \\ & + \beta_{n,j}^W V_{i_n,j,t} \epsilon_j^{W(1-\sigma_j^W)} \left( \frac{r_{i_n,t}^{\text{Water}}}{P_{i_n,j,t}^V V_{i_n,j,t}} \right)^{1-\sigma_j^W}. \end{aligned} \quad (78)$$

The parameters  $\beta_{n,j}^x$  are CDE distribution parameters,  $\sigma_j^x$  govern input-specific substitution curvature, and  $\epsilon_j^x$  govern input-specific expansion responses, for  $x \in \{\text{LAB}, L, W\}$ . For sectors that do not use land set  $\beta_{n,j}^L = 0$ ; for sectors that do not use water set  $\beta_{n,j}^W = 0$ .

**Upper production nest and source unit cost.** The primary-endowment composite and Armington intermediate inputs are combined through the upper Cobb–Douglas technology

$$Q_{i_n,j,t} = A_{n,j,t}^{\text{prod}} V_{i_n,j,t} \gamma_{n,j}^V \prod_{k \in \mathcal{J}} \left( M_{i_n,j,k,t}^{\text{int}} \right)^{\gamma_{n,j,k}}, \quad (79)$$

where

$$\gamma_{n,j}^V + \sum_{k \in \mathcal{J}} \gamma_{n,j,k} = 1. \quad (80)$$

Let  $P_{n,k,t}$  denote the country- $n$  purchaser price of the sector- $k$  composite. Define the Cobb–Douglas unit-cost constant

$$Y_{n,j} \equiv (\gamma_{n,j}^V)^{-\gamma_{n,j}^V} \prod_{k \in \mathcal{J}} \gamma_{n,j,k}^{-\gamma_{n,j,k}}. \quad (81)$$

The upper Cobb–Douglas nest implies the grid-sector source unit cost

$$c_{i_n,j,t} = \frac{Y_{n,j}}{A_{n,j,t}^{\text{prod}}} \left( P_{i_n,j,t}^V \right)^{\gamma_{n,j}^V} \prod_{k \in \mathcal{J}} (P_{n,k,t})^{\gamma_{n,j,k}}. \quad (82)$$

The upper unit cost itself is a grid-sector object. Plot heterogeneity appears only in the inner CDE land-price argument  $r_{i_n,j,t}^{\text{Land}}(\omega)/e_{i_n,j,t}(\omega)$ . Since all other objects in the CDE relation are common across plots, equilibrium requires this ratio to be common across active plots.

Each country has one national collection hub. Let  $\bar{d}_{i_n,j} \geq 1$  denote the iceberg collection cost from grid  $i_n$  to the country- $n$  hub, and let  $P_{n,j,t}$  denote the sector- $j$  source price at that hub.

Perfect competition in collection implies

$$c_{i_{n,j},t} \bar{d}_{i_{n,j}} = P_{nn,j,t}, \quad i \in \mathcal{I}_n, \quad (83)$$

or equivalently

$$c_{i_{n,j},t} = \frac{P_{nn,j,t}}{\bar{d}_{i_{n,j}}}. \quad (84)$$

The price  $P_{nn,j,t}$  is the domestic source price used in the trade system;  $P_{n,j,t}$  below is the sector- $j$  Armington purchaser price in country  $n$ . Solving the upper Cobb–Douglas unit-cost equation then gives the primary-endowment price

$$P_{i_{n,j},t}^V = \gamma_{n,j}^V \left[ A_{n,j,t}^{\text{prod}} c_{i_{n,j},t} \prod_{k \in \mathcal{J}} \left( \frac{\gamma_{n,j,k}}{P_{n,k,t}} \right)^{\gamma_{n,j,k}} \right]^{1/\gamma_{n,j}^V}. \quad (85)$$

The value of gross source output is

$$Y_{i_{n,j},t} \equiv c_{i_{n,j},t} Q_{i_{n,j},t} = \frac{P_{nn,j,t}}{\bar{d}_{i_{n,j}}} Q_{i_{n,j},t}. \quad (86)$$

**Return to physical land implied by the two production nests.** Because  $P_{i_{n,j},t}^V$ , the wage, and the water price are common across plots, isolate the land term in (78). The return per efficiency unit of land is

$$\frac{r_{i_{n,j},t}^{\text{Land}}(\omega)}{e_{i_{n,j},t}(\omega)} \equiv c_{i_{n,j},t} \bar{h}_{i_{n,j},t}. \quad (87)$$

Therefore the return per efficiency unit of land,  $r_{i_{n,j},t}^{\text{Land}}(\omega)/e_{i_{n,j},t}(\omega)$ , is common across plots and equals  $c_{i_{n,j},t} \bar{h}_{i_{n,j},t}$ .

The production-side term  $\bar{h}_{i_{n,j},t}$  is

$$\bar{h}_{i_{n,j},t} \equiv \frac{P_{i_{n,j},t}^V V_{i_{n,j},t}}{c_{i_{n,j},t}} \times \left[ \frac{\left\{ 1 - \beta_{n,j}^{\text{LAB}} V_{i_{n,j},t} \epsilon_j^{\text{LAB}} (1 - \sigma_j^{\text{LAB}}) \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1 - \sigma_j^{\text{LAB}}} - \beta_{n,j}^{\text{W}} V_{i_{n,j},t} \epsilon_j^{\text{W}} (1 - \sigma_j^{\text{W}}) \left( \frac{r_{i_{n,t}}^{\text{Water}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1 - \sigma_j^{\text{W}}} \right\}}{\left\{ \beta_{n,j}^{\text{L}} V_{i_{n,j},t} \epsilon_j^{\text{L}} (1 - \sigma_j^{\text{L}}) \right\}} \right]^{\frac{1}{1 - \sigma_j^{\text{L}}}}. \quad (88)$$

Equations (85) and (88) give the two-nest expression for  $\bar{h}_{i_{n,j},t}$ , with  $\sigma_j^{\text{L}} \neq 1$ .

The physical-plot return is therefore

$$r_{i_{n,j},t}^{\text{Land}}(\omega) = e_{i_{n,j},t}(\omega) c_{i_{n,j},t} \bar{h}_{i_{n,j},t}. \quad (89)$$

This is the direct analogue of the standard decomposition of a plot return into plot productivity and a common market-return component.

**Land productivity, setup cost, and plot choice.** Let  $\mathcal{J}^L \subseteq \mathcal{J}$  denote the set of active land-using sectors. Within grid  $i$  of country  $n$ , land consists of a continuum of physical plots  $\omega \in [0, 1]$ . For sector  $j \in \mathcal{J}^L$ ,  $e_{i,n,j,t}(\omega)$  is the plot-specific productivity draw, with

$$\mathbb{E}[e_{i,n,j,t}(\omega)] = E_{i,n,j,t}. \quad (90)$$

Mean land productivity has two components:

$$E_{i,n,j,t} = \bar{E}_{i,n,j} g_j(M_{i,t}, S_{i,t}). \quad (91)$$

Here  $\bar{E}_{i,n,j}$  is time invariant and captures persistent grid-sector differences in land quality and suitability, including soil, geographic, and other location-specific characteristics. The function  $g_j(M_{i,t}, S_{i,t})$  captures the time-varying contribution of local hydrological conditions to land productivity. Root-zone moisture  $M_{i,t}$  measures the water condition immediately available in the soil, while shallow groundwater  $S_{i,t}$  captures the contribution of near-surface groundwater conditions.

Deep groundwater  $D_{i,t}$  does not enter land productivity directly. Its economic value arises when it is extracted through  $\text{Div}_{i,t}^{\text{DP}}$  and becomes part of the local water endowment available to production. This distinction separates the effect of hydrological conditions on land efficiency from the use of extracted water as a production input.

Activating a plot requires a plot-specific setup requirement measured in labor units. Let  $e_{i,n,0}(\omega)$  denote this requirement, with

$$\mathbb{E}[e_{i,n,0}(\omega)] = E_{i,n,0}. \quad (92)$$

If the plot remains idle or unmanaged, this setup labor is not used. Its outside-option value is therefore

$$r_{i,n,0,t}^{\text{Land}}(\omega) \equiv e_{i,n,0}(\omega) w_{n,t}^{\text{LAB}}. \quad (93)$$

For later use, define the unconditional mean plot returns

$$\mathbb{E}[r_{i,n,0,t}^{\text{Land}}(\omega)] \equiv r_{i,n,0,t}^{\text{Land}} = E_{i,n,0} w_{n,t}^{\text{LAB}}, \quad (94)$$

and, for each active use  $j \in \mathcal{J}^L$ ,

$$\begin{aligned} \mathbb{E}[r_{i,n,j,t}^{\text{Land}}(\omega)] &\equiv r_{i,n,j,t}^{\text{Land}} \\ &= E_{i,n,j,t} c_{i,n,j,t} \bar{h}_{i,n,j,t}. \end{aligned} \quad (95)$$

The land manager therefore chooses

$$j_{i_n,t}^*(\omega) \in \arg \max_{g \in \{0\} \cup \mathcal{J}^L} r_{i_n,g,t}^{\text{Land}}(\omega). \quad (96)$$

Equivalently, an active use is selected only if its gross plot return covers the plot-specific setup cost:

$$\max \left\{ 0, \left\{ r_{i_n,j,t}^{\text{Land}}(\omega) - w_{n,t}^{\text{LAB}} e_{i_n,0}(\omega) \right\}_{j \in \mathcal{J}^L} \right\}. \quad (97)$$

**Nested Fréchet land allocation.** Collect these active-use productivity and setup-requirement draws in

$$\mathbf{e}_{i_n,t}(\omega) \equiv \left[ \left\{ e_{i_n,j,t}(\omega) \right\}_{j \in \mathcal{J}^L}, e_{i_n,0}(\omega) \right]. \quad (98)$$

They follow the two-layer nested Fréchet distribution

$$\begin{aligned} \Pr(\mathbf{e}_{i_n,t}(\omega) \leq \mathbf{e}_{i_n,t}) &= \exp \left\{ -[\Gamma(1 - 1/\nu_1)]^{-\nu_1} \left[ \left( \frac{e_{i_n,0}}{E_{i_n,0}} \right)^{-\nu_1} \right. \right. \\ &\quad \left. \left. + \left( \left[ \sum_{j \in \mathcal{J}^L} \left( \frac{e_{i_n,j}}{E_{i_n,j,t}} \right)^{-\nu_2} \right]^{-1/\nu_2} \right)^{-\nu_1} \right] \right\}, \end{aligned} \quad (99)$$

with  $1 < \nu_1 \leq \nu_2$ . The lower-layer shape parameter  $\nu_2$  governs substitution across active land uses, while  $\nu_1$  governs the active-versus-idle margin.

Because each plot-level return is the corresponding Fréchet draw multiplied by a deterministic grid-sector return component, the same nested Fréchet structure can be written directly in terms of the mean returns defined in (94)–(95). Conditional on activation, the land share of sector  $j$  is therefore

$$\pi_{i_n,j|A,t}^L = \frac{\left( r_{i_n,j,t}^{\text{Land}} \right)^{\nu_2}}{\sum_{k \in \mathcal{J}^L} \left( r_{i_n,k,t}^{\text{Land}} \right)^{\nu_2}}. \quad (100)$$

The upper-layer active and idle shares are

$$\begin{aligned} \pi_{i_n,A,t}^L &= \frac{\left[ \sum_{k \in \mathcal{J}^L} \left( r_{i_n,k,t}^{\text{Land}} \right)^{\nu_2} \right]^{\nu_1/\nu_2}}{\left( r_{i_n,0,t}^{\text{Land}} \right)^{\nu_1} + \left[ \sum_{k \in \mathcal{J}^L} \left( r_{i_n,k,t}^{\text{Land}} \right)^{\nu_2} \right]^{\nu_1/\nu_2}}, \\ \pi_{i_n,0,t}^L &= \frac{\left( r_{i_n,0,t}^{\text{Land}} \right)^{\nu_1}}{\left( r_{i_n,0,t}^{\text{Land}} \right)^{\nu_1} + \left[ \sum_{k \in \mathcal{J}^L} \left( r_{i_n,k,t}^{\text{Land}} \right)^{\nu_2} \right]^{\nu_1/\nu_2}}. \end{aligned} \quad (101)$$

The unconditional active-sector share is

$$\pi_{i_{n,j},t}^L = \pi_{i_n,A,t}^L \pi_{i_{n,j}|A,t}^L. \quad (102)$$

Physical land allocations are

$$F_{i_{n,j},t}^{\text{Land}} = \pi_{i_{n,j},t}^L \bar{F}_i^{\text{Land}}, \quad F_{i_n,0,t}^{\text{Land}} = \pi_{i_n,0,t}^L \bar{F}_i^{\text{Land}}, \quad (103)$$

and hence

$$F_{i_n,0,t}^{\text{Land}} + \sum_{j \in \mathcal{J}^L} F_{i_{n,j},t}^{\text{Land}} = \bar{F}_i^{\text{Land}}. \quad (104)$$

**Selected productivity, land payments, and grid output.** Let  $\Omega_{i_{n,j},t}$  denote the set of plots selected into active sector  $j$ . The nested Fréchet distribution gives

$$\mathbb{E}[e_{i_{n,j},t}(\omega) \mid \omega \in \Omega_{i_{n,j},t}] = E_{i_{n,j},t} \left( \pi_{i_n,A,t}^L \right)^{-1/v_1} \left( \pi_{i_{n,j}|A,t}^L \right)^{-1/v_2}. \quad (105)$$

The gross payment generated by active land in sector  $j$  is

$$RL_{i_{n,j},t} \equiv \mathbb{E} \left[ r_{i_{n,j},t}^{\text{Land}}(\omega) \mid \omega \in \Omega_{i_{n,j},t} \right] F_{i_{n,j},t}^{\text{Land}}. \quad (106)$$

Using (89),

$$RL_{i_{n,j},t} = c_{i_{n,j},t} \hbar_{i_{n,j},t} \mathbb{E}[e_{i_{n,j},t}(\omega) \mid \omega \in \Omega_{i_{n,j},t}] F_{i_{n,j},t}^{\text{Land}}. \quad (107)$$

Define land's value share in primary-endowment expenditure by

$$s_{i_{n,j},t}^L \equiv \frac{RL_{i_{n,j},t}}{P_{i_{n,j},t}^V V_{i_{n,j},t}}. \quad (108)$$

The upper Cobb–Douglas nest implies

$$P_{i_{n,j},t}^V V_{i_{n,j},t} = \gamma_{n,j}^V Y_{i_{n,j},t}. \quad (109)$$

Therefore

$$RL_{i_{n,j},t} = \gamma_{n,j}^V s_{i_{n,j},t}^L Y_{i_{n,j},t} = \gamma_{n,j}^V s_{i_{n,j},t}^L c_{i_{n,j},t} Q_{i_{n,j},t}. \quad (110)$$

Combining (107) with (110) cancels  $c_{i_{n,j},t}$  and yields

$$Q_{i_{n,j},t} = \frac{\hbar_{i_{n,j},t}}{\gamma_{n,j}^V s_{i_{n,j},t}^L} \mathbb{E}[e_{i_{n,j},t}(\omega) \mid \omega \in \Omega_{i_{n,j},t}] F_{i_{n,j},t}^{\text{Land}}. \quad (111)$$

Substituting the Fréchet conditional mean and land allocation gives

$$Q_{i_{n,j},t} = \frac{\hbar_{i_{n,j},t}}{\gamma_{n,j}^V s_{i_{n,j},t}^L} \bar{F}_i^{\text{Land}} E_{i_{n,j},t} \times \left( \pi_{i_{n,j},A,t}^L \right)^{\frac{\nu_1-1}{\nu_1}} \left( \pi_{i_{n,j}|A,t}^L \right)^{\frac{\nu_2-1}{\nu_2}}. \quad (112)$$

**CDE value shares and primary-factor demands.** The CDE value shares are

$$s_{i_{n,j},t}^{\text{LAB}} = \frac{\beta_{n,j}^{\text{LAB}} (1 - \sigma_j^{\text{LAB}}) \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^{\text{LAB}}} V_{i_{n,j},t}^{(1 - \sigma_j^{\text{LAB}})(\epsilon_j^{\text{LAB}} - 1)}}{\beta_{n,j}^{\text{LAB}} (1 - \sigma_j^{\text{LAB}}) \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^{\text{LAB}}} V_{i_{n,j},t}^{(1 - \sigma_j^{\text{LAB}})(\epsilon_j^{\text{LAB}} - 1)} + \beta_{n,j}^L (1 - \sigma_j^L) \left( \frac{c_{i_{n,j},t} \hbar_{i_{n,j},t}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^L} V_{i_{n,j},t}^{(1 - \sigma_j^L)(\epsilon_j^L - 1)} + \beta_{n,j}^W (1 - \sigma_j^W) \left( \frac{r_{i_{n,j},t}^{\text{Water}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^W} V_{i_{n,j},t}^{(1 - \sigma_j^W)(\epsilon_j^W - 1)}}. \quad (113)$$

$$s_{i_{n,j},t}^L = \frac{\beta_{n,j}^L (1 - \sigma_j^L) \left( \frac{c_{i_{n,j},t} \hbar_{i_{n,j},t}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^L} V_{i_{n,j},t}^{(1 - \sigma_j^L)(\epsilon_j^L - 1)}}{\beta_{n,j}^{\text{LAB}} (1 - \sigma_j^{\text{LAB}}) \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^{\text{LAB}}} V_{i_{n,j},t}^{(1 - \sigma_j^{\text{LAB}})(\epsilon_j^{\text{LAB}} - 1)} + \beta_{n,j}^L (1 - \sigma_j^L) \left( \frac{c_{i_{n,j},t} \hbar_{i_{n,j},t}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^L} V_{i_{n,j},t}^{(1 - \sigma_j^L)(\epsilon_j^L - 1)} + \beta_{n,j}^W (1 - \sigma_j^W) \left( \frac{r_{i_{n,j},t}^{\text{Water}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^W} V_{i_{n,j},t}^{(1 - \sigma_j^W)(\epsilon_j^W - 1)}}. \quad (114)$$

$$s_{i_{n,j},t}^W = \frac{\beta_{n,j}^W (1 - \sigma_j^W) \left( \frac{r_{i_{n,j},t}^{\text{Water}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^W} V_{i_{n,j},t}^{(1 - \sigma_j^W)(\epsilon_j^W - 1)}}{\beta_{n,j}^{\text{LAB}} (1 - \sigma_j^{\text{LAB}}) \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^{\text{LAB}}} V_{i_{n,j},t}^{(1 - \sigma_j^{\text{LAB}})(\epsilon_j^{\text{LAB}} - 1)} + \beta_{n,j}^L (1 - \sigma_j^L) \left( \frac{c_{i_{n,j},t} \hbar_{i_{n,j},t}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^L} V_{i_{n,j},t}^{(1 - \sigma_j^L)(\epsilon_j^L - 1)} + \beta_{n,j}^W (1 - \sigma_j^W) \left( \frac{r_{i_{n,j},t}^{\text{Water}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^W} V_{i_{n,j},t}^{(1 - \sigma_j^W)(\epsilon_j^W - 1)}}. \quad (115)$$

By construction,

$$s_{i_{n,j},t}^{\text{LAB}} + s_{i_{n,j},t}^L + s_{i_{n,j},t}^W = 1. \quad (116)$$

The share definitions are

$$s_{i_n,j,t}^{\text{LAB}} = \frac{w_{n,t}^{\text{LAB}} F_{i_n,j,t}^{\text{LAB}}}{P_{i_n,j,t}^V V_{i_n,j,t}}, \quad s_{i_n,j,t}^L = \frac{RL_{i_n,j,t}}{P_{i_n,j,t}^V V_{i_n,j,t}}, \quad (117)$$

and

$$s_{i_n,j,t}^W = \frac{r_{i_n,t}^{\text{Water}} F_{i_n,j,t}^{\text{Water}}}{P_{i_n,j,t}^V V_{i_n,j,t}}. \quad (118)$$

Using (109), direct production-labor demand is

$$w_{n,t}^{\text{LAB}} F_{i_n,j,t}^{\text{LAB}} = \gamma_{n,j}^V s_{i_n,j,t}^{\text{LAB}} Y_{i_n,j,t}, \quad (119)$$

gross active-land payment is

$$RL_{i_n,j,t} = \gamma_{n,j}^V s_{i_n,j,t}^L Y_{i_n,j,t}, \quad (120)$$

and water demand is

$$r_{i_n,t}^{\text{Water}} F_{i_n,j,t}^{\text{Water}} = \gamma_{n,j}^V s_{i_n,j,t}^W Y_{i_n,j,t}. \quad (121)$$

For every intermediate-input sector  $k$ ,

$$P_{n,k,t} M_{i_n,j,k,t}^{\text{int}} = \gamma_{n,j,k} Y_{i_n,j,t}. \quad (122)$$

For a land-using sector, source output value therefore satisfies

$$Y_{i_n,j,t} = w_{n,t}^{\text{LAB}} F_{i_n,j,t}^{\text{LAB}} + RL_{i_n,j,t} + r_{i_n,t}^{\text{Water}} F_{i_n,j,t}^{\text{Water}} + \sum_{k \in \mathcal{J}} P_{n,k,t} M_{i_n,j,k,t}^{\text{int}}. \quad (123)$$

The first labor payment is labor used directly in production. Setup labor associated with activating land is a separate use of the national labor endowment.

**Setup labor and net land revenue.** The setup labor used by active land is

$$S_{i_n,\text{agr},t} = E_{i_n,0} \bar{F}_i^{\text{Land}} \left[ 1 - \left( \pi_{i_n,0,t}^L \right)^{\frac{v_1-1}{v_1}} \right]. \quad (124)$$

The corresponding setup requirement attached to plots that remain idle is

$$S_{i_n,0,t} = E_{i_n,0} \bar{F}_i^{\text{Land}} \left( \pi_{i_n,0,t}^L \right)^{\frac{v_1-1}{v_1}}. \quad (125)$$

Only  $S_{i_n,\text{agr},t}$  is actually used as labor, and

$$S_{i_n,\text{agr},t} + S_{i_n,0,t} = E_{i_n,0} \bar{F}_i^{\text{Land}}. \quad (126)$$

The conditional avoided setup value on plots that remain idle is

$$\mathbb{E}\left[e_{i_n,0}(\omega)w_{n,t}^{\text{LAB}} \mid \omega \in \Omega_{i_n,0,t}\right] F_{i_n,0,t}^{\text{Land}} = w_{n,t}^{\text{LAB}} S_{i_n,0,t}. \quad (127)$$

This is an outside-option value, not labor actually hired by idle land.

Total gross active-land payment in grid  $i$  is

$$RL_{i_n,t} \equiv \sum_{j \in \mathcal{J}^L} RL_{i_n,j,t}. \quad (128)$$

Net land revenue is

$$\begin{aligned} NRL_{i_n,t} &\equiv RL_{i_n,t} - w_{n,t}^{\text{LAB}} S_{i_n,\text{agr},t} \\ &= \sum_{j \in \mathcal{J}^L} \gamma_{n,j}^V s_{i_n,j,t}^L Y_{i_n,j,t} - w_{n,t}^{\text{LAB}} S_{i_n,\text{agr},t}. \end{aligned} \quad (129)$$

Using (127) and (126), the same object can be written in the full active-plus-idle accounting form

$$\begin{aligned} NRL_{i_n,t} &= \sum_{j \in \mathcal{J}^L} \mathbb{E}\left[r_{i_n,j,t}^{\text{Land}}(\omega) \mid \omega \in \Omega_{i_n,j,t}\right] F_{i_n,j,t}^{\text{Land}} \\ &\quad + \mathbb{E}\left[e_{i_n,0}(\omega)w_{n,t}^{\text{LAB}} \mid \omega \in \Omega_{i_n,0,t}\right] F_{i_n,0,t}^{\text{Land}} \\ &\quad - w_{n,t}^{\text{LAB}} E_{i_n,0} \bar{F}_i^{\text{Land}}. \end{aligned} \quad (130)$$

The two forms are identical: idle plots retain the avoided setup value  $w_{n,t}^{\text{LAB}} S_{i_n,0,t}$ , while the full plot-specific setup requirement equals  $E_{i_n,0} \bar{F}_i^{\text{Land}}$ ; their difference leaves exactly the setup labor actually used on active land.

Country-level net land revenue is

$$NRL_{n,t} \equiv \sum_{i \in \mathcal{I}_n} NRL_{i_n,t}. \quad (131)$$

**National-hub supply.** Domestic collection converts gross grid output into the quantity available at the national hub. Of the  $Q_{i_n,j,t}$  units shipped from grid  $i_n$ ,  $Q_{i_n,j,t} / \bar{d}_{i_n,j}$  units arrive at the hub. Country-sector supply is

$$Q_{n,j,t} \equiv \sum_{i \in \mathcal{I}_n} \frac{Q_{i_n,j,t}}{\bar{d}_{i_n,j}}. \quad (132)$$

Using (83), the corresponding value is

$$Y_{n,j,t} \equiv P_{nn,j,t} Q_{n,j,t} = \sum_{i \in \mathcal{I}_n} c_{i_n,j,t} Q_{i_n,j,t} = \sum_{i \in \mathcal{I}_n} Y_{i_n,j,t}. \quad (133)$$

The collection step therefore removes the grid index before international trade:  $Q_{n,j,t}$  and  $P_{nn,j,t}$  are the country-level quantity and source price generated by grid-level production.

For a general origin country  $m$ , the same condition gives

$$P_{mm,j,t} = \bar{d}_{i_m,j} c_{i_m,j,t}, \quad i \in \mathcal{I}_m.$$

International delivery then gives  $P_{nm,j,t} = d_{nm,j} P_{mm,j,t}$ . The Armington aggregation below combines the domestic price  $P_{nn,j,t}$  with the delivered foreign-source prices to determine the purchaser price  $P_{n,j,t}$ . Thus the price sequence is

$$c_{i_m,j,t} \longrightarrow P_{mm,j,t} \longrightarrow P_{nm,j,t} \longrightarrow P_{n,j,t}.$$

## 2.4 Consumption and Armington Trade

Country  $n$  first substitutes between its domestic sector- $j$  variety and an imported composite, and then allocates the imported composite across foreign origins. Superscript  $D$  denotes the domestic-versus-import margin and superscript  $M$  the allocation across import sources.

For origin country  $m$ ,  $P_{mm,j,t}$  is the sector- $j$  source price at the national hub. Let  $d_{nm,j} \geq 1$  denote the iceberg cost of delivery from origin  $m$  to destination  $n$ , with  $d_{nn,j} = 1$ . The delivered bilateral price is

$$P_{nm,j,t} = d_{nm,j} P_{mm,j,t}, \quad d_{nn,j} = 1. \quad (134)$$

For  $m = n$ ,  $P_{nn,j,t}$  is both the national-hub source price and the domestic delivered price. The two Armington nests below aggregate  $\{P_{nm,j,t}\}_{m \in \mathcal{N}}$  into the purchaser price  $P_{n,j,t}$ .

**Lower nest: imported bundle across foreign sources.** Let  $\sigma_j^M > 1$  denote the elasticity of substitution among imported sector- $j$  varieties from different foreign origins, and let  $\zeta_{nm,j}^M > 0$  denote the corresponding conditional source weights, normalized so that

$$\sum_{m \neq n} \zeta_{nm,j}^M = 1. \quad (135)$$

The imported quantity composite is

$$Q_{n,j,t}^{\text{Imp}} = \left[ \sum_{m \neq n} (\zeta_{nm,j}^M)^{1/\sigma_j^M} Q_{nm,j,t}^{\frac{\sigma_j^M - 1}{\sigma_j^M}} \right]^{\frac{\sigma_j^M}{\sigma_j^M - 1}}, \quad (136)$$

with imported-composite price

$$P_{n,j,t}^{\text{Imp}} = \left[ \sum_{m \neq n} \zeta_{nm,j}^M P_{nm,j,t}^{1 - \sigma_j^M} \right]^{\frac{1}{1 - \sigma_j^M}}. \quad (137)$$

Conditional on importing sector  $j$ , the expenditure share sourced from foreign country  $m$  is

$$\lambda_{nm,j,t}^M = \frac{\zeta_{nm,j}^M P_{nm,j,t}^{1-\sigma_j^M}}{\sum_{\ell \neq n} \zeta_{n\ell,j}^M P_{n\ell,j,t}^{1-\sigma_j^M}}, \quad m \neq n, \quad (138)$$

so that

$$\sum_{m \neq n} \lambda_{nm,j,t}^M = 1. \quad (139)$$

**Upper nest: domestic versus imported composite.** Let  $\sigma_j^D > 1$  denote the elasticity of substitution between the domestic sector- $j$  variety and the imported sector- $j$  composite, and let  $\zeta_{n,j}^D \in (0, 1)$  denote the domestic CES weight. The sector- $j$  Armington composite used in country  $n$  is

$$Q_{n,j,t}^A = \left[ (\zeta_{n,j}^D)^{1/\sigma_j^D} Q_{nn,j,t}^{\frac{\sigma_j^D-1}{\sigma_j^D}} + (1 - \zeta_{n,j}^D)^{1/\sigma_j^D} Q_{n,j,t}^{\text{Imp} \frac{\sigma_j^D-1}{\sigma_j^D}} \right]^{\frac{\sigma_j^D}{\sigma_j^D-1}}. \quad (140)$$

Its country-sector composite purchaser price is

$$P_{n,j,t} = \left[ \zeta_{n,j}^D P_{nn,j,t}^{1-\sigma_j^D} + (1 - \zeta_{n,j}^D) P_{n,j,t}^{\text{Imp} 1-\sigma_j^D} \right]^{\frac{1}{1-\sigma_j^D}}. \quad (141)$$

Define the domestic expenditure share at the upper nest as

$$\lambda_{n,j,t}^D \equiv \frac{\zeta_{n,j}^D P_{nn,j,t}^{1-\sigma_j^D}}{P_{n,j,t}^{1-\sigma_j^D}}. \quad (142)$$

The corresponding import-composite share is  $1 - \lambda_{n,j,t}^D$ . Combining the two nests gives the overall bilateral expenditure shares:

$$\lambda_{nn,j,t} = \lambda_{n,j,t}^D, \quad \lambda_{nm,j,t} = (1 - \lambda_{n,j,t}^D) \lambda_{nm,j,t}^M, \quad m \neq n. \quad (143)$$

Therefore  $\lambda_{n,j,t}^D$  determines the domestic-versus-import margin, while  $\lambda_{nm,j,t}^M$  is the conditional source share within imports. The overall bilateral share  $\lambda_{nm,j,t}$  remains the object that maps destination expenditure into bilateral expenditure.

Let bilateral expenditure be

$$X_{nm,j,t} \equiv P_{nm,j,t} Q_{nm,j,t}, \quad (144)$$

and total sector- $j$  absorption expenditure be

$$X_{n,j,t} \equiv \sum_{m=1}^N X_{nm,j,t} = P_{n,j,t} Q_{n,j,t}^A. \quad (145)$$

Then

$$X_{nm,j,t} = \lambda_{nm,j,t} X_{n,j,t}. \quad (146)$$

Households have Cobb–Douglas preferences across sectoral composites:

$$C_{n,t} = \prod_{j \in \mathcal{J}} C_{n,j,t}^{\alpha_{n,j}^C}, \quad \sum_{j \in \mathcal{J}} \alpha_{n,j}^C = 1. \quad (147)$$

The consumption price index is

$$P_{n,t}^C = \prod_{j \in \mathcal{J}} \left( \frac{P_{n,j,t}}{\alpha_{n,j}^C} \right)^{\alpha_{n,j}^C}, \quad (148)$$

and final demand satisfies

$$P_{n,j,t} C_{n,j,t} = \alpha_{n,j}^C P_{n,t}^C C_{n,t}. \quad (149)$$

Intermediate use of sector- $k$  composites by sector  $j$  is

$$M_{n,j,k,t}^{int} \equiv \sum_{i \in \mathcal{I}_n} M_{i_{n,j},k,t}^{int}, \quad (150)$$

and, because  $\gamma_{n,j,k}$  is common across grids,

$$P_{n,k,t} M_{n,j,k,t}^{int} = \gamma_{n,j,k} Y_{n,j,t}. \quad (151)$$

Sector- $k$  absorption is the sum of final and intermediate uses:

$$Q_{n,k,t}^A = C_{n,k,t} + \sum_{j \in \mathcal{J}} M_{n,j,k,t}^{int}. \quad (152)$$

Equivalently, in values,

$$X_{n,k,t} = \alpha_{n,k}^C P_{n,t}^C C_{n,t} + \sum_{j \in \mathcal{J}} \gamma_{n,j,k} Y_{n,j,t}. \quad (153)$$

## 2.5 Market Clearing and Equilibrium

Land and water clear separately in every physical grid. Labor is mobile across sectors and grids within a country and has two distinct uses: direct production and land setup. The national labor-resource constraint is

$$\bar{F}_{n,t}^{LAB} = \underbrace{\sum_{j \in \mathcal{J}} \sum_{i \in \mathcal{I}_n} F_{i_{n,j},t}^{LAB}}_{\text{labor used directly in production}} + \underbrace{\sum_{i \in \mathcal{I}_n} S_{i_{n},agr,t}}_{\text{labor used for land setup}}. \quad (154)$$

The same national wage  $w_{n,t}^{LAB}$  applies to both uses. Only the first term enters the CDE production structure. The second term is real labor used to activate land and therefore enters labor-market clearing, but it is not a primary input inside the CDE production composite.

Local water-market clearing implies

$$r_{i_n,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} = \sum_{j \in \mathcal{J}} \gamma_{n,j}^V s_{i_n,j,t}^W Y_{i_n,j,t}, \quad \forall i \in \mathcal{I}_n. \quad (155)$$

Country- $n$  water-rent income is

$$WR_{n,t} \equiv \sum_{i \in \mathcal{I}_n} r_{i_n,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}}. \quad (156)$$

Actual gross land payments satisfy

$$RL_{i_n,t} = \sum_{j \in \mathcal{J}^L} \gamma_{n,j}^V s_{i_n,j,t}^L Y_{i_n,j,t}. \quad (157)$$

At the national hub,

$$Y_{m,j,t} = P_{mm,j,t} Q_{m,j,t} = \sum_{n=1}^N X_{nm,j,t}, \quad \forall m, j. \quad (158)$$

Using  $X_{nm,j,t} = P_{nm,j,t} Q_{nm,j,t}$  and  $P_{nm,j,t} = d_{nm,j} P_{mm,j,t}$ , the same condition can be written in quantities as

$$Q_{m,j,t} = \sum_{n=1}^N d_{nm,j} Q_{nm,j,t}, \quad \forall m, j. \quad (159)$$

Let  $D_{n,t}$  denote country  $n$ 's exogenous net transfer from the rest of the world, with

$$\sum_{n=1}^N D_{n,t} = 0. \quad (160)$$

The country trade balance is

$$\sum_{m=1}^N \sum_{j \in \mathcal{J}} X_{nm,j,t} = \sum_{m=1}^N \sum_{j \in \mathcal{J}} X_{mn,j,t} + D_{n,t}. \quad (161)$$

The representative household budget constraint is

$$\begin{aligned} P_{n,t}^C C_{n,t} &= w_{n,t}^{\text{LAB}} \bar{F}_{n,t}^{\text{LAB}} + NRL_{n,t} \\ &\quad + \sum_{i \in \mathcal{I}_n} r_{i_n,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} + D_{n,t} \\ &= w_{n,t}^{\text{LAB}} \bar{F}_{n,t}^{\text{LAB}} + NRL_{n,t} \\ &\quad + \sum_{i \in \mathcal{I}_n} r_{i_n,t}^{\text{Water}} \mathcal{A}_i \text{Div}_{i,t} + D_{n,t}. \end{aligned} \quad (162)$$

Because national labor includes both direct production labor and setup labor, while net land revenue subtracts the setup wage bill, the setup payment cancels once in aggregate factor-income

accounting. Factor income can therefore be written as

$$\begin{aligned}
& w_{n,t}^{\text{LAB}} \bar{F}_{n,t}^{\text{LAB}} + \text{NRL}_{n,t} + \sum_{i \in \mathcal{I}_n} r_{i,n,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} \\
&= \sum_{j \in \mathcal{J}} \gamma_{n,j}^V Y_{n,j,t} \\
&= w_{n,t}^{\text{LAB}} \bar{F}_{n,t}^{\text{LAB}} + \text{NRL}_{n,t} + \sum_{i \in \mathcal{I}_n} r_{i,n,t}^{\text{Water}} \mathcal{A}_i \text{Div}_{i,t}.
\end{aligned} \tag{163}$$

The labor and land accounting can be made explicit by combining (154) with the definition of net land revenue. Since  $\text{NRL}_{i,n,t} = \text{RL}_{i,n,t} - w_{n,t}^{\text{LAB}} S_{i,n,\text{agr},t}$ ,

$$\begin{aligned}
& w_{n,t}^{\text{LAB}} \bar{F}_{n,t}^{\text{LAB}} + \text{NRL}_{n,t} \\
&= \sum_{i \in \mathcal{I}_n} \sum_{j \in \mathcal{J}} w_{n,t}^{\text{LAB}} F_{i,n,j,t}^{\text{LAB}} + \sum_{i \in \mathcal{I}_n} w_{n,t}^{\text{LAB}} S_{i,n,\text{agr},t} \\
&\quad + \sum_{i \in \mathcal{I}_n} \left[ \text{RL}_{i,n,t} - w_{n,t}^{\text{LAB}} S_{i,n,\text{agr},t} \right] \\
&= \sum_{i \in \mathcal{I}_n} \sum_{j \in \mathcal{J}} w_{n,t}^{\text{LAB}} F_{i,n,j,t}^{\text{LAB}} + \sum_{i \in \mathcal{I}_n} \text{RL}_{i,n,t}.
\end{aligned} \tag{164}$$

Thus setup labor is counted once as wage income and is subtracted once from gross active-land payments when net land revenue is formed. It is not an additional production payment. Adding water rents then gives exactly the primary-factor payments generated by production, as in (163).

For completeness, the same net land-revenue object can be written directly in terms of the underlying active-land payments and setup-labor requirement. At the grid level,

$$\begin{aligned}
\text{NRL}_{i,n,t} &\equiv \sum_{j \in \mathcal{J}^L} \mathbb{E} \left[ r_{i,n,j,t}^{\text{Land}}(\omega) \mid \omega \in \Omega_{i,n,j,t} \right] F_{i,n,j,t}^{\text{Land}} \\
&\quad - w_{n,t}^{\text{LAB}} E_{i,n,0} \bar{F}_i^{\text{Land}} \left[ 1 - \left( \pi_{i,n,0,t}^L \right)^{\frac{v_1-1}{v_1}} \right] \\
&= \sum_{j \in \mathcal{J}^L} \gamma_{n,j}^V s_{i,n,j,t}^L Y_{i,n,j,t} \\
&\quad - w_{n,t}^{\text{LAB}} E_{i,n,0} \bar{F}_i^{\text{Land}} \left[ 1 - \left( \pi_{i,n,0,t}^L \right)^{\frac{v_1-1}{v_1}} \right].
\end{aligned} \tag{165}$$

Aggregating across grids within country  $n$  gives

$$\text{NRL}_{n,t} \equiv \sum_{i \in \mathcal{I}_n} \text{NRL}_{i,n,t}. \tag{166}$$

## 2.6 Intertemporal Water Extraction and Scarcity Value

This subsection closes the dynamic water-extraction problem. The physical water system derived above determines the evolution of Surface water, Shallow groundwater, and Deep groundwater.

The economic model takes these physical laws as given and chooses current extraction from the three sources.

The central distinction is between (i) the *current* cost of extracting water from a particular source and (ii) the *intertemporal* opportunity cost of reducing the physical stock available in the future. All dynamic conditions below are derived from the maximization problem and the three source-specific physical laws; no Hotelling or scarcity-rent equation is imposed separately.

**Timing, water aggregation, and the local water price.** For the extraction problem,  $X_{2,i,t-1}$ ,  $S_{i,t-1}$ , and  $D_{i,t-1}$  are the water stocks inherited at the beginning of period  $t$ . The economic agent chooses

$$\text{Div}_{i,t}^{\text{sur}}, \quad \text{Div}_{i,t}^{\text{Sh}}, \quad \text{Div}_{i,t}^{\text{Dp}},$$

and the hydrological equations determine the corresponding end-of-period stocks  $X_{2,i,t}$ ,  $S_{i,t}$ , and  $D_{i,t}$ .

As established above, the three physical sources are perfect substitutes in the within-period production economy. Total extraction is therefore

$$\text{Div}_{i,t} = \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}}, \quad (167)$$

and the corresponding local water endowment supplied to production is

$$\bar{F}_{i,t}^{\text{Water}} = \mathcal{A}_i \text{Div}_{i,t}. \quad (168)$$

Hence there is one local economic water price,  $r_{i,n,t}^{\text{Water}}$ , for all three sources within grid  $i$ . The static production equilibrium determines this price through local water-market clearing. In particular,

$$r_{i,n,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} = \sum_{j \in \mathcal{J}} \gamma_{n,j}^V s_{i,n,j,t}^W Y_{i,n,j,t}. \quad (169)$$

Conditional on the equilibrium price, current gross water revenue in grid  $i$  is therefore

$$r_{i,n,t}^{\text{Water}} \mathcal{A}_i \left( \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right). \quad (170)$$

Perfect substitution in production implies one contemporaneous water price. It does not, by itself, imply that the three physical water stocks have the same continuation value, because the three stocks obey different physical laws of motion. This distinction becomes explicit below.

**Reduced-form source-specific extraction costs.** We use a reduced-form stock-dependent extraction technology. This captures the groundwater-economics mechanism that depletion raises future extraction difficulty without introducing an additional pumping-depth state into the base-line model.

For Surface water, total extraction cost is

$$C_{i,t}^{\text{sur}} = \frac{\mathcal{A}_i}{GF_i^{\text{sur}}} \left\{ [\bar{c}_i^{\text{sur}} + \alpha_i^{\text{sur}} (X_{2,i,0} - X_{2,i,t-1})] \text{Div}_{i,t}^{\text{sur}} + \frac{\beta_i^{\text{sur}}}{2} (\text{Div}_{i,t}^{\text{sur}})^2 \right\}. \quad (171)$$

For Shallow groundwater,

$$C_{i,t}^{\text{Sh}} = \frac{\mathcal{A}_i}{GF_i^{\text{Sh}}} \left\{ [\bar{c}_i^{\text{Sh}} + \alpha_i^{\text{Sh}} (S_{i,0} - S_{i,t-1})] \text{Div}_{i,t}^{\text{Sh}} + \frac{\beta_i^{\text{Sh}}}{2} (\text{Div}_{i,t}^{\text{Sh}})^2 \right\}. \quad (172)$$

For Deep groundwater,

$$C_{i,t}^{\text{Dp}} = \frac{\mathcal{A}_i}{GF_i^{\text{Dp}}} \left\{ [\bar{c}_i^{\text{Dp}} + \alpha_i^{\text{Dp}} (D_{i,0} - D_{i,t-1})] \text{Div}_{i,t}^{\text{Dp}} + \frac{\beta_i^{\text{Dp}}}{2} (\text{Div}_{i,t}^{\text{Dp}})^2 \right\}. \quad (173)$$

The parameters have the same interpretation across sources.  $\bar{c}_i^a$  is the baseline unit extraction cost;  $\alpha_i^a \geq 0$  controls how strongly extraction cost rises when the corresponding stock falls below its initial level;  $\beta_i^a > 0$  controls within-period convexity; and  $GF_i^a > 0$  is the source-specific extraction-efficiency shifter.

For Surface water, the parameters absorb spatial differences in capture, access, and conveyance conditions. For Shallow and Deep groundwater, the stock term parsimoniously captures the increasing difficulty of pumping a depleted aquifer. Thus all three sources can have positive and increasing marginal extraction costs while retaining a common functional form.

The current marginal extraction cost per unit of physical water volume is defined from the total cost itself:

$$MC_{i,t}^a \equiv \frac{1}{\mathcal{A}_i} \frac{\partial C_{i,t}^a}{\partial \text{Div}_{i,t}^a}. \quad (174)$$

Equations (171)–(173) therefore imply

$$MC_{i,t}^{\text{sur}} = \frac{1}{GF_i^{\text{sur}}} [\bar{c}_i^{\text{sur}} + \alpha_i^{\text{sur}} (X_{2,i,0} - X_{2,i,t-1}) + \beta_i^{\text{sur}} \text{Div}_{i,t}^{\text{sur}}], \quad (175)$$

$$MC_{i,t}^{\text{Sh}} = \frac{1}{GF_i^{\text{Sh}}} [\bar{c}_i^{\text{Sh}} + \alpha_i^{\text{Sh}} (S_{i,0} - S_{i,t-1}) + \beta_i^{\text{Sh}} \text{Div}_{i,t}^{\text{Sh}}], \quad (176)$$

$$MC_{i,t}^{\text{Dp}} = \frac{1}{GF_i^{\text{Dp}}} [\bar{c}_i^{\text{Dp}} + \alpha_i^{\text{Dp}} (D_{i,0} - D_{i,t-1}) + \beta_i^{\text{Dp}} \text{Div}_{i,t}^{\text{Dp}}]. \quad (177)$$

The stock derivatives of total cost, which matter for the intertemporal conditions below, are

$$-\frac{1}{\mathcal{A}_i} \frac{\partial C_{i,t}^{\text{sur}}}{\partial X_{2,i,t-1}} = \frac{\alpha_i^{\text{sur}}}{GF_i^{\text{sur}}} \text{Div}_{i,t}^{\text{sur}}, \quad (178)$$

$$-\frac{1}{\mathcal{A}_i} \frac{\partial C_{i,t}^{\text{Sh}}}{\partial S_{i,t-1}} = \frac{\alpha_i^{\text{Sh}}}{GF_i^{\text{Sh}}} \text{Div}_{i,t}^{\text{Sh}}, \quad (179)$$

$$-\frac{1}{\mathcal{A}_i} \frac{\partial C_{i,t}^{\text{Dp}}}{\partial D_{i,t-1}} = \frac{\alpha_i^{\text{Dp}}}{GF_i^{\text{Dp}}} \text{Div}_{i,t}^{\text{Dp}}. \quad (180)$$

Hence an additional unit of inherited stock lowers current extraction cost whenever  $\alpha_i^a > 0$ .

**Discounting and the dynamic objective.** Let the date-0 present-value weight on period  $t$  be

$$\mathcal{D}_t^{\text{PV}} \equiv \prod_{\tau=1}^t \frac{1}{1+r_{\tau}^K}, \quad \mathcal{D}_0^{\text{PV}} = 1. \quad (181)$$

The notation  $\mathcal{D}_t^{\text{PV}}$  is used here to avoid conflict with the hydrological variable  $\Delta_t$ , which denotes the number of days in period  $t$  elsewhere in the model.

Taking the local equilibrium water-price sequence as given, the extraction problem is

$$\begin{aligned} \max_{\{\text{Div}_{i,t}^{\text{sur}}, \text{Div}_{i,t}^{\text{Sh}}, \text{Div}_{i,t}^{\text{Dp}}\}} \quad & \sum_{t=1}^T \mathcal{D}_t^{\text{PV}} \sum_{n \in \mathcal{N}} \sum_{i \in \mathcal{I}_n} \left\{ r_{i,n,t}^{\text{Water}} \mathcal{A}_i \left[ \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right] \right. \\ & \left. - C_{i,t}^{\text{sur}} - C_{i,t}^{\text{Sh}} - C_{i,t}^{\text{Dp}} \right\}. \end{aligned} \quad (182)$$

The initial stocks  $X_{2,i,0}$ ,  $S_{i,0}$ , and  $D_{i,0}$  are given. The controls are continuous and non-negative.

**Exact scalar dynamic constraints.** The dynamic problem is constrained by the same three physical laws derived above. We repeat them here because they are the equations from which the intertemporal first-order conditions are obtained.

First define the fraction of total extraction that returns directly to Surface water:

$$\rho_i^R \equiv v_i^{\text{surf}} (1 - v_i). \quad (183)$$

Since  $\mathcal{R}_{i,t} = (1 - v_i) \text{Div}_{i,t}$ , the Surface return-flow term satisfies

$$v_i^{\text{surf}} \mathcal{R}_{i,t} = \rho_i^R \left( \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right). \quad (184)$$

Surface water evolves according to

$$\begin{aligned}
X_{2,i,t} = & (1 - \kappa_i^{\text{rout}})X_{2,i,t-1} + q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} \\
& + \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} \\
& + \rho_i^R \left( \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right) \\
& - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}}.
\end{aligned} \tag{185}$$

Shallow groundwater evolves according to

$$S_{i,t} = S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t} - \text{Div}_{i,t}^{\text{Sh}}. \tag{186}$$

Deep groundwater evolves according to

$$D_{i,t} = D_{i,t-1} + f_{i,t}^{\text{Dp}} - \text{Div}_{i,t}^{\text{Dp}}. \tag{187}$$

The physical linkages entering these equations are also inherited directly from the hydrological block. In particular,

$$b_{i,t} = \beta_i^{\text{base}} S_{i,t-1}, \tag{188}$$

$$s_{i,t} = \zeta_i^{\text{bed}} (1 - \kappa_i^{\text{rout}}) X_{2,i,t-1}, \tag{189}$$

$$f_{i,t}^{\text{Sh}} = \min \{ (1 - \zeta_i) u_{i,t}^{\text{net}}, (\bar{S}_i - S_{i,t-1})_+ \}, \tag{190}$$

$$f_{i,t}^{\text{Dp}} = \min \{ \zeta_i u_{i,t}^{\text{net}}, (\bar{D}_i - D_{i,t-1})_+ \}, \tag{191}$$

$$q_{i,t} = u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}}. \tag{192}$$

The feasibility conditions are

$$\text{Div}_{i,t}^{\text{sur}} \geq 0, \quad \text{Div}_{i,t}^{\text{Sh}} \geq 0, \quad \text{Div}_{i,t}^{\text{Dp}} \geq 0, \tag{193}$$

and

$$X_{2,i,t} \geq 0, \quad S_{i,t} \geq 0, \quad D_{i,t} \geq 0. \tag{194}$$

In particular,

$$0 \leq \text{Div}_{i,t}^{\text{Sh}} \leq S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t}, \tag{195}$$

$$0 \leq \text{Div}_{i,t}^{\text{Dp}} \leq D_{i,t-1} + f_{i,t}^{\text{Dp}}. \tag{196}$$

Surface feasibility is imposed by (185) together with  $X_{2,i,t} \geq 0$ .

**Lagrangian and physical stock values.** Because there are three distinct physical state equations, the constrained dynamic problem has three physical co-state multipliers. Let

$$\lambda_{i,t}^X, \quad \lambda_{i,t}^S, \quad \lambda_{i,t}^D$$

denote the current-value marginal values of one additional unit of Surface, Shallow, and Deep water, respectively. The constraints are multiplied by  $\mathcal{A}_i$ , so these multipliers are measured per unit of physical water volume and can be compared directly with the local water price and marginal extraction costs.

For notational transparency, the Lagrangian is written using the explicit scalar constraints rather than a vector law of motion:

$$\begin{aligned} \mathcal{L} = \sum_{t=1}^T \mathcal{D}_t^{\text{PV}} \sum_{n \in \mathcal{N}} \sum_{i \in \mathcal{I}_n} \bigg\{ & r_{i,n,t}^{\text{Water}} \mathcal{A}_i \left( \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right) \\ & - C_{i,t}^{\text{sur}} - C_{i,t}^{\text{Sh}} - C_{i,t}^{\text{Dp}} \\ & + \mathcal{A}_i \lambda_{i,t}^X \left[ (1 - \kappa_i^{\text{rout}}) X_{2,i,t-1} + q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} \right. \\ & \quad + \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} \\ & \quad + \rho_i^R \left( \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}} \right) \\ & \quad \left. - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}} - X_{2,i,t} \right] \\ & + \mathcal{A}_i \lambda_{i,t}^S \left[ S_{i,t-1} + f_{i,t}^{\text{Sh}} + s_{i,t} - b_{i,t} - \text{Div}_{i,t}^{\text{Sh}} - S_{i,t} \right] \\ & \left. + \mathcal{A}_i \lambda_{i,t}^D \left[ D_{i,t-1} + f_{i,t}^{\text{Dp}} - \text{Div}_{i,t}^{\text{Dp}} - D_{i,t} \right] \right\}. \end{aligned} \tag{197}$$

The multipliers are therefore generated by the constrained maximization problem. They are not imposed resource rents.

**Extraction first-order conditions.** For an interior Surface-water extraction choice, differentiate (197) with respect to  $\text{Div}_{i,t}^{\text{sur}}$ . Using (174) gives

$$r_{i,n,t}^{\text{Water}} - MC_{i,t}^{\text{sur}} + \rho_i^R \lambda_{i,t}^X - \lambda_{i,t}^X = 0. \tag{198}$$

Hence

$$r_{i,n,t}^{\text{Water}} = MC_{i,t}^{\text{sur}} + (1 - \rho_i^R) \lambda_{i,t}^X. \tag{199}$$

For Shallow groundwater,

$$r_{i,n,t}^{\text{Water}} - MC_{i,t}^{\text{Sh}} + \rho_i^R \lambda_{i,t}^X - \lambda_{i,t}^S = 0, \quad (200)$$

so

$$r_{i,n,t}^{\text{Water}} = MC_{i,t}^{\text{Sh}} + \lambda_{i,t}^S - \rho_i^R \lambda_{i,t}^X. \quad (201)$$

For Deep groundwater,

$$r_{i,n,t}^{\text{Water}} - MC_{i,t}^{\text{Dp}} + \rho_i^R \lambda_{i,t}^X - \lambda_{i,t}^D = 0, \quad (202)$$

so

$$r_{i,n,t}^{\text{Water}} = MC_{i,t}^{\text{Dp}} + \lambda_{i,t}^D - \rho_i^R \lambda_{i,t}^X. \quad (203)$$

These equations derive the dynamic scarcity component directly. In particular, define the source-specific marginal user costs implied by the three FOCs:

$$UC_{i,t}^{\text{sur}} \equiv (1 - \rho_i^R) \lambda_{i,t}^X, \quad (204)$$

$$UC_{i,t}^{\text{Sh}} \equiv \lambda_{i,t}^S - \rho_i^R \lambda_{i,t}^X, \quad (205)$$

$$UC_{i,t}^{\text{Dp}} \equiv \lambda_{i,t}^D - \rho_i^R \lambda_{i,t}^X. \quad (206)$$

These are definitions of expressions already obtained from the FOCs; they are not additional behavioral assumptions. Therefore every active source satisfies

$$r_{i,n,t}^{\text{Water}} = MC_{i,t}^a + UC_{i,t}^a. \quad (207)$$

**Cross-source allocation and the law of one water price.** Because the three extracted waters are perfect substitutes in production, they face the same local water price. If all three sources are used simultaneously at an interior solution,

$$\begin{aligned} r_{i,n,t}^{\text{Water}} &= MC_{i,t}^{\text{sur}} + UC_{i,t}^{\text{sur}} \\ &= MC_{i,t}^{\text{Sh}} + UC_{i,t}^{\text{Sh}} \\ &= MC_{i,t}^{\text{Dp}} + UC_{i,t}^{\text{Dp}}. \end{aligned} \quad (208)$$

Thus perfect substitution equalizes *full marginal costs*, not necessarily the physical co-state multipliers themselves. The reason is that one unit left in Surface, Shallow, or Deep water does not have the same future physical consequences.

For a source that is not used, the corresponding Kuhn–Tucker condition is

$$r_{i,n,t}^{\text{Water}} \leq MC_{i,t}^a + UC_{i,t}^a, \quad \text{Div}_{i,t}^a = 0. \quad (209)$$

Hence a relatively costly source, such as Deep groundwater, can remain inactive until the common local water price becomes high enough to cover both its current marginal extraction

cost and its marginal user cost.

**Co-state equations implied by the hydrological laws.** We now derive the evolution of the physical stock values. These equations are the water-resource analogues of Euler equations.

The recharge functions are piecewise differentiable because of the storage capacity constraints. Away from the kink, define the derivatives implied by (190) and (191):

$$\phi_{i,t}^{\text{Sh}} \equiv \frac{\partial f_{i,t}^{\text{Sh}}}{\partial S_{i,t-1}} = \begin{cases} 0, & (1 - \zeta_i)u_{i,t}^{\text{net}} < \bar{S}_i - S_{i,t-1}, \\ -1, & (1 - \zeta_i)u_{i,t}^{\text{net}} > \bar{S}_i - S_{i,t-1}, \end{cases} \quad (210)$$

and

$$\phi_{i,t}^{\text{Dp}} \equiv \frac{\partial f_{i,t}^{\text{Dp}}}{\partial D_{i,t-1}} = \begin{cases} 0, & \zeta_i u_{i,t}^{\text{net}} < \bar{D}_i - D_{i,t-1}, \\ -1, & \zeta_i u_{i,t}^{\text{net}} > \bar{D}_i - D_{i,t-1}, \end{cases} \quad (211)$$

At the kink, the corresponding Kuhn–Tucker/subgradient condition applies. Because runoff is the residual,

$$\frac{\partial q_{i,t}}{\partial S_{i,t-1}} = -\phi_{i,t}^{\text{Sh}}, \quad \frac{\partial q_{i,t}}{\partial D_{i,t-1}} = -\phi_{i,t}^{\text{Dp}}. \quad (212)$$

Baseflow and seepage imply

$$\frac{\partial b_{i,t}}{\partial S_{i,t-1}} = \beta_i^{\text{base}}, \quad \frac{\partial s_{i,t}}{\partial X_{2,i,t-1}} = \zeta_i^{\text{bed}}(1 - \kappa_i^{\text{rout}}). \quad (213)$$

Under the explicit hydrological formulas currently specified above, differentiating the Lagrangian with respect to  $X_{2,i,t}$  gives

$$\begin{aligned} (1 + r_{t+1}^K)\lambda_{i,t}^X &= \frac{\alpha_i^{\text{sur}}}{GF_i^{\text{sur}}} \text{Div}_{i,t+1}^{\text{sur}} \\ &\quad + (1 - \kappa_i^{\text{rout}})(1 - \zeta_i^{\text{bed}})\lambda_{i,t+1}^X \\ &\quad + \zeta_i^{\text{bed}}(1 - \kappa_i^{\text{rout}})\lambda_{i,t+1}^S \\ &\quad + \kappa_i^{\text{rout}} \sum_j w_{i \rightarrow j} \lambda_{j,t+1}^X. \end{aligned} \quad (214)$$

The four terms on the right-hand side are, respectively: the reduction in next-period Surface extraction cost generated by preserving more Surface water; retained local Surface value; value transferred to Shallow water through seepage; and value transferred downstream through the routing network.

Differentiating with respect to  $S_{i,t}$  gives

$$\begin{aligned}
(1 + r_{t+1}^K) \lambda_{i,t}^S &= \frac{\alpha_i^{\text{Sh}}}{GF_i^{\text{Sh}}} \text{Div}_{i,t+1}^{\text{Sh}} \\
&+ \left( \beta_i^{\text{base}} - \phi_{i,t+1}^{\text{Sh}} \right) \lambda_{i,t+1}^X \\
&+ \left[ 1 + \phi_{i,t+1}^{\text{Sh}} - \beta_i^{\text{base}} \right] \lambda_{i,t+1}^S.
\end{aligned} \tag{215}$$

Preserving Shallow groundwater therefore has three effects in this specification: it lowers future Shallow extraction cost, changes the amount of water reaching Surface water through baseflow and the runoff response, and changes the Shallow stock carried forward.

Finally, differentiating with respect to  $D_{i,t}$  gives

$$\begin{aligned}
(1 + r_{t+1}^K) \lambda_{i,t}^D &= \frac{\alpha_i^{\text{Dp}}}{GF_i^{\text{Dp}}} \text{Div}_{i,t+1}^{\text{Dp}} \\
&- \phi_{i,t+1}^{\text{Dp}} \lambda_{i,t+1}^X \\
&+ \left[ 1 + \phi_{i,t+1}^{\text{Dp}} \right] \lambda_{i,t+1}^D.
\end{aligned} \tag{216}$$

The Deep co-state equation therefore combines the future cost-saving effect of a larger Deep stock, the effect of Deep recharge capacity on residual Surface runoff, and the Deep stock carried into the following period.

Equations (214)–(216) are the general intertemporal scarcity conditions generated by the model. They are not imposed Hotelling rules.

**Deep groundwater as an exhaustible-resource special case.** Deep groundwater provides the cleanest connection to the standard exhaustible-resource problem. If Deep recharge is negligible,

$$f_{i,t}^{\text{Dp}} \simeq 0, \tag{217}$$

then

$$D_{i,t} \simeq D_{i,t-1} - \text{Div}_{i,t}^{\text{Dp}}. \tag{218}$$

In this case  $\phi_{i,t}^{\text{Dp}} = 0$ , and (216) becomes

$$(1 + r_{t+1}^K) \lambda_{i,t}^D = \frac{\alpha_i^{\text{Dp}}}{GF_i^{\text{Dp}}} \text{Div}_{i,t+1}^{\text{Dp}} + \lambda_{i,t+1}^D. \tag{219}$$

Equivalently,

$$\lambda_{i,t+1}^D = (1 + r_{t+1}^K) \lambda_{i,t}^D - \frac{\alpha_i^{\text{Dp}}}{GF_i^{\text{Dp}}} \text{Div}_{i,t+1}^{\text{Dp}}. \tag{220}$$

This is the stock-dependent-cost version of the Deep-water Hotelling condition. The second

term appears because preserving more Deep groundwater also lowers future physical extraction cost.

If, in addition, extraction cost is independent of the remaining Deep stock,

$$\alpha_i^{\text{Dp}} = 0, \quad (221)$$

then the previous equation reduces exactly to

$$\lambda_{i,t+1}^D = (1 + r_{t+1}^K) \lambda_{i,t}^D. \quad (222)$$

Thus the standard discrete-time Hotelling rule is obtained from the model as a special case; it is not imposed as a primitive equation.

If return flows are also shut down for this special-case comparison,  $\rho_i^R = 0$ , the Deep extraction FOC becomes

$$r_{i_n,t}^{\text{Water}} - MC_{i,t}^{\text{Dp}} = \lambda_{i,t}^D. \quad (223)$$

Combining (222) and (223) gives the familiar net-rent form:

$$\begin{aligned} r_{i_n,t+1}^{\text{Water}} - MC_{i,t+1}^{\text{Dp}} \\ = (1 + r_{t+1}^K) [r_{i_n,t}^{\text{Water}} - MC_{i,t}^{\text{Dp}}]. \end{aligned} \quad (224)$$

**One water price versus one physical shadow value.** The perfect-substitution assumption requires one local water price because one unit extracted from any source supplies the same within-period water input. It does *not* generally imply

$$\lambda_{i,t}^X = \lambda_{i,t}^S = \lambda_{i,t}^D.$$

Those three multipliers are attached to three distinct physical state constraints, and equations (214)–(216) show explicitly that their continuation values differ whenever routing, seepage, baseflow, recharge, or source-specific stock effects differ.

A genuinely common physical scarcity multiplier arises only under the stronger special case in which the three physical stocks can be collapsed into one aggregate stock with a common law of motion. For example, if an aggregate stock  $Z_{i,t}$  satisfies

$$Z_{i,t} = Z_{i,t-1} + I_{i,t} - \text{Div}_{i,t}^{\text{sur}} - \text{Div}_{i,t}^{\text{Sh}} - \text{Div}_{i,t}^{\text{Dp}}, \quad (225)$$

and source identity has no separate effect on future physical availability, then a single multiplier on (225) is sufficient.

The full Deep Waters model intentionally retains the three physical state equations, so the relevant law of one price is instead (208): active sources have the same *full marginal cost* because they sell into one local water market.

**Fixed extraction infrastructure and other extensions.** The baseline problem is continuous. It treats existing wells, intakes, and conveyance infrastructure as already available and therefore focuses on the intensive margin of how much water to extract.

If source  $a$  requires a true activation cost, one can extend the total cost to

$$C_{i,t}^{a,\text{all}} = F_i^a z_{i,t}^a + C_{i,t}^a, \quad \text{Div}_{i,t}^a \leq \overline{\text{Div}}_i^a z_{i,t}^a, \quad z_{i,t}^a \in \{0, 1\}. \quad (226)$$

The fixed cost affects whether the source is activated. Conditional on an active source,

$$\frac{\partial \left( F_i^a z_{i,t}^a \right)}{\partial \text{Div}_{i,t}^a} = 0,$$

so it does not enter the interior marginal extraction condition. This is why the continuous baseline omits the discrete activation margin.

The source-specific efficiency shifter  $GF_i^a$  can also be made time-varying if future applications require exogenous extraction-technology progress. Doing so changes the level and time path of marginal extraction costs but does not change the structure of the dynamic first-order conditions derived above.

**Summary of the dynamic mechanism.** The complete extraction block can be read in four steps.

First, current hydrological stocks determine current source-specific extraction costs through (171)–(173).

Second, the three extraction quantities are perfect substitutes in production and jointly determine the local water endowment through (167)–(168).

Third, current extraction changes future Surface, Shallow, and Deep stocks through the exact physical laws (185)–(187).

Fourth, the Lagrangian combines these two margins. The extraction FOCs deliver the common-price/full-marginal-cost conditions (199)–(203), while the co-state equations (214)–(216) determine the evolution of the scarcity values. The standard Hotelling rule then follows as the limiting Deep-water case with negligible recharge and no stock-dependent extraction cost.

## 2.7 Geographic Model with Economic Structure

The geographic model combines the physical water system with the economic structure. Within each period  $t$ : (1) exogenous precipitation and temperature are realized and the snow module runs; (2) total liquid input is partitioned into abstraction, recharge, and runoff; (3) water stocks are updated; (4) runoff is routed downstream; (5) given the current hydrological state, the economic equilibrium is solved—water owners choose extraction, while firms produce and trade; (6) extraction depletes stocks and generates return flows. The key feedback loop is that extraction in step 5 affects the hydrological state governing step 2 in the next period. This intertemporal

externality is the hydrological analogue of the fossil fuel extraction problem in [Arkolakis and Walsh \(2023\)](#).

### 3 The Water Balance Model

This section presents the physical core of the paper: a dynamic spatial-temporal model that tracks water across multiple states and locations. We present the main equations here; proofs of mass conservation, properties of the routing operator, and the alternative curve-number specification are in the appendix.

#### 3.1 Water Motion

The model considers a grid of  $I$  discrete locations, each holding water in  $S$  states, over discrete periods  $t$ . We define  $\mathbf{W}_{ti}$  as the  $S \times 1$  vector of water quantities at location  $i$ , time  $t$ , across all states. Throughout, we work in water-equivalent depth units (inches per period). The state of the entire system at time  $t$  is the  $S \times I$  matrix  $\mathbf{W}_t$ .

The system evolves from an initial endowment  $\mathbf{W}_0 = \mathbf{E}_0$  according to:

$$\mathbf{W}_t \equiv \mathbf{W}_{t-1} + \Omega_t, \quad (227)$$

where the flow matrix  $\Omega_t$  decomposes into within-location state transitions  $\mathbf{T}_t$  and across-location spatial transitions  $\mathbf{S}_t$ :

$$\Omega_t \equiv \mathbf{T}_t + \mathbf{S}_t. \quad (228)$$

The within-location transitions are constructed from location-specific state-transition matrices  $\mathbf{B}_{it}$  ( $S \times S$ ), which capture gross movements of water between states. Net within-location changes are:

$$\boldsymbol{\tau}_{it} \equiv \mathbf{B}_{it}^\top \mathbf{1} - \mathbf{B}_{it} \mathbf{1} \in \mathbb{R}^S, \quad \mathbf{T}_t \equiv [\boldsymbol{\tau}_{1t} \ \boldsymbol{\tau}_{2t} \ \cdots \ \boldsymbol{\tau}_{It}] \in \mathbb{R}^{S \times I}. \quad (229)$$

The state-transition matrix at each location encodes the physical processes governing water transformation. Rows and columns correspond to the water states (atmosphere, surface/root-zone, shallow groundwater, deep groundwater, deep aquifer, agricultural land), with entries representing precipitation, evapotranspiration, runoff, recharge, baseflow, infiltration, and irrigation withdrawals:

\*\*\*The glacier runoff term  $g_{i,t}$  requires a formal definition.\*\*\*

$$B_{it} \equiv \begin{pmatrix} 0 & r_{i,t}^{\text{rain}} + q_{i,t} + g_{i,t} & r_{i,t}^{\text{snow}} & 0 & 0 & r_{i,t} \\ e_{i,t}^{\text{open}} & 0 & 0 & 0 & 0 & \text{Div}_{i,t}^{\text{sur}} \\ e_{i,t} & q_{i,t} & 0 & f_{i,t}^{\text{Sh}} & 0 & \text{Div}_{i,t}^{\text{Sh}} \\ 0 & e_{i,t} & b_{i,t} & 0 & 0 & \text{Div}_{i,t}^{\text{Sh}} \\ 0 & 0 & 0 & 0 & 0 & \text{Div}_{i,t}^{\text{Dp}} \\ e_{i,t} & q_{i,t} & f_{i,t} & 0 & 0 & 0 \end{pmatrix}$$

**Warning note.** The withdrawal terms in the last column of the third and fourth rows are currently both written as  $\text{Div}_{i,t}^{\text{Sh}}$ . This duplication should be carefully checked against the intended water-state transitions before the final specification is fixed.

In the empirical implementation, the hydrologic model tracks root-zone moisture  $M_{i,t}$ , shallow groundwater storage  $S_{i,t}$ , deep groundwater storage  $D_{i,t}$ , and snow water equivalent  $\text{SWE}_{i,t}$ . Surface runoff  $q_{i,t}$  and routed inflow  $l_{i,t}$  are computed as auxiliary diagnostic flows.

### 3.2 Snowpack and Effective Precipitation

Let  $r_{i,t}^{\text{rain}}$  and  $r_{i,t}^{\text{snow}}$  denote rainfall and snowfall (inches per period),  $T_{i,t}$  air temperature ( $^{\circ}\text{C}$ ), and  $\Delta_t$  the days in period  $t$ . Snowmelt follows a degree-day rule:

$$m_{i,t} = \min\left\{\text{SWE}_{i,t-1} + r_{i,t}^{\text{snow}}, \text{ddf} \cdot \max(T_{i,t} - T^{\text{melt}}, 0) \cdot \Delta_t\right\}, \quad (230)$$

with snowpack updating as  $\text{SWE}_{i,t} = \max\{\text{SWE}_{i,t-1} + r_{i,t}^{\text{snow}} - m_{i,t}, 0\}$ . Effective liquid water input is:

$$\tilde{r}_{i,t} \equiv r_{i,t}^{\text{rain}} + m_{i,t}. \quad (231)$$

### 3.3 Runoff, Infiltration, and Stock Updates

Total liquid input at cell  $i$  in period  $t$  combines local effective precipitation with routed inflow from upstream:

$$u_{i,t} \equiv \tilde{r}_{i,t} + l_{i,t-1}. \quad (232)$$

After initial losses  $a_{i,t} = \min\{u_{i,t}, \bar{a}_i\}$  (interception, depression storage), the remaining water  $u_{i,t}^{\text{net}} = u_{i,t} - a_{i,t}$  is allocated to shallow and deep recharge using the deep-percolation share  $\xi_i \in [0, 1]$ , each capped by available storage capacity:

$$f_{i,t}^{\text{Sh}} = \min\{(1 - \xi_i) u_{i,t}^{\text{net}}, (\bar{S}_i - S_{i,t-1})_+\}, \quad f_{i,t}^{\text{Dp}} = \min\{\xi_i u_{i,t}^{\text{net}}, (\bar{D}_i - D_{i,t-1})_+\}. \quad (233)$$

Surface runoff is the residual:

$$q_{i,t} \equiv u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}} \geq 0. \quad (234)$$

These equations imply the conservative within-cell mass balance  $u_{i,t} = a_{i,t} + f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}} + q_{i,t}$  (see Appendix A5 for proof). Root-zone moisture, shallow, and deep groundwater stocks update as:

$$M_{i,t} = \max\{M_{i,t-1} + a_{i,t} - e_{i,t}, 0\}, \quad S_{i,t} = S_{i,t-1} + f_{i,t}^{\text{Sh}}, \quad D_{i,t} = D_{i,t-1} + f_{i,t}^{\text{Dp}}. \quad (235)$$

### 3.4 Routing and Upstream Inflow

Locally generated runoff is routed one step downstream per period using a fixed flow-direction raster. With downstream routing weight  $w_{j \rightarrow i} \geq 0$  ( $\sum_i w_{j \rightarrow i} \leq 1$ ), routed inflow is:

$$l_{i,t} \equiv \sum_{j=1}^I w_{j \rightarrow i} q_{j,t}. \quad (236)$$

In vector form,  $\boldsymbol{l}_t = \mathbf{W}^\top \boldsymbol{q}_t$ , where  $\mathbf{W}$  is the row-stochastic routing matrix implied by the flow-direction grid. This creates a spatial multiplier: a unit of precipitation at an upstream location generates inflow at every downstream cell along the drainage path. Properties of the routing operator (acyclicity, nilpotency, Leontief-inverse interpretation) are developed in Appendix A6.

### 3.5 Surface Water Balance

For applications requiring explicit surface-water stocks (lakes, reservoirs, within-cell residence time), we introduce  $X_{2,i,t}$  with mass balance:

$$\begin{aligned} X_{2,i,t} = & \underbrace{(1 - \kappa_i^{\text{rout}}) X_{2,i,t-1}}_{\text{retained storage}} + \underbrace{q_{i,t}}_{\text{overland runoff}} + \underbrace{b_{i,t}}_{\text{baseflow}} + \underbrace{r_{i,t}^{\text{water}}}_{\text{precip on open water}} + \underbrace{\sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1}}_{\text{upstream inflow}} \\ & + \underbrace{v_i^{\text{surf}} \mathcal{R}_{i,t}}_{\text{return flows}} - \underbrace{\text{Div}_{i,t}^{\text{sur}}}_{\text{diversions}} - \underbrace{e_{i,t}^{\text{open}}}_{\text{open-water ET}} - \underbrace{s_{i,t}}_{\text{seepage}} - \underbrace{O_{i,t}^{\text{ocean}}}_{\text{terminal export}}, \end{aligned} \quad (237)$$

where  $\kappa_i^{\text{rout}} \in (0, 1]$  is the routing fraction (small for lakes/reservoirs, large for channels). Seepage  $s_{i,t} = \zeta_i^{\text{bed}}(1 - \kappa_i^{\text{rout}})X_{2,i,t-1}$  closes the loop with groundwater. The baseline runoff-routing specification is nested as a zero-residence special case.

## 4 Data

Our model integrates high-resolution hydrological, climatic, and geospatial datasets to parameterize the spatial-temporal dynamics of water stocks and flows. Table 1 summarizes the data sources; detailed processing steps are in Appendix A9.

Precipitation and land-surface forcings are from NASA's GLDAS Noah land surface model at 0.25° resolution: rainfall, snowfall, evapotranspiration, and near-surface air temperature. Soil moisture and snow water equivalent from the same source provide initial conditions. Subsur-

face storage capacities ( $\bar{S}_i, \bar{D}_i$ ) are constructed from the aquifer thickness dataset of [de Graaf et al. \(2015\)](#) combined with porosity information. The routing operator  $\mathbf{W}$  is built from HydroSHEDS flow-direction rasters. For the alternative curve-number specification, we use the GCN250 dataset ([Jaafar et al., 2019](#)). Validation relies on NASA GRACE/GRACE-FO satellite gravity measurements of total terrestrial water storage changes. Additional datasets include ESA CCI land cover, river channel width data, the World Water Table dataset, MODIS evapotranspiration, and MERRA-2 precipitation for robustness.

Table 1: Overview of Data Sources

| Input                               | Source                                         | Resolution     |
|-------------------------------------|------------------------------------------------|----------------|
| <i>Hydrological Forcings</i>        |                                                |                |
| Rainfall, snowfall, temperature, ET | NASA GLDAS Noah                                | 0.25°, monthly |
| Soil moisture, SWE (initialization) | NASA GLDAS Noah                                | 0.25°          |
| <i>Subsurface and Routing</i>       |                                                |                |
| Aquifer thickness                   | <a href="#">de Graaf et al. (2015)</a>         | 0.5°           |
| Soil hydraulic conductivity         | SoilKsat database                              | Point data     |
| Flow directions                     | HydroSHEDS                                     | 15 arc-sec     |
| Curve numbers                       | GCN250 ( <a href="#">Jaafar et al., 2019</a> ) | 250 m          |
| <i>Validation</i>                   |                                                |                |
| Total water storage anomalies       | NASA GRACE/GRACE-FO                            | ~ 300 km       |
| Land cover                          | ESA CCI                                        | 300 m          |
| <i>Economic</i>                     |                                                |                |
| Trade elasticities                  | Gravity estimates                              | Country-pair   |
| Factor shares                       | GTAP                                           | Country-sector |

## 5 Quantitative Analysis

### 5.1 Calibration

The model contains parameters taken directly from external data (storage capacities, curve numbers, flow directions) and parameters that must be calibrated (the degree-day factor  $ddf$ , the deep-percolation share  $\zeta_i$ , and the abstraction capacity  $\bar{a}_i$ ). We calibrate the latter by matching model-predicted monthly runoff to observations from gauged river basins. Economic parameters follow standard procedures: trade elasticities from the gravity literature, factor shares from GTAP, and trade costs from a gravity regression. Table 2 summarizes parameter choices.

### 5.2 Model Fit

We validate the model along three dimensions. First, model-predicted monthly soil moisture is compared against GLDAS soil moisture products at the grid-cell level. Second, model-predicted

Table 2: Overview of Parameter Choices

| Parameter                                 | Value/Source                 | Description                                   | Type                   |
|-------------------------------------------|------------------------------|-----------------------------------------------|------------------------|
| <i>Hydrological Parameters</i>            |                              |                                               |                        |
| ddf                                       | Calibrated                   | Degree-day melt factor                        | Calibrated             |
| $T^{\text{melt}}$                         | 0°C                          | Melt threshold temperature                    | Fixed                  |
| $\xi_i$                                   | Calibrated                   | Deep-percolation share                        | Calibrated             |
| $\bar{a}_i$                               | Soil/land cover data         | Abstraction capacity                          | Data                   |
| $\bar{S}_i, \bar{D}_i$                    | Hydrogeology data            | Storage capacities                            | Data                   |
| $w_{j \rightarrow i}$                     | HydroSHEDS                   | Routing weights                               | Data                   |
| <i>Economic Parameters</i>                |                              |                                               |                        |
| $\sigma_j^M, \sigma_j^D$                  | Gravity / external estimates | Armington substitution elasticities           | External               |
| $\alpha_{n,j}^C$                          | GTAP / national accounts     | Final-expenditure shares                      | Data                   |
| $\gamma_{n,j}^V, \gamma_{n,j,k}$          | GTAP / input-output data     | Upper-nest value shares                       | Data                   |
| $d_{nm,j}$                                | Gravity regression           | Bilateral iceberg trade costs, $d_{nn,j} = 1$ | Estimated              |
| $\beta_{n,j}^x, \sigma_j^x, \epsilon_j^x$ | Calibration / estimation     | CDE parameters, $x \in \{\text{LAB}, L, W\}$  | Calibrated / estimated |
| $\nu_1, \nu_2$                            | Land-allocation calibration  | Nested Fréchet shape parameters               | Calibrated             |

runoff aggregated to major river basins is compared against observed streamflow records. Third, model-predicted total water storage changes are compared against GRACE satellite anomalies. These exercises confirm that the hydrological core captures the salient spatial and temporal patterns in observed water stocks and flows.

### 5.3 Water Allocation Across the Globe

The calibrated model produces monthly maps of root-zone moisture, shallow and deep ground-water stocks, snowpack, surface runoff, and routed streamflow for each grid cell. The results reveal three salient patterns. First, the spatial distribution of water is highly unequal: tropical regions maintain high soil moisture year-round, while arid regions depend on groundwater accumulated over long horizons. Second, the routing network creates strong spatial correlations—downstream regions benefit from upstream runoff, following the topographic drainage network. Third, the seasonal cycle of snowpack is critical in mid- and high-latitude regions, where winter accumulation and spring melt dominate the annual water budget.

### 5.4 Groundwater Depletion and Welfare

Regions relying on non-renewable groundwater face substantial welfare losses as aquifers deplete. When extraction exceeds natural recharge, deep groundwater stocks decline monotonically, raising extraction costs. The welfare losses concentrate in regions with large agricultural sectors dependent on deep-well irrigation. In contrast, regions with abundant renewable surface water face stable costs over time. This divergence generates a dynamic reallocation of comparative advantage: water-intensive production gradually shifts toward regions with sustainable

supplies.

## **5.5 Spatial Spillovers Through the Routing Network**

The routing network generates important spatial spillovers. Upstream water use affects downstream availability through reduced runoff, baseflow, and return flows. The spillovers are largest along major river systems where populated regions lie downstream of major agricultural or industrial water users.

# **6 Deep Water Scenarios**

We use the calibrated model to project the hydrological-economic system forward under alternative assumptions about climate, extraction, and infrastructure.

## **6.1 Baseline Projections**

Under current extraction rates and climatic forcing, the model projects a growing divergence between two types of regions. Water-abundant regions with high renewable surface flow maintain stable or declining water costs, while regions mining non-renewable groundwater face steadily rising extraction costs as the water table drops. The baseline scenario reveals that several major aquifer systems—including parts of the High Plains (Ogallala), the North China Plain, and the Indo-Gangetic Basin—face depletion timelines of decades under current use patterns.

## **6.2 Climate Change Scenarios**

We perturb the precipitation forcing using downscaled climate projections and solve for the new distribution of water stocks and economic activity. Regions where precipitation declines face compounding losses: reduced direct rainfall, lower recharge, declining groundwater stocks, and reduced downstream inflow from upstream tributaries. Regions where precipitation increases may gain, but the gains depend on whether the additional water can be stored (in aquifers or reservoirs) or simply runs off.

## **6.3 Extraction Scenarios**

We vary extraction rates to trace out the dynamic path of groundwater depletion under alternative assumptions. A “business as usual” scenario maintains current extraction growth, while a “sustainable yield” scenario caps extraction at the long-run recharge rate. The difference in welfare paths between these scenarios quantifies the dynamic externality of current over-extraction.

## 7 Sustainable Water Policies

### 7.1 Water Pricing Reform

We evaluate the welfare effects of moving from subsidized or unpriced water to marginal-cost pricing that reflects the full extraction cost plus the dynamic externality of depletion. Marginal-cost pricing reallocates water from low-value to high-value uses, reducing overall extraction and slowing aquifer depletion. The aggregate welfare gains are positive, though with distributional consequences: agricultural regions that currently benefit from cheap groundwater face higher costs, while downstream regions benefit from improved streamflow.

### 7.2 Inter-Basin Transfer Infrastructure

We model inter-basin water transfers as additions to the routing matrix  $\mathbf{W}$  that create new links between donor and recipient basins. The counterfactual equilibrium shows that large-scale transfers generate substantial welfare gains in recipient regions, but at the cost of reduced flows in the donor basin and along its downstream network. The net welfare effect depends on the relative marginal value of water in the two basins and on the cost of the transfer infrastructure.

### 7.3 Grid Improvements and Storage Infrastructure

Analogous to the grid expansion counterfactuals in [Arkolakis and Walsh \(2023\)](#), we evaluate the impact of investments in water storage infrastructure (dams, managed aquifer recharge) and conveyance improvements. These investments are modeled as changes to the storage capacities  $\bar{S}_i$ ,  $\bar{D}_i$  and the routing matrix  $\mathbf{W}$ . The results show that strategic investments in storage and conveyance can generate welfare gains that justify their cost of construction, particularly in regions where the mismatch between the timing of precipitation and the timing of demand is large.

## 8 Conclusions

This paper develops a spatial theory of water allocation that integrates the physics of the hydrological cycle with a multi-region, multi-sector economic model. The framework tracks water across multiple states—root-zone moisture, shallow groundwater, deep groundwater, snowpack, and surface water—and routes it across space through a topography-based drainage network. Economic agents extract water at costs that depend on the local state of water stocks, creating a dynamic feedback between economic activity and water availability.

The quantitative analysis reveals several key findings. First, the spatial distribution of water is highly unequal and generates large differences in the cost of water across regions. Second, groundwater depletion is a first-order concern in regions that rely on non-renewable aquifer stocks for irrigation. Third, the routing network creates important spatial spillovers: upstream water use affects downstream availability, generating externalities that cross political boundaries.

Fourth, water pricing reform and infrastructure investments can generate substantial welfare gains, though with important distributional consequences.

The model provides a framework for evaluating a wide range of water policy questions: inter-basin transfers, groundwater regulation, dam construction, climate adaptation, and the interaction between water policy and trade policy. In ongoing work, we extend the model to incorporate endogenous land-use change, dynamic investment in water infrastructure, and the interaction between water scarcity and urbanization.

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# Online Appendix for “Deep Waters: A Spatial Theory of Water Allocation”

Costas Arkolakis, José-Antonio Espín-Sánchez, and Anton Yang

Yale University

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This appendix provides extended derivations, data documentation, algorithmic details, and additional results that support the main text. It is organized as follows. Section A1 collects all symbols and definitions. Section A3 provides a detailed overview of the hydrological cycle as modeled in the paper. Section A4 derives the alternative curve-number runoff specification. Section A5 proves the mass-balance identities. Section A6 develops the mathematical properties of the routing operator. Section A7 proves existence and uniqueness of the economic equilibrium. Section A8 presents the complete surface-water stock derivation. Section A9 documents data construction in detail. Section A10 describes the numerical implementation. Section A11 derives the welfare decomposition. Section A12 presents the full sensitivity analysis. Section A13 collects maps and additional figures.

## A1 Symbols and Definitions

Symbols in blue are externally measured inputs or fixed spatial covariates taken from datasets. Symbols in orange are parameters that are chosen or calibrated. Symbols in green are endogenous stocks, flows, or equilibrium objects computed by the model. Pure indices and set definitions are left in black.

| Symbol                         | Definition (units / domain)                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------|
| <i>Sets, Indices, and Time</i> |                                                                                                                  |
| $n, m$                         | Country indices, $n, m \in \mathcal{N} \equiv \{1, \dots, N\}$ ; in trade, $n$ is destination and $m$ is origin. |
| $i$                            | Physical water–land grid-cell index, $i \in \mathcal{I} \equiv \{1, \dots, I\}$ .                                |
| $n(i)$                         | Unique country containing grid cell $i$ .                                                                        |
| $\mathcal{I}_n$                | Set of physical grid cells contained in country $n$ .                                                            |
| $j, k$                         | Sector indices, $j, k \in \mathcal{J} \equiv \{1, \dots, J\}$ .                                                  |
| $\mathcal{J}^L$                | Subset of active land-using sectors.                                                                             |
| $\omega$                       | Physical land-plot index, $\omega \in [0, 1]$ .                                                                  |

*Continued on next page*

| Symbol                                                | Definition (units / domain)                                                                                            |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| $a$                                                   | Water-source index, $a \in \{\text{sur}, \text{Sh}, \text{Dp}\}$ .                                                     |
| $s$                                                   | Hydrological-state index.                                                                                              |
| $N$                                                   | Number of countries.                                                                                                   |
| $I$                                                   | Number of physical grid cells.                                                                                         |
| $J$                                                   | Number of economic sectors.                                                                                            |
| $S$                                                   | Number of hydrological states.                                                                                         |
| $t$                                                   | Time-period index, $t = 0, 1, \dots, T$ .                                                                              |
| $T$                                                   | Final time period.                                                                                                     |
| $\mathcal{T}$                                         | Set of terminal coastal/outlet cells where routed flow exits the domain.                                               |
| $\mathcal{U}(i)$                                      | Set of cells upstream of grid cell $i$ in the drainage network.                                                        |
| <i>Water Motion (State–Location Stocks and Flows)</i> |                                                                                                                        |
| $\mathbf{W}_{ti}$                                     | $S \times 1$ vector of water quantities across states at location $i$ and time $t$ (depth-equivalent).                 |
| $W_{ti}^s$                                            | Water quantity in state $s$ at location $i$ , time $t$ (depth-equivalent).                                             |
| $\mathbf{W}_t$                                        | $S \times I$ matrix collecting $\mathbf{W}_{ti}$ across all locations at time $t$ (depth-equivalent).                  |
| $\mathbf{E}_0$                                        | Initial endowment matrix for all locations and states; $\mathbf{W}_0 = \mathbf{E}_0$ .                                 |
| $\Omega_t$                                            | Net change in water stocks from $t - 1$ to $t$ , with $\mathbf{W}_t = \mathbf{W}_{t-1} + \Omega_t$ (depth-equivalent). |
| $\mathbf{T}_t$                                        | Within-location (across-state) transition component of $\Omega_t$ (depth-equivalent).                                  |
| $\mathbf{S}_t$                                        | Spatial (across-location) transition component of $\Omega_t$ (depth-equivalent).                                       |
| $\mathbf{B}_{it}$                                     | Gross within-location state-transition matrix at location $i$ , time $t$ ( $S \times S$ ).                             |
| $\boldsymbol{\tau}_{it}$                              | Net within-location state changes implied by $\mathbf{B}_{it}$ (vector in $\mathbb{R}^S$ ).                            |
| <i>Hydrology (Snowpack, Runoff, Recharge)</i>         |                                                                                                                        |
| $T_{i,t}$                                             | Near-surface air temperature in cell $i$ during period $t$ ( $^{\circ}\text{C}$ ).                                     |
| $\Delta_t$                                            | Number of days in period $t$ (days).                                                                                   |
| $r_{i,t}^{\text{rain}}$                               | Liquid rainfall depth in $i, t$ (inches per period).                                                                   |

*Continued on next page*

| Symbol                                             | Definition (units / domain)                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $r_{i,t}^{\text{snow}}$                            | Snowfall water-equivalent depth in $i, t$ (inches per period).                                      |
| $r_{i,t}$                                          | Total precipitation: $r_{i,t} = r_{i,t}^{\text{rain}} + r_{i,t}^{\text{snow}}$ (inches per period). |
| $\text{SWE}_{i,t}$                                 | Snow water equivalent (snowpack) stock in $i, t$ (inches).                                          |
| $m_{i,t}$                                          | Snowmelt released in $i, t$ (inches per period).                                                    |
| $\text{ddf}$                                       | Degree-day melt factor (inches per ( $^{\circ}\text{C}\cdot\text{day}$ )).                          |
| $T^{\text{melt}}$                                  | Melt threshold temperature (baseline $0^{\circ}\text{C}$ ).                                         |
| $\tilde{r}_{i,t}$                                  | Effective liquid input (rainfall + melt) (inches per period).                                       |
| $u_{i,t}$                                          | Total liquid input: $u_{i,t} = \tilde{r}_{i,t} + l_{i,t-1}$ (inches per period).                    |
| $u_{i,t}^{\text{net}}$                             | Post-abstraction water: $u_{i,t}^{\text{net}} = u_{i,t} - a_{i,t}$ (inches per period).             |
| $q_{i,t}$                                          | Surface runoff generated in $i, t$ (inches per period).                                             |
| $f_{i,t}^{\text{Sh}}, f_{i,t}^{\text{Dp}}$         | Shallow and deep recharge (inches per period).                                                      |
| $f_{i,t}^{\text{Sh,raw}}, f_{i,t}^{\text{Dp,raw}}$ | Potential shallow and deep recharge before capacity cap.                                            |
| $\tilde{\zeta}_i$                                  | Deep-percolation share (dimensionless, in $[0, 1]$ ).                                               |
| $\bar{a}_i$                                        | Abstraction capacity (inches per period).                                                           |
| $a_{i,t}$                                          | Realized abstraction, $\min\{u_{i,t}, \bar{a}_i\}$ (inches per period).                             |
| $\bar{S}_i, \bar{D}_i$                             | Shallow and deep storage capacities (inches).                                                       |
| $\Delta S_{i,t}^{\text{cap}}$                      | Available shallow space at start of $t$ : $(\bar{S}_i - S_{i,t-1})_+$ (inches).                     |
| $\Delta D_{i,t}^{\text{cap}}$                      | Available deep space at start of $t$ : $(\bar{D}_i - D_{i,t-1})_+$ (inches).                        |
| $M_{i,t}$                                          | Root-zone moisture stock (inches).                                                                  |
| $S_{i,t}$                                          | Shallow groundwater storage stock (inches).                                                         |
| $D_{i,t}$                                          | Deep groundwater storage stock (inches).                                                            |
| $e_{i,t}$                                          | Evapotranspiration depth (inches per period).                                                       |
| $l_{i,t}$                                          | Routed inflow (inches per period).                                                                  |
| $w_{j \rightarrow i}$                              | Downstream routing weight from $j$ to $i$ (dimensionless).                                          |
| $\mathbf{W}$                                       | Downstream routing matrix.                                                                          |
| <i>Surface Water</i>                               |                                                                                                     |
| $X_{2,i,t}$                                        | Surface-water storage stock (depth-equivalent or volume).                                           |
| $\kappa_i^{\text{rout}}$                           | Routing/release fraction (dimensionless, in $(0, 1]$ ).                                             |
| $\mathcal{A}_i$                                    | Physical area of grid cell $i$ ( $\text{m}^2$ ).                                                    |
| $\omega_i$                                         | Open-water area fraction (dimensionless).                                                           |
| $r_{i,t}^{\text{land}}, r_{i,t}^{\text{water}}$    | Precipitation over land and open-water portions.                                                    |

*Continued on next page*

| Symbol                                                 | Definition (units / domain)                                                             |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------|
| $\zeta_i^{\text{bed}}$                                 | Bed seepage coefficient (per-period rate).                                              |
| $h_i$                                                  | Fraction of incoming routed flow interacting with bed ( $\in [0, 1]$ ).                 |
| $b_{i,t}$                                              | Baseflow from shallow GW to surface water.                                              |
| $e_{i,t}^{\text{open}}$                                | Open-water evaporation.                                                                 |
| $s_{i,t}$                                              | Seepage to shallow groundwater.                                                         |
| $\text{Div}_{i,t}^{\text{sur}}$                        | Surface-water diversion.                                                                |
| $O_{i,t}^{\text{ocean}}$                               | Terminal outflow to ocean.                                                              |
| $v_i^{\text{surf}}$                                    | Share of return flows routed to surface water.                                          |
| $\mathcal{R}_{i,t}$                                    | Return flows: $(1 - v_i)\text{Div}_{i,t}$ .                                             |
| <i>Curve-Number Method (Alternative Specification)</i> |                                                                                         |
| $v_i$                                                  | Curve number (dimensionless, in $(0, 100]$ ).                                           |
| $\zeta_i$                                              | Retention parameter: $\zeta_i = 1000/v_i - 10$ (inches).                                |
| $\mathcal{I}_i$                                        | Initial abstraction threshold, $0.2 \zeta_i$ (inches).                                  |
| $f_{i,t}$                                              | Infiltration residual in CN approach (inches per period).                               |
| <i>Consumption and Armington Trade</i>                 |                                                                                         |
| $C_{n,t}$                                              | Household consumption composite in country $n$ .                                        |
| $C_{n,j,t}$                                            | Household consumption of the country-sector Armington composite $j$ .                   |
| $\alpha_{n,j}^C$                                       | Cobb–Douglas final-expenditure share on sector $j$ , with $\sum_j \alpha_{n,j}^C = 1$ . |
| $P_{n,t}^C$                                            | Consumption price index.                                                                |
| $Q_{n,j,t}^A$                                          | Country- $n$ , sector- $j$ Armington absorption composite.                              |
| $Q_{n,j,t}^{\text{Imp}}, P_{n,j,t}^{\text{Imp}}$       | Imported quantity composite and its price.                                              |
| $P_{nm,j,t}$                                           | Delivered price in destination $n$ of sector- $j$ goods from origin $m$ .               |
| $P_{n,j,t}$                                            | Country- $n$ Armington purchaser price for the sector- $j$ composite.                   |
| $d_{nm,j}$                                             | Bilateral iceberg trade cost from origin $m$ to destination $n$ , with $d_{nn,j} = 1$ . |
| $\sigma_j^M$                                           | Elasticity of substitution across foreign origins.                                      |
| $\sigma_j^D$                                           | Elasticity of substitution between the domestic variety and the import composite.       |
| $\zeta_{nm,j}^M, \zeta_{n,j}^D$                        | Armington distribution weights.                                                         |

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| Symbol                                                    | Definition (units / domain)                                                                                                                                                             |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lambda_{nm,j,t}^M, \lambda_{n,j,t}^D, \lambda_{nm,j,t}$ | Conditional foreign-source share, domestic share, and overall bilateral expenditure share.                                                                                              |
| $X_{nm,j,t}$                                              | Destination- $n$ expenditure on sector- $j$ goods from origin $m$ .                                                                                                                     |
| $X_{n,j,t}$                                               | Total country- $n$ absorption expenditure in sector $j$ .                                                                                                                               |
| <i>Technology, Production, and Land Allocation</i>        |                                                                                                                                                                                         |
| $\bar{F}_i^{\text{Land}}$                                 | Exogenous physical land endowment of grid $i$ ; land endowments are $i$ -dimensional.                                                                                                   |
| $\bar{F}_{n,t}^{\text{LAB}}$                              | Exogenous labor endowment of country $n$ at time $t$ ; labor endowments are $n$ -dimensional.                                                                                           |
| $\bar{F}_{i,t}^{\text{Water}}$                            | Current-period local water endowment of grid $i$ , taken as given by the static production equilibrium and determined by the extraction problem; water endowments are $i$ -dimensional. |
| $Q_{in,j,t}$                                              | Gross output of sector $j$ in grid $i$ of country $n$ at time $t$ .                                                                                                                     |
| $Q_{n,j,t}$                                               | Sector- $j$ quantity available at country $n$ 's national hub after domestic collection.                                                                                                |
| $Y_{in,j,t}$                                              | Value of gross source output: $Y_{in,j,t} = c_{in,j,t} Q_{in,j,t}$ .                                                                                                                    |
| $V_{in,j,t}$                                              | Primary-endowment composite of direct production labor, efficiency units of land, and local water.                                                                                      |
| $p_{in,j,t}^V$                                            | Grid-sector price of the primary-endowment composite; it carries no plot index.                                                                                                         |
| $\beta_{n,j}^{\text{LAB}}, \beta_{n,j}^L, \beta_{n,j}^W$  | CDE distribution parameters for labor, land, and water.                                                                                                                                 |
| $\sigma_j^{\text{LAB}}, \sigma_j^L, \sigma_j^W$           | Input-specific CDE substitution parameters.                                                                                                                                             |
| $\epsilon_j^{\text{LAB}}, \epsilon_j^L, \epsilon_j^W$     | Input-specific CDE expansion / non-homotheticity parameters.                                                                                                                            |
| $A_{n,j,t}^{\text{prod}}$                                 | Sectoral production shifter in the upper Cobb–Douglas nest.                                                                                                                             |
| $\gamma_{n,j}^V, \gamma_{n,j,k}$                          | Upper Cobb–Douglas shares on the primary-endowment composite and sector- $k$ intermediate inputs.                                                                                       |
| $c_{in,j,t}$                                              | Grid-sector source unit cost.                                                                                                                                                           |
| $P_{nn,j,t}$                                              | Sector- $j$ source price at country $n$ 's national hub; it is also the domestic delivered price because $d_{nn,j} = 1$ .                                                               |
| $w_{n,t}^{\text{LAB}}$                                    | National wage.                                                                                                                                                                          |
| $r_{i,t}^{\text{Water}}$                                  | Local water price in grid $i$ .                                                                                                                                                         |

*Continued on next page*

| Symbol                                                                                      | Definition (units / domain)                                                                                                   |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $e_{i_n,j,t}(\omega)$                                                                       | Plot-specific land productivity: efficiency units of land supplied by one physical unit of plot $\omega$ in active use $j$ .  |
| $E_{i_n,j,t}$                                                                               | Mean plot productivity, $\mathbb{E}[e_{i_n,j,t}(\omega)]$ .                                                                   |
| $r_{i_n,j,t}^{\text{Land}}(\omega)$                                                         | Gross return to one physical unit of plot $\omega$ in active sector $j$ .                                                     |
| $r_{i_n,j,t}^{\text{Land}}$                                                                 | Unconditional mean active-plot return, $\mathbb{E}[r_{i_n,j,t}^{\text{Land}}(\omega)]$ .                                      |
| $r_{i_n,0,t}^{\text{Land}}(\omega)$                                                         | Idle-land outside-option value: the wage value of the plot-specific setup requirement avoided when the plot remains inactive. |
| $r_{i_n,0,t}^{\text{Land}}$                                                                 | Unconditional mean idle-land outside-option return, $\mathbb{E}[r_{i_n,0,t}^{\text{Land}}(\omega)]$ .                         |
| $\tilde{h}_{i_n,j,t}$                                                                       | Common output-per-efficiency-land return component implied by the CDE and upper unit-cost systems.                            |
| $F_{i_n,j,t}^{\text{Land}}$                                                                 | Endogenous physical land used by active sector $j$ in grid $i$ .                                                              |
| $F_{i_n,j,t}^{\text{LAB}}$                                                                  | Endogenous labor used directly in production; excludes land-setup labor.                                                      |
| $F_{i_n,j,t}^{\text{Water}}$                                                                | Endogenous local water used directly in production.                                                                           |
| $s_{i_n,j,t}^{\text{LAB}}, s_{i_n,j,t}^{\text{L}}, s_{i_n,j,t}^{\text{W}}$                  | CDE value shares of direct production labor, land, and water in primary-endowment expenditure.                                |
| $RL_{i_n,j,t}$                                                                              | Gross payment generated by active land in grid-sector $(i_n, j)$ .                                                            |
| $e_{i_n,0}(\omega)$                                                                         | Plot-specific setup requirement for activating land, measured in labor units.                                                 |
| $E_{i_n,0}$                                                                                 | Mean setup requirement.                                                                                                       |
| $S_{i_n,\text{agr},t}$                                                                      | Labor actually used to satisfy active-land setup requirements.                                                                |
| $NRL_{i_n,t}$                                                                               | Net land revenue after subtracting the wage cost of setup labor.                                                              |
| <i>Water Extraction and Economic Interaction</i>                                            |                                                                                                                               |
| $\text{Div}_{i,t}$                                                                          | Total water extraction.                                                                                                       |
| $\text{Div}_{i,t}^{\text{sur}}, \text{Div}_{i,t}^{\text{Sh}}, \text{Div}_{i,t}^{\text{Dp}}$ | Surface, shallow GW, and deep GW extraction.                                                                                  |
| $c_i^{\text{sur}}, c_{i,t}^{\text{Sh}}, c_{i,t}^{\text{Dp}}$                                | Extraction costs from each source.                                                                                            |
| $v_i$                                                                                       | Consumptive-use fraction of extracted water.                                                                                  |
| <i>Welfare and Income</i>                                                                   |                                                                                                                               |
| $P_{n,t}^{\text{C}} C_{n,t}$                                                                | Nominal household expenditure in country $n$ .                                                                                |
| $NRL_{n,t}$                                                                                 | Country-level net land revenue.                                                                                               |

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| Symbol     | Definition (units / domain)                         |
|------------|-----------------------------------------------------|
| $WR_{n,t}$ | Country-level water-rent income.                    |
| $D_{n,t}$  | Exogenous net transfer, with $\sum_n D_{n,t} = 0$ . |

Table A.2: Full Model: Level Equations

| ID     | Level equation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Scope                                  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Hydro. | Water states, recharge, and routing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                        |
| H1     | $\mathbf{W}_t = \mathbf{W}_{t-1} + \mathbf{\Omega}_t, \quad \mathbf{\Omega}_t = \mathbf{T}_t + \mathbf{S}_t, \quad \boldsymbol{\tau}_{it} = \mathbf{B}_{it}^\top \mathbf{1} - \mathbf{B}_{it} \mathbf{1}, \quad \mathbf{T}_t = [\tau_{1t} \cdots \tau_{It}].$                                                                                                                                                                                                                                                                                                                     | $\forall i, t$                         |
| H2     | $m_{i,t} = \min\left\{SWE_{i,t-1} + r_{i,t}^{\text{snow}}, \text{ddf} \max\left(T_{i,t} - T^{\text{melt}}, 0\right) \Delta_t\right\},$<br>$SWE_{i,t} = \max\left\{SWE_{i,t-1} + r_{i,t}^{\text{snow}} - m_{i,t}, 0\right\}, \quad \tilde{r}_{i,t} = r_{i,t}^{\text{rain}} + m_{i,t}.$                                                                                                                                                                                                                                                                                             | $\forall i, t$                         |
| H3     | $u_{i,t} = \tilde{r}_{i,t} + l_{i,t-1}, \quad a_{i,t} = \min\{u_{i,t}, \tilde{a}_t\}, \quad u_{i,t}^{\text{net}} = u_{i,t} - a_{i,t},$<br>$f_{i,t}^{\text{Sh}} = \min\{(1 - \xi_i) u_{i,t}^{\text{net}}, (S_i - S_{i,t-1})_+\}, \quad f_{i,t}^{\text{Dp}} = \min\{\xi_i u_{i,t}^{\text{net}}, (\bar{D}_i - D_{i,t-1})_+\},$<br>$q_{i,t} = u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}}, \quad l_{i,t} = \sum_{\ell=1}^I w_{\ell \rightarrow i} q_{\ell,t}.$                                                                                                   | $\forall i, t$                         |
| H4     | $M_{i,t} = \max\{M_{i,t-1} + a_{i,t} - e_{i,t}, 0\}, \quad S_{i,t} = S_{i,t-1} + f_{i,t}^{\text{Sh}} - \text{Div}_{i,t}^{\text{Sh}}, \quad D_{i,t} = D_{i,t-1} + f_{i,t}^{\text{Dp}} - \text{Div}_{i,t}^{\text{Dp}}.$                                                                                                                                                                                                                                                                                                                                                             | $\forall i, t$                         |
| Water  | Water extraction and supply to production                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                        |
| W1     | $\text{Div}_{i,t} = \text{Div}_{i,t}^{\text{sur}} + \text{Div}_{i,t}^{\text{Sh}} + \text{Div}_{i,t}^{\text{Dp}}, \quad \bar{F}_{i,t}^{\text{Water}} = \mathcal{A}_i \text{Div}_{i,t}.$                                                                                                                                                                                                                                                                                                                                                                                            | $\forall i, t$                         |
| W2     | $\max_{\{\text{Div}_{i,t}^a\}} \sum_{t=0}^T \left( \prod_{\tau=1}^t \frac{1}{1 + r_\tau^K} \right) \sum_n \sum_{i \in \mathcal{I}_n} \left[ r_{in,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} \right.$<br>$\left. - \sum_{a \in \{\text{sur}, \text{Sh}, \text{Dp}\}} \text{ExtractCost}_i^a(\mathcal{A}_i \text{Div}_{i,t}^a, \mathbf{W}_{ti}; \mathbf{GF}_i^a) \right].$                                                                                                                                                                                                     | $\forall t$                            |
| W3     | $\text{Div}_{i,t}^a \geq 0, \quad \text{FullMC}_{i,t}^{\text{sur}} = \text{FullMC}_{i,t}^{\text{Sh}} = \text{FullMC}_{i,t}^{\text{Dp}} \quad \text{for sources used simultaneously.}$<br>$\frac{\partial [r_{in,t+1}^{\text{Water}} \bar{F}_{i,t+1}^{\text{Water}}]}{\partial \text{Div}_{i,t+1}} - \frac{\partial \text{ExtractCost}_i^a(\mathcal{A}_i \text{Div}_{i,t+1}^a, \mathbf{W}_{t+1,i}; \mathbf{GF}_i^a)}{\partial \text{Div}_{i,t+1}^a}$                                                                                                                               | $\forall i, t, a$                      |
| W4     | $= (1 + r_{t+1}^K) \left[ \frac{\partial [r_{in,t+1}^{\text{Water}} \bar{F}_{i,t+1}^{\text{Water}}]}{\partial \text{Div}_{i,t}} - \frac{\partial \text{ExtractCost}_i^a(\mathcal{A}_i \text{Div}_{i,t}^a, \mathbf{W}_{ti}; \mathbf{GF}_i^a)}{\partial \text{Div}_{i,t}^a} \right],$<br>$a \in \{\text{Sh}, \text{Dp}\}.$                                                                                                                                                                                                                                                          | $\forall i, t$                         |
| Prod.  | Production, CDE primary factors, and source prices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                        |
| P0     | $p_{in,j,t}^{\text{LAB}} \equiv w_{n,t}^{\text{LAB}}, \quad p_{in,j,t}^L \equiv c_{in,j,t} \bar{n}_{in,j,t}, \quad p_{in,j,t}^W \equiv r_{in,t}^{\text{Water}} \quad (\text{table shorthand for the three CDE primary-factor prices}).$                                                                                                                                                                                                                                                                                                                                           | $\forall n, i, j, t$                   |
| P1     | $Q_{in,j,t} = A_{n,j,t}^{\text{prod}} V_{in,j,t}^{\gamma_{n,j}} \prod_{k \in \mathcal{J}} \left( M_{in,j,k,t}^{\text{int}} \right)^{\gamma_{n,j,k}}, \quad \gamma_{n,j}^V + \sum_{k \in \mathcal{J}} \gamma_{n,j,k} = 1.$                                                                                                                                                                                                                                                                                                                                                         | $\forall n, i \in \mathcal{I}_n, j, t$ |
| P2     | $c_{in,j,t} = \frac{Y_{n,j}}{A_{n,j,t}^{\text{prod}}} \left( P_{in,j,t}^V \right)^{\gamma_{n,j}} \prod_{k \in \mathcal{J}} (P_{n,k,t})^{\gamma_{n,j,k}}, \quad c_{in,j,t} \bar{d}_{in,j} = P_{m,j,t}.$                                                                                                                                                                                                                                                                                                                                                                            | $\forall n, i, j, t$                   |
| P3     | $1 = \sum_{x \in \{\text{LAB}, L, W\}} \beta_{n,j}^x V_{in,j,t}^{\epsilon_j^x (1 - \sigma_j^x)} \left( \frac{p_{in,j,t}^x}{P_{in,j,t}^V V_{in,j,t}} \right)^{1 - \sigma_j^x},$<br>$s_{in,j,t}^x = \frac{\beta_{n,j}^x (1 - \sigma_j^x) \left( \frac{p_{in,j,t}^x}{P_{in,j,t}^V} \right)^{1 - \sigma_j^x} V_{in,j,t}^{(1 - \sigma_j^x)(\epsilon_j^x - 1)}}{\sum_{y \in \{\text{LAB}, L, W\}} \beta_{n,j}^y (1 - \sigma_j^y) \left( \frac{p_{in,j,t}^y}{P_{in,j,t}^V} \right)^{1 - \sigma_j^y} V_{in,j,t}^{(1 - \sigma_j^y)(\epsilon_j^y - 1)}}, \quad x \in \{\text{LAB}, L, W\}.$ | $\forall n, i, j, t$                   |

Continued on next page

Table A.2 – continued from previous page

| ID     | Level equation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scope                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| P4     | $w_{n,t}^{\text{LAB}} F_{n,j,t}^{\text{LAB}} = \gamma_{n,j}^V s_{n,j,t}^{\text{LAB}} Y_{n,j,t}, \quad RL_{n,j,t} = \gamma_{n,j}^V s_{n,j,t}^L Y_{n,j,t},$<br>$r_{n,t}^{\text{Water}} F_{n,j,t}^{\text{Water}} = \gamma_{n,j}^V s_{n,j,t}^W Y_{n,j,t}, \quad P_{n,k,t} M_{n,j,k,t}^{\text{int}} = \gamma_{n,j,k} Y_{n,j,t}.$                                                                                                                                                                                      | $\forall n, i, j, k, t$                |
| Land   | Land productivity, Fréchet allocation, and grid output                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |
| L1     | $E_{n,j,t} = E_{n,j} g_j(M_{i,t}, S_{i,t}, D_{i,t}), \quad r_{n,j,t}^{\text{Land}} = E_{n,j,t} c_{n,j,t} h_{n,j,t}, \quad r_{n,0,t}^{\text{Land}} = E_{n,0} w_{n,t}^{\text{LAB}}.$                                                                                                                                                                                                                                                                                                                               | $\forall n, i, j \in \mathcal{J}^L, t$ |
| L2     | $\pi_{n,j A,t}^L = \frac{\left(r_{n,j,t}^{\text{Land}}\right)^{v_2}}{\sum_{k \in \mathcal{J}^L} \left(r_{n,k,t}^{\text{Land}}\right)^{v_2}},$<br>$\pi_{n,A,t}^L = \frac{\left[\sum_{k \in \mathcal{J}^L} \left(r_{n,k,t}^{\text{Land}}\right)^{v_2}\right]^{v_1/v_2}}{\left(r_{n,0,t}^{\text{Land}}\right)^{v_1} + \left[\sum_{k \in \mathcal{J}^L} \left(r_{n,k,t}^{\text{Land}}\right)^{v_2}\right]^{v_1/v_2}}, \quad \pi_{n,0,t}^L = 1 - \pi_{n,A,t}^L, \quad \pi_{n,j,t}^L = \pi_{n,A,t}^L \pi_{n,j A,t}^L.$ | $\forall n, i, j \in \mathcal{J}^L, t$ |
| L3     | $F_{n,j,t}^{\text{Land}} = \pi_{n,j,t}^L \bar{F}_i^{\text{Land}}, \quad F_{n,0,t}^{\text{Land}} = \pi_{n,0,t}^L \bar{F}_i^{\text{Land}}, \quad F_{n,0,t}^{\text{Land}} + \sum_{j \in \mathcal{J}^L} F_{n,j,t}^{\text{Land}} = \bar{F}_i^{\text{Land}}.$                                                                                                                                                                                                                                                          | $\forall n, i, t$                      |
| L4     | $\mathbb{E}[e_{n,j,t}(\omega) \mid \omega \in \Omega_{i,n,j,t}] = E_{n,j,t} \left(\pi_{n,A,t}^L\right)^{-1/v_1} \left(\pi_{n,j A,t}^L\right)^{-1/v_2},$<br>$Q_{n,j,t} = \frac{h_{n,j,t}}{\gamma_{n,j}^V s_{n,j,t}^L} F_i^{\text{Land}} E_{n,j,t} \left(\pi_{n,A,t}^L\right)^{(v_1-1)/v_1} \left(\pi_{n,j A,t}^L\right)^{(v_2-1)/v_2}.$                                                                                                                                                                           | $\forall n, i, j \in \mathcal{J}^L, t$ |
| L5     | $S_{n,\text{agr},t} = E_{n,0} \bar{F}_i^{\text{Land}} \left[1 - \left(\pi_{n,0,t}^L\right)^{(v_1-1)/v_1}\right],$<br>$NRL_{n,t} = \sum_{j \in \mathcal{J}^L} RL_{n,j,t} - w_{n,t}^{\text{LAB}} S_{n,\text{agr},t}, \quad NRL_{n,t} = \sum_{i \in \mathcal{I}_n} NRL_{i,n,t}.$                                                                                                                                                                                                                                    | $\forall n, i, t$                      |
| Hub    | National collection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |
| N1     | $Q_{n,j,t} = \sum_{i \in \mathcal{I}_n} \frac{Q_{i,n,j,t}}{d_{i,n,j}}, \quad Y_{n,j,t} = P_{nm,j,t} Q_{n,j,t} = \sum_{i \in \mathcal{I}_n} Y_{i,n,j,t}, \quad Y_{i,n,j,t} = c_{i,n,j,t} Q_{i,n,j,t}.$                                                                                                                                                                                                                                                                                                            | $\forall n, j, t$                      |
| Trade  | Two-level Armington trade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                        |
| T1     | $P_{nm,j,t} = d_{nm,j} P_{mm,j,t}, \quad d_{nn,j} = 1.$                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\forall n, m, j, t$                   |
| T2     | $P_{n,j,t}^{\text{Imp}} = \left[\sum_{m \neq n} \zeta_{nm,j}^M P_{nm,j,t}^{1-\sigma_j^M}\right]^{1/(1-\sigma_j^M)}, \quad \lambda_{nm,j,t}^M = \frac{\zeta_{nm,j}^M P_{nm,j,t}^{1-\sigma_j^M}}{\sum_{\ell \neq n} \zeta_{n\ell,j}^M P_{n\ell,j,t}^{1-\sigma_j^M}}.$                                                                                                                                                                                                                                              | $\forall n, m \neq n, j, t$            |
| T3     | $P_{n,j,t} = \left[\zeta_{n,j}^D P_{nn,j,t}^{1-\sigma_j^D} + (1 - \zeta_{n,j}^D) \left(P_{n,j,t}^{\text{Imp}}\right)^{1-\sigma_j^D}\right]^{1/(1-\sigma_j^D)},$<br>$\lambda_{n,j,t}^D = \frac{\zeta_{n,j}^D P_{nn,j,t}^{1-\sigma_j^D}}{P_{n,j,t}^{1-\sigma_j^D}}, \quad \lambda_{nn,j,t} = \lambda_{n,j,t}^D, \quad \lambda_{nm,j,t} = \left(1 - \lambda_{n,j,t}^D\right) \lambda_{nm,j,t}^M, \quad m \neq n.$                                                                                                   | $\forall n, m, j, t$                   |
| T4     | $X_{nm,j,t} = P_{nm,j,t} Q_{nm,j,t} = \lambda_{nm,j,t} X_{n,j,t}, \quad X_{n,j,t} = \sum_m X_{nm,j,t} = P_{n,j,t} Q_{n,j,t}^A.$                                                                                                                                                                                                                                                                                                                                                                                  | $\forall n, m, j, t$                   |
| HH     | Household consumption and absorption                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                        |
| C1     | $C_{n,t} = \prod_{j \in \mathcal{J}} C_{n,j,t}^{\alpha_{n,j}^C}, \quad P_{n,t}^C = \prod_{j \in \mathcal{J}} \left(\frac{P_{n,j,t}}{\alpha_{n,j}^C}\right)^{\alpha_{n,j}^C}, \quad P_{n,j,t} C_{n,j,t} = \alpha_{n,j}^C P_{n,t}^C C_{n,t}.$                                                                                                                                                                                                                                                                      | $\forall n, j, t$                      |
| C2     | $M_{n,j,k,t}^{\text{int}} = \sum_{i \in \mathcal{I}_n} M_{i,n,j,k,t}^{\text{int}}, \quad Q_{n,k,t}^A = C_{n,k,t} + \sum_{j \in \mathcal{J}} M_{n,j,k,t}^{\text{int}},$<br>$X_{n,k,t} = \alpha_{n,k}^C P_{n,t}^C C_{n,t} + \sum_{j \in \mathcal{J}} \gamma_{n,j,k} Y_{n,j,t}.$                                                                                                                                                                                                                                    | $\forall n, j, k, t$                   |
| Equil. | Market clearing, income, and normalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                        |
| E1     | $F_{n,t}^{\text{LAB}} = \sum_{j \in \mathcal{J}} \sum_{i \in \mathcal{I}_n} F_{i,n,j,t}^{\text{LAB}} + \sum_{i \in \mathcal{I}_n} S_{i,n,\text{agr},t}.$                                                                                                                                                                                                                                                                                                                                                         | $\forall n, t$                         |

Continued on next page

Table A.2 – continued from previous page

| ID | Level equation                                                                                                                                                                                                          | Scope                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| E2 | $\sum_{j \in \mathcal{J}} F_{i_{n,j},t}^{\text{Water}} = \bar{F}_{i,t}^{\text{Water}}$ (equivalently $r_{i_{n,t}}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} = \sum_j \gamma_{n,j}^V s_{i_{n,j},t}^W Y_{i_{n,j},t}$ ). | $\forall n, i \in \mathcal{I}_n, t$ |
| E3 | $Y_{m,j,t} = P_{mm,j,t} Q_{m,j,t} = \sum_{n=1}^N X_{nm,j,t}, \quad Q_{m,j,t} = \sum_{n=1}^N d_{nm,j} Q_{nm,j,t}.$                                                                                                       | $\forall m, j, t$                   |
| E4 | $\sum_m \sum_j X_{nm,j,t} = \sum_m \sum_j X_{mm,j,t} + D_{n,t}, \quad \sum_n D_{n,t} = 0.$                                                                                                                              | $\forall n, t$                      |
| E5 | $P_{n,t}^C C_{n,t} = w_{n,t}^{\text{LAB}} F_{n,t}^{\text{LAB}} + \text{NRL}_{n,t} + \sum_{i \in \mathcal{I}_n} r_{i_{n,t}}^{\text{Water}} F_{i,t}^{\text{Water}} + D_{n,t}.$                                            | $\forall n, t$                      |
| E6 | $w_{n^*,t}^{\text{LAB}} = 1$ (one nominal normalization; replace with the normalization used in the numerical code).                                                                                                    | $\forall t$                         |

## A2 CDE Primary-Endowment Structure: Detailed Equations

This appendix collects the primal and dual equations used in Section 2.3. The three primary inputs are direct production labor, efficiency units of land, and water. Plot heterogeneity enters only through land productivity  $e_{i_{n,j},t}(\omega)$  and the associated physical-land return  $r_{i_{n,j},t}^{\text{Land}}(\omega)$ . The wage, local water price, primary-endowment quantity index, and primary-endowment price do not carry a plot index.

### A2.1 Implicit CDE production frontier

On the quantity side, the primary-endowment composite is represented by

$$\mathcal{F}_{n,j}^V \left( V_{i_{n,j},t}, F_{i_{n,j},t}^{\text{LAB}}, e_{i_{n,j},t}(\omega) F_{i_{n,j},t}^{\text{Land}}, F_{i_{n,j},t}^{\text{Water}} \right) = 0. \quad (\text{A.1})$$

The three quantity arguments are units of direct production labor, efficiency units of land, and units of local water.

### A2.2 CDE cost relation for a candidate plot

For a candidate plot  $\omega$ , the price of one efficiency unit of land is

$$\frac{r_{i_{n,j},t}^{\text{Land}}(\omega)}{e_{i_{n,j},t}(\omega)}. \quad (\text{A.2})$$

The inner CDE relation is therefore the following. The composite price  $P_{i_{n,j},t}^V$  is common across plots, while the land-return/productivity ratio is the only candidate-plot-specific argument:

$$\begin{aligned}
1 = & \beta_{n,j}^{\text{LAB}} V_{i_{n,j},t} \epsilon_j^{\text{LAB}(1-\sigma_j^{\text{LAB}})} \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1-\sigma_j^{\text{LAB}}} \\
& + \beta_{n,j}^L V_{i_{n,j},t} \epsilon_j^{L(1-\sigma_j^L)} \left( \frac{r_{i_{n,j},t}^{\text{Land}}(\omega) / e_{i_{n,j},t}(\omega)}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1-\sigma_j^L} \\
& + \beta_{n,j}^W V_{i_{n,j},t} \epsilon_j^{W(1-\sigma_j^W)} \left( \frac{r_{i_{n,t}}^{\text{Water}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1-\sigma_j^W}.
\end{aligned} \tag{A.3}$$

### A2.3 Upper Cobb–Douglas cost and the primary-endowment price

The upper production nest implies

$$c_{i_{n,j},t} = \frac{Y_{n,j}}{A_{n,j,t}^{\text{prod}}} \left( P_{i_{n,j},t}^V \right)^{\gamma_{n,j}^V} \prod_{k \in \mathcal{J}} (P_{n,k,t})^{\gamma_{n,j,k}}. \tag{A.4}$$

Perfect competition at the national collection hub gives

$$c_{i_{n,j},t} = \frac{P_{nn,j,t}}{\bar{d}_{i_{n,j}}}, \tag{A.5}$$

and the upper unit-cost equation gives the primary-endowment price

$$P_{i_{n,j},t}^V = \gamma_{n,j}^V \left[ A_{n,j,t}^{\text{prod}} c_{i_{n,j},t} \prod_{k \in \mathcal{J}} \left( \frac{\gamma_{n,j,k}}{P_{n,k,t}} \right)^{\gamma_{n,j,k}} \right]^{1/\gamma_{n,j}^V}. \tag{A.6}$$

### A2.4 Explicit land-return inversion

With the common  $P_{i_{n,j},t}^V$  from (A.6), isolate the land term in (A.3). This gives

$$\begin{aligned}
\frac{r_{i_{n,j},t}^{\text{Land}}(\omega)}{e_{i_{n,j},t}(\omega)} = & P_{i_{n,j},t}^V V_{i_{n,j},t} \\
1 - & \beta_{n,j}^{\text{LAB}} V_{i_{n,j},t} \epsilon_j^{\text{LAB}(1-\sigma_j^{\text{LAB}})} \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1-\sigma_j^{\text{LAB}}} \\
& - \beta_{n,j}^W V_{i_{n,j},t} \epsilon_j^{W(1-\sigma_j^W)} \left( \frac{r_{i_{n,t}}^{\text{Water}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1-\sigma_j^W} \\
& \times \left[ \frac{\beta_{n,j}^L V_{i_{n,j},t} \epsilon_j^{L(1-\sigma_j^L)}}{\beta_{n,j}^L V_{i_{n,j},t} \epsilon_j^{L(1-\sigma_j^L)}} \right]^{\frac{1}{1-\sigma_j^L}}.
\end{aligned} \tag{A.7}$$

The right-hand side does not depend on  $\omega$ . Hence plot heterogeneity enters the physical-land return only through  $e_{i_{n,j},t}(\omega)$ .

Using source unit cost, the same result can be written as

$$\frac{r_{i_{n,j},t}^{\text{Land}}(\omega)}{e_{i_{n,j},t}(\omega)} = c_{i_{n,j},t} \tilde{h}_{i_{n,j},t}, \quad (\text{A.8})$$

where

$$\begin{aligned} \tilde{h}_{i_{n,j},t} &= \frac{P_{i_{n,j},t}^V V_{i_{n,j},t}}{c_{i_{n,j},t}} \\ &\quad \times \left[ \frac{1 - \beta_{n,j}^{\text{LAB}} V_{i_{n,j},t} \epsilon_j^{\text{LAB}} (1 - \sigma_j^{\text{LAB}}) \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1 - \sigma_j^{\text{LAB}}} - \beta_{n,j}^W V_{i_{n,j},t} \epsilon_j^W (1 - \sigma_j^W) \left( \frac{r_{i_{n,t}}^{\text{Water}}}{P_{i_{n,j},t}^V V_{i_{n,j},t}} \right)^{1 - \sigma_j^W}}{\beta_{n,j}^L V_{i_{n,j},t} \epsilon_j^L (1 - \sigma_j^L)} \right]^{\frac{1}{1 - \sigma_j^L}}. \end{aligned} \quad (\text{A.9})$$

Substituting (A.6) into (A.9) gives a fully explicit expression for  $\tilde{h}_{i_{n,j},t}$  in terms of  $c_{i_{n,j},t}$ ,  $V_{i_{n,j},t}$ ,  $w_{n,t}^{\text{LAB}}$ ,  $r_{i_{n,t}}^{\text{Water}}$ ,  $\{P_{n,k,t}\}_{k \in \mathcal{J}}$ , and technology parameters.

## A2.5 CDE value shares

After the land-price ratio is common across plots, the price of one efficiency unit of land is  $c_{i_{n,j},t} \tilde{h}_{i_{n,j},t}$ . The exact value-share equations are those reported in (113)–(115). In particular, using land as the reference input, the common denominator cancels and gives

$$\begin{aligned} \frac{s_{i_{n,j},t}^{\text{LAB}}}{s_{i_{n,j},t}^L} &= \frac{\beta_{n,j}^{\text{LAB}} (1 - \sigma_j^{\text{LAB}})}{\beta_{n,j}^L (1 - \sigma_j^L)} \left( \frac{w_{n,t}^{\text{LAB}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^{\text{LAB}}} \left( \frac{c_{i_{n,j},t} \tilde{h}_{i_{n,j},t}}{P_{i_{n,j},t}^V} \right)^{\sigma_j^L - 1} \\ &\quad \times V_{i_{n,j},t}^{(1 - \sigma_j^{\text{LAB}})(\epsilon_j^{\text{LAB}} - 1) - (1 - \sigma_j^L)(\epsilon_j^L - 1)}. \end{aligned} \quad (\text{A.10})$$

Similarly,

$$\begin{aligned} \frac{s_{i_{n,j},t}^W}{s_{i_{n,j},t}^L} &= \frac{\beta_{n,j}^W (1 - \sigma_j^W)}{\beta_{n,j}^L (1 - \sigma_j^L)} \left( \frac{r_{i_{n,t}}^{\text{Water}}}{P_{i_{n,j},t}^V} \right)^{1 - \sigma_j^W} \left( \frac{c_{i_{n,j},t} \tilde{h}_{i_{n,j},t}}{P_{i_{n,j},t}^V} \right)^{\sigma_j^L - 1} \\ &\quad \times V_{i_{n,j},t}^{(1 - \sigma_j^W)(\epsilon_j^W - 1) - (1 - \sigma_j^L)(\epsilon_j^L - 1)}. \end{aligned} \quad (\text{A.11})$$

These equations make clear that the CDE share system is evaluated after the plot-specific land return has been reduced to the common efficiency-unit price  $c_{i_{n,j},t} \tilde{h}_{i_{n,j},t}$ .

## A2.6 Factor quantities and accounting

The upper Cobb–Douglas expenditure identity is

$$P_{i_{n,j},t}^V V_{i_{n,j},t} = \gamma_{n,j}^V Y_{i_{n,j},t}. \quad (\text{A.12})$$

Hence

$$F_{i_{n,j},t}^{\text{LAB}} = \frac{\gamma_{n,j}^V s_{i_{n,j},t}^{\text{LAB}} Y_{i_{n,j},t}}{w_{n,t}^{\text{LAB}}}, \quad F_{i_{n,j},t}^{\text{Water}} = \frac{\gamma_{n,j}^V s_{i_{n,j},t}^{\text{W}} Y_{i_{n,j},t}}{r_{i_{n,t}}^{\text{Water}}}. \quad (\text{A.13})$$

The land payment is

$$RL_{i_{n,j},t} = \gamma_{n,j}^V s_{i_{n,j},t}^L Y_{i_{n,j},t} = \mathbb{E} \left[ r_{i_{n,j},t}^{\text{Land}}(\omega) \mid \omega \in \Omega_{i_{n,j},t} \right] F_{i_{n,j},t}^{\text{Land}}. \quad (\text{A.14})$$

## A3 The Hydrological Cycle: Detailed Overview

This section provides a self-contained description of the hydrological cycle as represented in our model. Figure A.1 shows a schematic of the full system, and Figure A.2 shows the data processing pipeline.

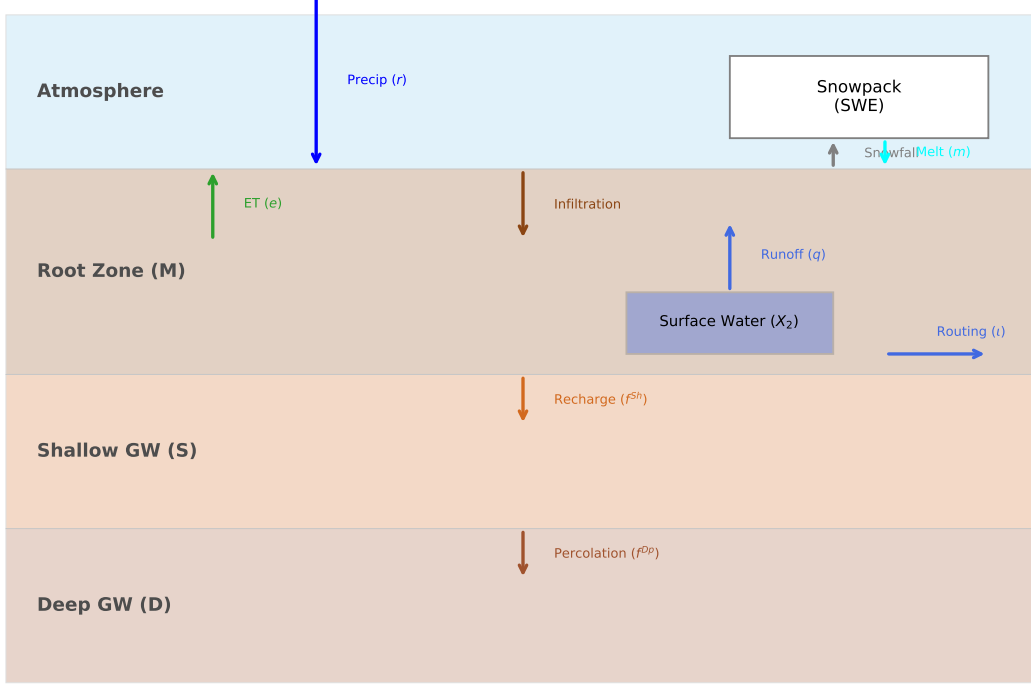


Figure A.1: Schematic of the hydrological cycle in the model. Precipitation enters from the atmosphere and is partitioned into snowfall (stored as snowpack SWE) and rainfall. Effective liquid input infiltrates into the root zone (M), recharges shallow (S) and deep (D) groundwater, or generates surface runoff (q). Evapotranspiration (ET) returns water to the atmosphere. Surface runoff is routed downstream through the flow-direction network.

### A3.1 Physical States and Transitions

The model tracks water in five states at each grid cell  $i$  and time  $t$ : root-zone moisture  $M_{i,t}$ , shallow groundwater  $S_{i,t}$ , deep groundwater  $D_{i,t}$ , snowpack  $SWE_{i,t}$ , and (when the extended specification is used) surface water  $X_{2,i,t}$ . The state vector at cell  $i$  is

$$\mathbf{W}_{ii} = \begin{pmatrix} M_{i,t} \\ S_{i,t} \\ D_{i,t} \\ SWE_{i,t} \\ X_{2,i,t} \end{pmatrix}. \quad (\text{A.15})$$

The global state is the  $S \times I$  matrix  $\mathbf{W}_t = [\mathbf{W}_{t1} \cdots \mathbf{W}_{tI}]$ . The law of motion is  $\mathbf{W}_t = \mathbf{W}_{t-1} + \Omega_t$ , where  $\Omega_t = \mathbf{T}_t + \mathbf{S}_t$  decomposes net changes into within-location state transitions and across-location spatial flows.

### A3.2 Within-Location Transitions

At each location  $i$ , the gross state-transition matrix  $\mathbf{B}_{it}$  is an  $S \times S$  non-negative matrix. Entry  $B_{it}^{(s,s')}$  represents the gross volume of water moving from state  $s$  to state  $s'$  within cell  $i$  during period  $t$ . The key entries are:

- **Atmosphere  $\rightarrow$  Root Zone / Snowpack:** Precipitation enters as rainfall (to the root zone) or snowfall (to the snowpack). The partition depends on temperature:

$$r_{i,t}^{\text{rain}} = r_{i,t} \cdot \mathbf{1}[T_{i,t} > T^{\text{melt}}], \quad r_{i,t}^{\text{snow}} = r_{i,t} \cdot \mathbf{1}[T_{i,t} \leq T^{\text{melt}}]. \quad (\text{A.16})$$

In practice, the GLDAS forcing data provides rainfall and snowfall as separate variables, so this partition is given exogenously.

- **Snowpack  $\rightarrow$  Root Zone:** Melt  $m_{i,t}$  follows the degree-day rule (equation (5) in the main text). The melt cannot exceed available snowpack.
- **Root Zone  $\rightarrow$  Shallow GW:** Shallow recharge  $f_{i,t}^{\text{Sh}}$  is the minimum of the allocated share  $(1 - \xi_i)u_{i,t}^{\text{net}}$  and available storage capacity  $(\bar{S}_i - S_{i,t-1})_+$ .
- **Root Zone  $\rightarrow$  Deep GW:** Deep recharge  $f_{i,t}^{\text{Dp}}$  is the minimum of  $\xi_i u_{i,t}^{\text{net}}$  and  $(\bar{D}_i - D_{i,t-1})_+$ .
- **Root Zone  $\rightarrow$  Surface (Runoff):** Residual water that cannot be stored:  $q_{i,t} = u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}}$ .
- **Root Zone / Surface  $\rightarrow$  Atmosphere:** Evapotranspiration  $e_{i,t}$  removes water from the root zone.
- **Shallow GW  $\rightarrow$  Surface Water:** Baseflow  $b_{i,t}$  provides a persistent contribution to stream-flow.
- **Surface Water  $\rightarrow$  Shallow GW:** Seepage  $s_{i,t}$  closes the loop.

The net within-location change for cell  $i$  is  $\tau_{it} = \mathbf{B}_{it}^\top \mathbf{1} - \mathbf{B}_{it} \mathbf{1}$ , where  $\mathbf{B}_{it}^\top \mathbf{1}$  gives the column sums (total inflows to each state) and  $\mathbf{B}_{it} \mathbf{1}$  gives the row sums (total outflows from each state).

### A3.3 Across-Location Transitions: Spatial Routing

Surface runoff is the only flux that moves between locations. The spatial transition matrix  $\mathbf{S}_t$  has non-zero entries only in the surface-water row. The routing operator  $\mathbf{W}$  is a sparse  $I \times I$  matrix derived from D8 flow directions, with  $w_{j \rightarrow i} = 1$  if cell  $j$  drains into cell  $i$ , and zero otherwise. Routed inflow at cell  $i$  is  $\iota_{i,t} = \sum_j w_{j \rightarrow i} q_{j,t}$ .

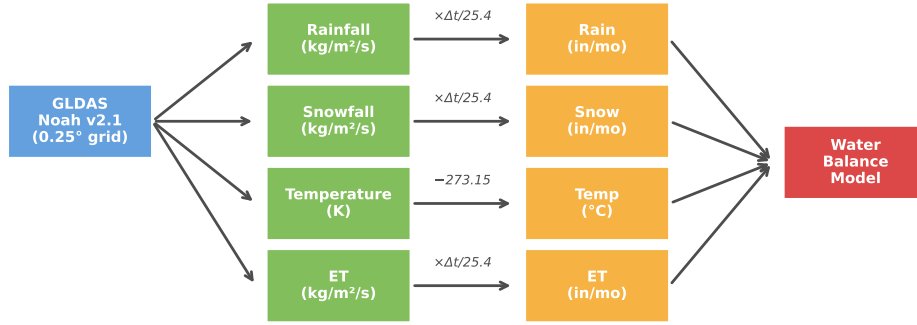


Figure A.2: Processing pipeline for GLDAS Noah forcing variables. Raw flux data in SI units ( $\text{kg m}^{-2} \text{s}^{-1}$ ) are converted to depth units (inches/month) for direct input to the water balance model. Temperature is converted from Kelvin to Celsius.

## A4 Alternative Runoff: Curve Numbers

This appendix derives the alternative curve-number (CN) runoff specification and compares it with the baseline storage-limited residual approach.

### A4.1 The SCS/NRCS Curve-Number Method

The curve-number method was developed by the USDA Soil Conservation Service ([USDA Soil Conservation Service, 1985](#)) and is widely used in hydrological engineering. It maps current-period water input to surface runoff using a single land-surface parameter—the curve number  $v_i \in (0, 100]$ —and a fixed initial-abstraction ratio.

The retention parameter (maximum soil retention) is

$$\varsigma_i = 10 \left( \frac{100}{v_i} - 1 \right) = \frac{1000}{v_i} - 10 \quad (\text{inches}). \quad (\text{A.17})$$

The initial abstraction threshold is  $\mathcal{J}_i = 0.2 \varsigma_i$ . The 0.2 ratio is the standard SCS assumption; empirical studies have suggested values between 0.05 and 0.2 ([Hawkins et al., 2009](#)). Realized abstraction is  $a_{i,t} = \min\{u_{i,t}, \mathcal{J}_i\}$ , and the runoff response function is

$$q_{i,t} = \begin{cases} 0 & \text{if } u_{i,t} \leq \mathcal{J}_i, \\ \frac{(u_{i,t} - \mathcal{J}_i)^2}{u_{i,t} - \mathcal{J}_i + \varsigma_i} = \frac{(u_{i,t} - \mathcal{J}_i)^2}{u_{i,t} + 0.8 \varsigma_i} & \text{if } u_{i,t} > \mathcal{J}_i. \end{cases} \quad (\text{A.18})$$

## A4.2 Properties of the CN Response Function

The CN runoff function has several useful properties. First, for  $u > \mathcal{J}$ :

$$\frac{\partial q}{\partial u} = \frac{(u - \mathcal{J})(u - \mathcal{J} + 2\varsigma)}{(u - \mathcal{J} + \varsigma)^2} \in (0, 1), \quad (\text{A.19})$$

so runoff is a strictly increasing, concave function of input. Second, as  $u \rightarrow \infty$ ,  $q/u \rightarrow 1$ : all additional input becomes runoff. Third, as  $\nu \rightarrow 100$ ,  $\varsigma \rightarrow 0$  and  $q \rightarrow u - a$ : the surface is impervious and all post-abstraction water runs off. As  $\nu \rightarrow 0$ ,  $\varsigma \rightarrow \infty$  and  $q \rightarrow 0$ : the surface retains all water.

Figure A.3 plots the CN response function for representative values of  $\nu$ . Higher curve numbers (corresponding to impervious surfaces, clay soils, or urban land cover) generate more runoff for a given input. Lower curve numbers (sandy soils, forest cover) retain more water in the subsurface.

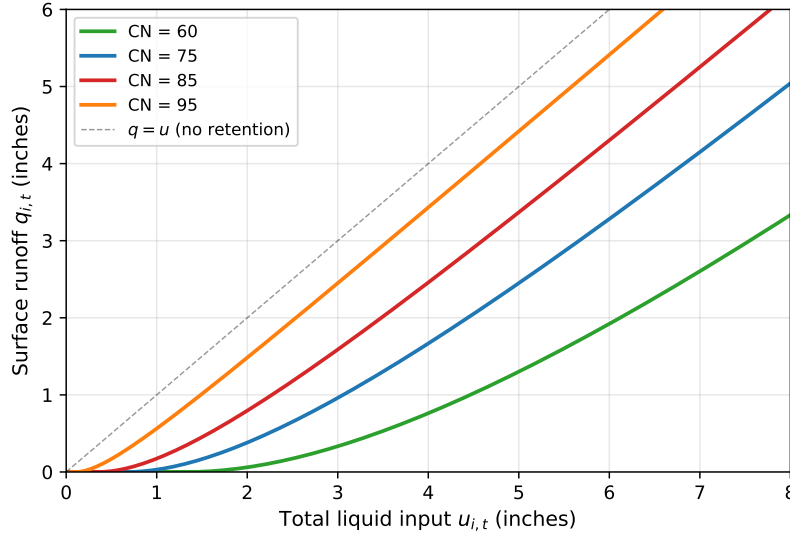


Figure A.3: SCS/NRCS runoff response functions for curve numbers  $\nu \in \{60, 75, 85, 95\}$ . Higher curve numbers generate more runoff for a given liquid input. The dashed line shows the no-retention benchmark  $q = u$ .

## A4.3 Curve Numbers by Land Cover and Soil Group

The GCN250 dataset (Jaafar et al., 2019) provides curve numbers at 250 m resolution based on land cover (from ESA CCI) and hydrologic soil group (from SoilGrids). Table A.3 shows representative curve numbers for common land-cover and soil-group combinations.

Table A.3: Representative Curve Numbers by Land Cover and Hydrologic Soil Group

| Land Cover                  | Group A | Group B | Group C | Group D |
|-----------------------------|---------|---------|---------|---------|
| Forest (good condition)     | 25      | 55      | 70      | 77      |
| Grassland (fair condition)  | 49      | 69      | 79      | 84      |
| Cropland (straight row)     | 67      | 78      | 85      | 89      |
| Cropland (contour/terrace)  | 59      | 72      | 81      | 86      |
| Urban (residential, 1/4 ac) | 54      | 70      | 80      | 85      |
| Urban (commercial/business) | 89      | 92      | 94      | 95      |
| Impervious surfaces         | 98      | 98      | 98      | 98      |

*Notes:* Group A = deep sand, low runoff potential. Group B = moderate depth, moderate infiltration. Group C = shallow soil, low infiltration. Group D = clay, high runoff potential. Values from [USDA Soil Conservation Service \(1985\)](#), Table 2-2a.

#### A4.4 Comparison: Residual Runoff Versus Curve Number

Table A.4 summarizes the structural differences between the two approaches.

Table A.4: Structural Comparison of Runoff Specifications

| Feature                       | Storage-Limited Residual                           | Curve Number                                         |
|-------------------------------|----------------------------------------------------|------------------------------------------------------|
| Runoff mechanism              | Excess over storage capacity                       | Reduced-form response function                       |
| Role of antecedent conditions | Strong: runoff depends on $(S_{i,t-1}, D_{i,t-1})$ | Weak: runoff depends only on current input and $v_i$ |
| Key parameters                | $\bar{S}_i, \bar{D}_i, \xi_i, \bar{a}_i$           | $v_i$ (and derived $\zeta_i, \mathcal{J}_i$ )        |
| Infiltration                  | Primary: allocated first, capped by capacity       | Residual: $f = u - a - q$                            |
| Data requirements             | Hydrogeology (aquifer thickness, porosity)         | Land cover, soil hydrologic group                    |
| Sensitivity to storage state  | High                                               | Low                                                  |

## A5 Mass Balance Proofs

**Proposition 1** (Within-Cell Mass Balance). *Under either the residual-runoff or curve-number specification, for all  $i$  and  $t$ :*

$$u_{i,t} = a_{i,t} + f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}} + q_{i,t}. \quad (\text{A.20})$$

*Proof. Case 1: Residual-runoff specification.* We have  $u_{i,t}^{\text{net}} = u_{i,t} - a_{i,t}$  and  $q_{i,t} = u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}}$ . Substituting the first equation into the second:

$$q_{i,t} = (u_{i,t} - a_{i,t}) - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}} \quad (\text{A.21})$$

$$\implies u_{i,t} = a_{i,t} + f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}} + q_{i,t}. \quad (\text{A.22})$$

*Case 2: Curve-number specification.* Infiltration is  $f_{i,t} = u_{i,t} - a_{i,t} - q_{i,t}$ , and  $f_{i,t} = f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}}$  by the recharge split. Therefore  $u_{i,t} - a_{i,t} - q_{i,t} = f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}}$ , which rearranges to (A.20).  $\square$

**Proposition 2** (Stock Change Identity). *Period-over-period stock changes satisfy:*

$$\Delta M_{i,t} + \Delta S_{i,t} + \Delta D_{i,t} + \Delta \text{SWE}_{i,t} = r_{i,t} + l_{i,t-1} - e_{i,t} - q_{i,t}, \quad (\text{A.23})$$

where  $\Delta$  denotes the change from  $t-1$  to  $t$ .

*Proof.* From the stock update equations:

$$\Delta M_{i,t} = a_{i,t} - e_{i,t}, \quad (\text{A.24})$$

$$\Delta S_{i,t} = f_{i,t}^{\text{Sh}}, \quad (\text{A.25})$$

$$\Delta D_{i,t} = f_{i,t}^{\text{Dp}}, \quad (\text{A.26})$$

$$\Delta \text{SWE}_{i,t} = r_{i,t}^{\text{snow}} - m_{i,t}. \quad (\text{A.27})$$

Summing and using  $a_{i,t} + f_{i,t}^{\text{Sh}} + f_{i,t}^{\text{Dp}} = u_{i,t} - q_{i,t}$  (from Proposition 1),  $u_{i,t} = \tilde{r}_{i,t} + l_{i,t-1} = r_{i,t}^{\text{rain}} + m_{i,t} + l_{i,t-1}$ , and  $r_{i,t} = r_{i,t}^{\text{rain}} + r_{i,t}^{\text{snow}}$ :

$$\begin{aligned} \sum \Delta &= (r_{i,t}^{\text{rain}} + m_{i,t} + l_{i,t-1} - q_{i,t}) - e_{i,t} + r_{i,t}^{\text{snow}} - m_{i,t} \\ &= r_{i,t} + l_{i,t-1} - e_{i,t} - q_{i,t}. \end{aligned} \quad (\text{A.28}) \quad \square$$

**Proposition 3** (Domain-Wide Mass Balance). *Summing over all cells and accounting for routing:*

$$\sum_{i=1}^I r_{i,t} = \sum_{i=1}^I e_{i,t} + \sum_{i \in \mathcal{T}} O_{i,t}^{\text{ocean}} + \sum_{i=1}^I (\Delta M_{i,t} + \Delta S_{i,t} + \Delta D_{i,t} + \Delta \text{SWE}_{i,t}). \quad (\text{A.29})$$

*Proof.* Summing Proposition 2 over all cells  $i$ :

$$\sum_i (\Delta M + \Delta S + \Delta D + \Delta \text{SWE}) = \sum_i r_{i,t} + \sum_i l_{i,t-1} - \sum_i e_{i,t} - \sum_i q_{i,t}. \quad (\text{A.30})$$

The routing operator is conservative:  $\sum_i l_{i,t-1} = \sum_i \sum_j w_{j \rightarrow i} q_{j,t-1} = \sum_j q_{j,t-1} \sum_i w_{j \rightarrow i}$ . For non-terminal cells,  $\sum_i w_{j \rightarrow i} = 1$ ; for terminal cells  $j \in \mathcal{T}$ ,  $\sum_i w_{j \rightarrow i} < 1$ , with the deficit equal to the fraction exported. In the one-period routing framework, the net effect is that  $\sum_i l_{i,t} = \sum_i q_{i,t} - \sum_{j \in \mathcal{T}} O_{j,t}^{\text{ocean}}$ , which yields (A.29).  $\square$

## A6 Properties of the Routing Operator

The routing matrix  $\mathbf{W}$  plays a role in the hydrological model analogous to the input-output matrix in Leontief analysis. This section develops its mathematical properties.

### A6.1 Construction from D8 Flow Directions

The D8 algorithm assigns each raster cell one of eight flow directions (N, NE, E, SE, S, SW, W, NW), pointing toward the neighbor with the steepest descent. The routing matrix  $\mathbf{W}$  is constructed by setting  $w_{j \rightarrow i} = 1$  if  $j$ 's D8 direction points to cell  $i$ , and  $w_{j \rightarrow i} = 0$  otherwise. Each row of  $\mathbf{W}$  sums to at most one (exactly one for interior cells, less than one for terminal cells at the domain boundary or coastline).

Figure A.4 illustrates the D8 flow-direction assignment on a  $5 \times 5$  elevation grid. Cells are colored by elevation, and arrows show the steepest-descent direction.

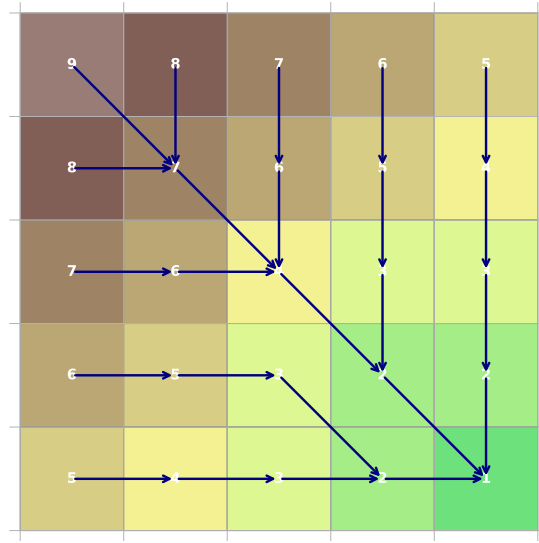


Figure A.4: D8 flow directions on a  $5 \times 5$  grid. Numbers show elevation values. Arrows point in the direction of steepest descent, defining the routing matrix  $\mathbf{W}$ .

### A6.2 Acyclicity and Nilpotency

**Lemma 1** (Acyclicity). *Under D8 flow directions with a well-defined elevation surface (no flat areas or pits after hydrological conditioning), the routing matrix  $\mathbf{W}$  represents a directed acyclic graph (DAG).*

After topological sorting by decreasing elevation,  $\mathbf{W}$  is strictly lower-triangular.

*Proof.* Each D8 direction points from a cell to a neighbor with strictly lower elevation. Therefore, following any sequence of routing steps yields a monotonically decreasing elevation path, which cannot revisit a cell.  $\square$

**Corollary 1** (Nilpotency).  $\mathbf{W}$  is nilpotent: there exists  $K \leq I$  such that  $\mathbf{W}^K = \mathbf{0}$ , where  $K$  is the length of the longest flow path in the drainage network.

### A6.3 The Spatial Multiplier (Leontief Inverse)

Total flow through cell  $i$ , including all upstream contributions, is given by the flow-accumulation vector:

$$\mathbf{F} = (\mathbf{I} - \mathbf{W}^\top)^{-1} \mathbf{q}. \quad (\text{A.31})$$

Because  $\mathbf{W}$  is nilpotent, the Neumann series converges in finitely many terms:

$$(\mathbf{I} - \mathbf{W}^\top)^{-1} = \sum_{k=0}^{K-1} (\mathbf{W}^\top)^k = \mathbf{I} + \mathbf{W}^\top + (\mathbf{W}^\top)^2 + \dots + (\mathbf{W}^\top)^{K-1}. \quad (\text{A.32})$$

The term  $(\mathbf{W}^\top)^k \mathbf{q}$  gives the runoff contribution arriving at each cell after exactly  $k$  routing steps. The  $(i, j)$  entry of  $(\mathbf{I} - \mathbf{W}^\top)^{-1}$  gives the total (direct plus indirect) impact of a unit runoff injection at cell  $j$  on the flow through cell  $i$ .

**Lemma 2** (Spatial Multiplier Interpretation). Entry  $(i, j)$  of  $(\mathbf{I} - \mathbf{W}^\top)^{-1}$  is positive if and only if  $j \in \mathcal{U}(i) \cup \{i\}$ , i.e.,  $j$  is upstream of  $i$  or  $j = i$ . The entry equals one if  $j = i$ , and it equals the number of distinct flow paths from  $j$  to  $i$  (weighted by routing fractions) otherwise.

This is the hydrological analogue of the Leontief inverse in input-output economics:  $\mathbf{W}$  is the direct-requirements matrix, and  $(\mathbf{I} - \mathbf{W}^\top)^{-1}$  gives total requirements.

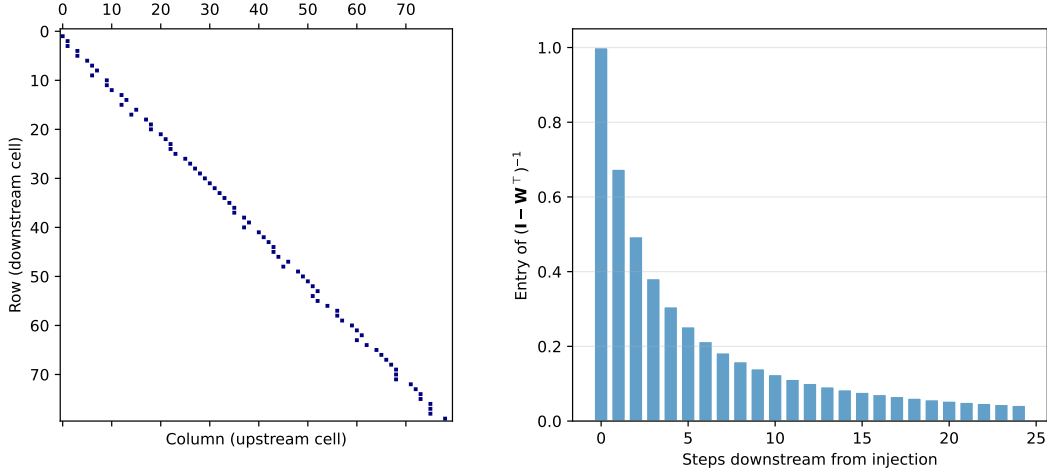


Figure A.5: Left: sparsity pattern of the routing matrix  $\mathbf{W}$  (80 cells, topologically sorted). Right: decay of Leontief-inverse entries along a flow path—the contribution of upstream runoff diminishes with routing distance.

#### A6.4 Computational Complexity

Because  $\mathbf{W}$  is sparse (each row has at most one non-zero entry) and the graph is acyclic, the flow-accumulation computation  $\mathbf{F} = (\mathbf{I} - \mathbf{W}^\top)^{-1} \mathbf{q}$  can be performed in  $O(I)$  time using a single topological-sort pass. This is the standard flow-accumulation algorithm used in GIS software. For our global grid at  $0.25^\circ$  resolution ( $I \approx 1,036,800$  land cells), the computation takes less than one second per time step.

### A7 Equilibrium Definition and Closure

This section collects the closure of the within-period economic equilibrium. Given the hydrological state and the extraction vector inherited by period  $t$ , a competitive equilibrium is a collection of national wages  $\{w_{n,t}^{\text{LAB}}\}_n$ , local water prices  $\{r_{i,n,t}^{\text{Water}}\}_{i \in \mathcal{I}_{n,n}}$ , domestic hub prices  $\{P_{nn,j,t}\}_{n,j}$ , Armington purchaser prices  $\{P_{n,j,t}\}_{n,j}$ , source unit costs  $\{c_{i,n,j,t}\}_{i,j}$ , primary-endowment prices and quantities  $\{P_{i,n,j,t}^V, V_{i,n,j,t}\}_{i,j}$ , land allocations, production quantities, and bilateral expenditures that jointly satisfy the production, land-allocation, trade, and market-clearing equations in Section 2.

In particular, closure requires: (i) the CDE and upper Cobb–Douglas production conditions; (ii) the plot-level land-return equation together with the nested Fréchet allocation and physical land clearing; (iii) local water-market clearing in every physical grid; (iv) national labor-market clearing including both direct production labor and setup labor; (v) national-hub goods-market clearing and the two-level Armington demand system; and (vi) the country trade balance and household budget constraint. The dynamic extraction problem then links successive within-period equilibria through the hydrological law of motion.

A formal existence-and-uniqueness proof for the full CDE–Fréchet economy requires regular-

ity conditions on the CDE expenditure system and the dynamic water-cost problem beyond the standard Armington structure. We therefore treat the full proof as a separate theoretical exercise rather than importing the standard one-factor Armington argument unchanged.

## A8 Surface Water Balance: Full Derivation

This section provides the step-by-step construction of the surface-water stock equation.

### A8.1 Precipitation Partition

Let  $\mathcal{A}_i$  be grid-cell area and  $\omega_i \in [0, 1]$  the open-water fraction. Total precipitation  $r_{i,t}$  is partitioned:

$$r_{i,t}^{\text{land}} = (1 - \omega_i) r_{i,t}, \quad r_{i,t}^{\text{water}} = \omega_i r_{i,t}. \quad (\text{A.33})$$

Only  $r_{i,t}^{\text{land}}$  enters the infiltration-runoff partition;  $r_{i,t}^{\text{water}}$  enters the surface-water stock directly.

### A8.2 Linear Reservoir Release

The surface-water stock releases a fraction  $\kappa_i^{\text{rout}} \in (0, 1]$  of its storage each period. The parameter captures within-cell residence time: small  $\kappa$  for lakes and reservoirs (long residence), large  $\kappa$  for river channels (short residence). Retained storage is  $(1 - \kappa_i^{\text{rout}})X_{2,i,t-1}$ .

### A8.3 Inflow Components

Surface water receives inflow from four sources:

1. **Overland runoff**  $q_{i,t}$ : generated by the storage-limited or CN mechanism.
2. **Baseflow**  $b_{i,t}$ : slow drainage from shallow groundwater. In the simplest specification,  $b_{i,t} = \beta_i^{\text{base}} S_{i,t-1}$  where  $\beta_i^{\text{base}}$  is a recession coefficient.
3. **Direct precipitation**  $r_{i,t}^{\text{water}}$ : rainfall on the open-water surface.
4. **Upstream routed inflow**:  $\sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1}$ .

### A8.4 Outflow Components

1. **Open-water evaporation**  $e_{i,t}^{\text{open}}$ : evaporation from the water surface, typically larger than terrestrial ET in arid regions.
2. **Seepage**  $s_{i,t} = \varsigma_i^{\text{bed}} (1 - \kappa_i^{\text{rout}}) X_{2,i,t-1}$ : infiltration through the river/lake bed into shallow groundwater.
3. **Diversions**  $\text{Div}_{i,t}^{\text{sur}}$ : human withdrawals for irrigation, municipal, or industrial use.

4. **Terminal export**  $O_{i,t}^{\text{ocean}}$ : outflow at coastal cells.
5. **Return flows**  $v_i^{\text{surf}} \mathcal{R}_{i,t}$ : a fraction of water extracted and used is returned to the surface system.

### A8.5 Assembled Balance Equation

Combining all terms:

$$\begin{aligned}
 X_{2,i,t} = & (1 - \kappa_i^{\text{rout}}) X_{2,i,t-1} + q_{i,t} + b_{i,t} + r_{i,t}^{\text{water}} + \sum_j w_{j \rightarrow i} \kappa_j^{\text{rout}} X_{2,j,t-1} \\
 & + v_i^{\text{surf}} \mathcal{R}_{i,t} - \text{Div}_{i,t}^{\text{sur}} - e_{i,t}^{\text{open}} - s_{i,t} - O_{i,t}^{\text{ocean}}.
 \end{aligned} \tag{A.34}$$

This is equation (13) in the main text. The baseline runoff-routing model is nested by setting  $\kappa_i^{\text{rout}} = 1$ ,  $\omega_i = 0$ , and  $\varsigma_i^{\text{bed}} = 0$  for all  $i$ , in which case  $X_{2,i,t}$  reduces to transient pass-through flow.

## A9 Data Construction Details

This section documents the processing steps for each dataset used in the model. All data are harmonized to a common  $0.25^\circ \times 0.25^\circ$  grid covering global land areas (excluding Antarctica).

### A9.1 GLDAS Noah Land Surface Model

The NASA Global Land Data Assimilation System (GLDAS) Noah model version 2.1 provides land-surface variables on a  $0.25^\circ$  global grid at 3-hourly resolution from January 2000 to present (Rodell et al., 2004). We use the following variables:

Table A.5: GLDAS Noah Variables Used

| Variable Name         | Description              | Native Units                     | Model Units        |
|-----------------------|--------------------------|----------------------------------|--------------------|
| Rainf_tavg            | Rainfall rate            | $\text{kg m}^{-2} \text{s}^{-1}$ | in/mo              |
| Snowf_tavg            | Snowfall rate            | $\text{kg m}^{-2} \text{s}^{-1}$ | in/mo              |
| Tair_f_inst           | Air temperature          | K                                | $^{\circ}\text{C}$ |
| Evap_tavg             | Evapotranspiration       | $\text{kg m}^{-2} \text{s}^{-1}$ | in/mo              |
| SoilMoi0_10cm_inst    | Soil moisture 0–10 cm    | $\text{kg m}^{-2}$               | in                 |
| SoilMoi10_40cm_inst   | Soil moisture 10–40 cm   | $\text{kg m}^{-2}$               | in                 |
| SoilMoi40_100cm_inst  | Soil moisture 40–100 cm  | $\text{kg m}^{-2}$               | in                 |
| SoilMoi100_200cm_inst | Soil moisture 100–200 cm | $\text{kg m}^{-2}$               | in                 |
| SWE_inst              | Snow water equivalent    | $\text{kg m}^{-2}$               | in                 |

Notes: Flux conversions: multiply by  $\Delta_t$  (seconds in month) and divide by 25.4 to convert from  $\text{kg m}^{-2} \text{s}^{-1}$  to in/mo. State conversions: divide  $\text{kg m}^{-2}$  by 25.4 to obtain inches.

**Unit conversions.** The conversion from GLDAS native units to model units follows:

$$r_{i,t}^{\text{rain}} [\text{in/mo}] = \text{Rainf\_tavg}_{i,t} [\text{kg m}^{-2} \text{s}^{-1}] \times \Delta_t [\text{s/mo}] \times \frac{1}{25.4}, \quad (\text{A.35})$$

where  $\Delta_t$  is the number of seconds in month  $t$  (varying between 2,419,200 for February and 2,678,400 for 31-day months). The same conversion applies to snowfall and ET fluxes. Temperature is converted as  $T_{i,t} [^{\circ}\text{C}] = \text{Tair\_f\_inst}_{i,t} [\text{K}] - 273.15$ . Soil moisture states are converted from  $\text{kg m}^{-2}$  (equivalent to mm) to inches by dividing by 25.4.

**Temporal aggregation.** The 3-hourly GLDAS data are aggregated to monthly frequency. Flux variables are time-averaged (mean of all 3-hourly values in a month), while state variables use end-of-month instantaneous values.

**Soil moisture initialization.** The initial root-zone moisture  $M_{i,0}$  is constructed by summing the four soil moisture layers:

$$M_{i,0} [\text{in}] = \frac{1}{25.4} \sum_{k=1}^4 \text{SoilMoi}_{i,0}^{(k)} [\text{kg m}^{-2}]. \quad (\text{A.36})$$

## A9.2 HydroSHEDS Flow Direction Grid

The HydroSHEDS database (Lehner et al., 2008) provides hydrologically conditioned flow-direction rasters at 15 arc-second resolution ( $\approx 500$  m at the equator). We use the D8 flow-direction layer, which assigns each cell one of eight cardinal/diagonal directions based on the steepest descent.

**Aggregation to 0.25°.** The 15 arc-second grid is coarsened to 0.25° by: (1) identifying the dominant flow direction within each 0.25° cell (mode of the  $60 \times 60$  sub-cells); (2) constructing  $w_{j \rightarrow i}$  from the resulting coarsened direction field. In practice, we use the flow accumulation at the fine resolution to identify the main channel within each coarse cell and assign the coarse-cell direction to follow the main channel.

**Terminal cells.** Cells at the coastline or domain boundary are flagged as terminal. For these cells,  $\sum_i w_{j \rightarrow i} < 1$ , with the deficit representing outflow to the ocean.

### A9.3 Subsurface Storage Capacity

**Shallow storage.** Shallow aquifer storage capacity  $\bar{S}_i$  is constructed as:

$$\bar{S}_i [\text{in}] = \text{depth}_i^{\text{shallow}} [\text{m}] \times \phi_i^{\text{eff}} \times \frac{1000}{25.4}, \quad (\text{A.37})$$

where  $\text{depth}_i^{\text{shallow}}$  is the shallow aquifer thickness and  $\phi_i^{\text{eff}}$  is effective porosity (specific yield). Aquifer thickness is from [de Graaf et al. \(2015\)](#), who construct a global groundwater model at 0.5° resolution. Effective porosity is estimated from soil texture using the relationships in [Rawls et al. \(1982\)](#): 0.25 for sand, 0.15 for loam, 0.08 for clay, with intermediate values for mixed textures.

**Deep storage.** Deep aquifer capacity  $\bar{D}_i$  follows the same formula using deep aquifer thickness from [de Graaf et al. \(2015\)](#). Where the deep aquifer is absent or the dataset reports zero thickness, we set  $\bar{D}_i = 0$  and  $\xi_i = 0$  (all recharge goes to the shallow layer).

**Abstraction capacity.** The interception/depression storage parameter  $\bar{a}_i$  is estimated from land cover (ESA CCI) and soil type:

$$\bar{a}_i = \bar{a}_i^{\text{veg}} + \bar{a}_i^{\text{depr}}, \quad (\text{A.38})$$

where  $\bar{a}_i^{\text{veg}}$  is canopy interception storage (higher for forests, lower for grasslands) and  $\bar{a}_i^{\text{depr}}$  is surface depression storage (higher for flat terrain, lower for steep slopes).

### A9.4 GCN250 Curve Number Dataset

The GCN250 dataset ([Jaafar et al., 2019](#)) provides spatially explicit curve numbers at 250 m resolution, derived from ESA CCI land cover and SoilGrids hydrologic soil group classifications. We aggregate to the 0.25° computational grid using area-weighted averaging:

$$v_i = \frac{\sum_{k \in \text{cell } i} A_k \cdot v_k}{\sum_{k \in \text{cell } i} A_k}, \quad (\text{A.39})$$

where the sum is over all 250 m pixels  $k$  within the 0.25° cell  $i$ . Values are clipped to  $[30, 100]$  to avoid numerical issues with very low curve numbers.

## A9.5 GRACE/GRACE-FO Satellite Gravimetry

The Gravity Recovery and Climate Experiment (GRACE) and its follow-on mission (GRACE-FO) provide monthly estimates of terrestrial water storage (TWS) anomalies from April 2002 to present (Tapley et al., 2004; Landerer et al., 2020). We use the JPL mascon (mass concentration) solution, which provides TWS anomalies in cm of water-equivalent at  $\sim 3^\circ$  effective resolution.

**Regridding.** Mascon solutions are regridded to  $0.25^\circ$  via bilinear interpolation. The native  $3^\circ$  resolution means that sub-grid variability is not captured; GRACE data are used for validation of basin-scale and regional-scale TWS trends, not cell-level comparisons.

**Unit conversion.** GRACE TWS anomalies in cm are converted to inches by dividing by 2.54.

**Comparison with model.** The model-predicted TWS change is:

$$\Delta TWS_{i,t}^{\text{model}} = \Delta M_{i,t} + \Delta S_{i,t} + \Delta D_{i,t} + \Delta SWE_{i,t} + \Delta X_{2,i,t}. \quad (\text{A.40})$$

We compare monthly anomalies (deviations from the long-run mean) at the basin scale.

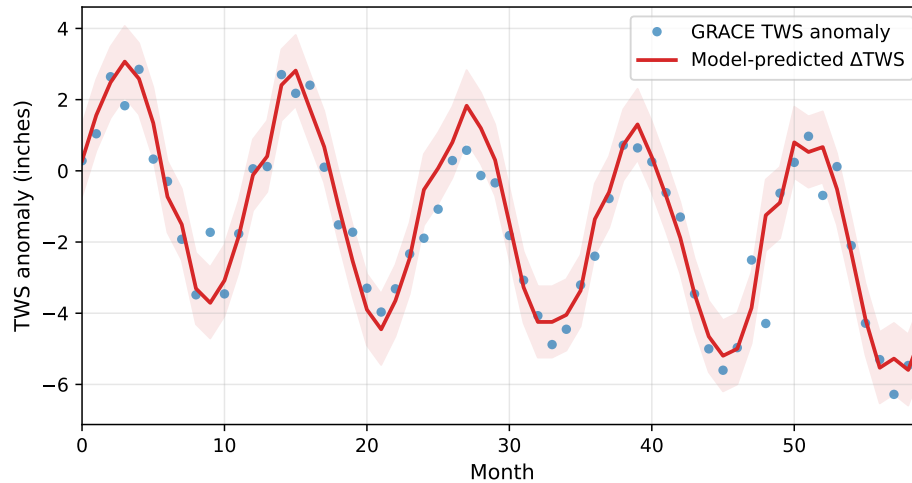


Figure A.6: Model-predicted total water storage anomalies versus GRACE satellite observations (illustrative basin-average time series). The shaded band shows the model's  $\pm 1$  standard deviation envelope.

## A9.6 ESA CCI Land Cover

The ESA Climate Change Initiative (CCI) land cover product provides annual land-cover maps at 300m resolution, classified into 37 categories following the FAO Land Cover Classification System. We use this dataset for three purposes: (1) estimating the open-water fraction  $\omega_i$  at each

cell; (2) informing canopy interception and abstraction parameters; (3) cross-referencing with GCN250 for curve-number validation.

## A9.7 Additional Datasets

Table A.6: Additional Datasets

| Dataset           | Resolution                     | Source                  | Use in Model                                                         |
|-------------------|--------------------------------|-------------------------|----------------------------------------------------------------------|
| SoilKsat          | Point data                     | Dai et al. (2013)       | Saturated hydraulic conductivity for estimating seepage coefficients |
| MERRA-2           | $0.5^\circ \times 0.625^\circ$ | NASA GMAO               | Precipitation robustness checks                                      |
| MODIS ET          | 500 m, 8-day                   | Running et al. (2017)   | ET validation                                                        |
| World Water Table | $\sim 0.5^\circ$               | Fan et al. (2013)       | Initial groundwater levels                                           |
| River Width       | Varies                         | Allen & Pavelsky (2018) | Channel width for routing parameter calibration                      |
| GTAP 10           | Country-sector                 | Aguiar et al. (2019)    | Factor shares $(\alpha, \beta, \gamma)$                              |

## A10 Numerical Implementation

### A10.1 Algorithm Overview

The coupled hydro-economic model is solved using an outer iteration that alternates between: (i) a forward pass of the hydrological time-stepper (given extraction rates), and (ii) a static economic equilibrium solver at each time step (given hydrological states). The two stages iterate until extraction rates converge.

1. **Initialize.** Set initial water stocks from GLDAS data. Set initial extraction rates to zero (or to observed irrigation withdrawals if available).
2. **Hydrological forward pass.** For  $t = 1, \dots, T$ :
  - (a) Compute snowmelt  $m_{i,t}$  and update  $\text{SWE}_{i,t}$ .
  - (b) Compute effective liquid input  $\tilde{r}_{i,t} = r_{i,t}^{\text{rain}} + m_{i,t}$ .
  - (c) Add routed inflow:  $u_{i,t} = \tilde{r}_{i,t} + \iota_{i,t-1}$ .
  - (d) Compute abstraction:  $a_{i,t} = \min\{u_{i,t}, \bar{a}_i\}$ ,  $u_{i,t}^{\text{net}} = u_{i,t} - a_{i,t}$ .
  - (e) Allocate recharge:  $f_{i,t}^{\text{Sh}} = \min\{(1 - \xi_i)u^{\text{net}}, (\bar{S}_i - S_{i,t-1})_+\}$ ,  $f_{i,t}^{\text{Dp}} = \min\{\xi_i u^{\text{net}}, (\bar{D}_i - D_{i,t-1})_+\}$ .
  - (f) Compute runoff:  $q_{i,t} = u_{i,t}^{\text{net}} - f_{i,t}^{\text{Sh}} - f_{i,t}^{\text{Dp}}$ .

- (g) Update stocks:  $M_{i,t}$ ,  $S_{i,t}$ ,  $D_{i,t}$  (subtract extraction if coupled).
  - (h) Route runoff:  $t_{i,t} = \sum_j w_{j \rightarrow i} q_{j,t}$  (single topological-sort pass).
3. **Economic equilibrium.** At each  $t$ , given  $(M_{i,t}, S_{i,t}, D_{i,t})$ , solve for wages, prices, trade flows, and extraction.
  4. **Convergence check.** Compare new extraction rates with previous iteration. If  $\max_i |\text{Div}_{i,t}^{(k+1)} - \text{Div}_{i,t}^{(k)}| < 10^{-8}$  for all  $t$ , stop. Otherwise, return to step 2 with updated extraction.

## A10.2 Vectorized Hydrological Computation

All within-cell operations (steps 2a–2g) are vectorized over the spatial dimension using NumPy array operations. For a grid of  $I \approx 10^6$  cells and  $T = 240$  months (20 years), the hydrological forward pass takes approximately 2 minutes on a single CPU core. The routing step (2h) uses the topological ordering of cells by elevation, allowing a single-pass computation without matrix inversion.

## A10.3 Economic Equilibrium Solver

At each date  $t$ , conditional on the hydrological state and a candidate extraction vector, we solve the static equilibrium jointly for national wages, local water prices, domestic hub prices, Armington prices, land allocations, and production. The numerical routine iterates on the market-clearing residuals for the national labor markets, the local water markets, and the national-hub goods markets. Within each iteration, the production system is evaluated in the same order as in the model: source unit cost and  $P^V$ , the return per efficiency unit of land, nested Fréchet land shares and selected productivity, grid output, Armington prices and expenditure shares, and finally factor and goods-market residuals. Damping is used when updating prices so that the sup-norm of all market-clearing residuals decreases monotonically near the solution.

## A10.4 Software and Script Organization

The hydrological core is implemented in Python (NumPy, SciPy). The main modules are:

- `step_runoff_residual.py`: implements the storage-limited residual approach. Takes forcing arrays and stock arrays as input; returns updated stocks, runoff field, and diagnostics.
- `step_infiltration_residual.py`: implements the curve-number approach with the same interface.
- `simulation_tests.ipynb`: Jupyter notebook that loads GLDAS forcing data, initializes stocks, runs the model forward, routes runoff, and produces validation plots.

Data preprocessing scripts handle format conversion (NetCDF to NumPy arrays), regridding, and unit transformation. The economic equilibrium solver is implemented in MATLAB (`economic_model.m`).

## A11 Welfare and Counterfactual Changes

### A11.1 Exact Welfare Measure

With Cobb–Douglas preferences over sectoral Armington composites, household welfare in country  $n$  is the consumption quantity index  $C_{n,t}$ . Using the household budget constraint,

$$C_{n,t} = \frac{w_{n,t}^{\text{LAB}} \bar{F}_{n,t}^{\text{LAB}} + \text{NRL}_{n,t} + \sum_{i \in \mathcal{I}_n} r_{i,n,t}^{\text{Water}} \bar{F}_{i,t}^{\text{Water}} + D_{n,t}}{P_{n,t}^C}. \quad (\text{A.41})$$

Thus a counterfactual welfare change can always be decomposed exactly into the change in nominal household resources and the change in the consumption price index.

### A11.2 Hat Algebra for the Two-Level Armington Structure

For any variable  $x$ , define  $\hat{x} \equiv x^1/x^0$ . Holding Armington weights fixed, the lower imported-bundle price satisfies

$$\hat{P}_{n,j,t}^{\text{Imp}} = \left[ \sum_{m \neq n} \lambda_{nm,j,t}^{M,0} \left( \hat{P}_{nm,j,t} \right)^{1-\sigma_j^M} \right]^{1/(1-\sigma_j^M)}, \quad (\text{A.42})$$

and the conditional foreign-source expenditure shares satisfy

$$\hat{\lambda}_{nm,j,t}^M = \frac{\left( \hat{P}_{nm,j,t} \right)^{1-\sigma_j^M}}{\sum_{\ell \neq n} \lambda_{n\ell,j,t}^{M,0} \left( \hat{P}_{n\ell,j,t} \right)^{1-\sigma_j^M}}, \quad m \neq n. \quad (\text{A.43})$$

At the upper domestic-versus-import nest,

$$\hat{P}_{n,j,t} = \left[ \lambda_{n,j,t}^{D,0} \left( \hat{P}_{n,j,t} \right)^{1-\sigma_j^D} + \left( 1 - \lambda_{n,j,t}^{D,0} \right) \left( \hat{P}_{n,j,t}^{\text{Imp}} \right)^{1-\sigma_j^D} \right]^{1/(1-\sigma_j^D)}. \quad (\text{A.44})$$

Because  $d_{nn,j} = 1$  and  $P_{nm,j,t} = d_{nm,j} P_{mm,j,t}$ , the same bilateral-price notation applies to domestic and foreign sources throughout the counterfactual system.

## A12 Sensitivity Analysis

This section presents systematic sensitivity analysis for the key parameters of the hydrological model.

## A12.1 Degree-Day Factor

The degree-day factor governs the speed of snowmelt. Higher values produce faster spring melt, concentrating effective liquid input in a shorter window and potentially increasing peak runoff. Figure A.7 shows the sensitivity of snowpack evolution and effective liquid input to the choice of ddf for a representative mid-latitude cell.

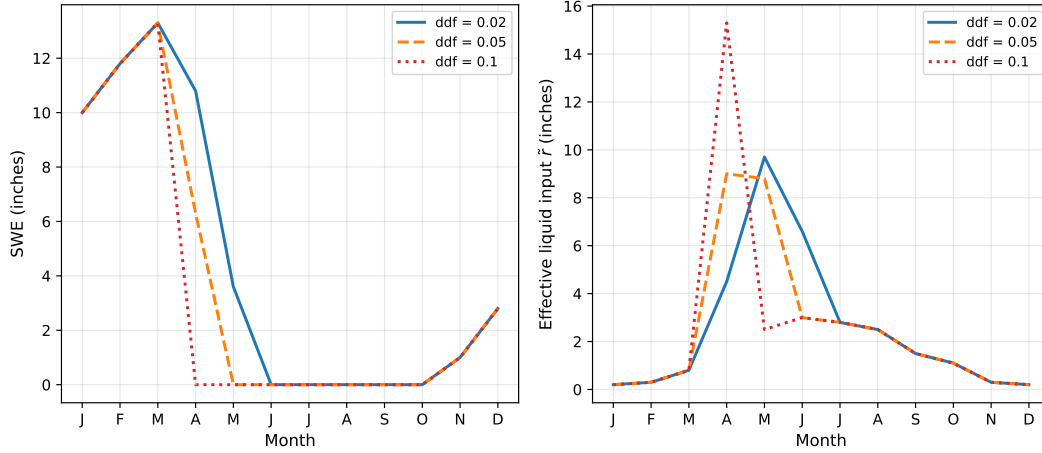


Figure A.7: Left: snowpack (SWE) evolution over one year for different degree-day factors. Right: effective liquid input  $\tilde{r}$  (rainfall + melt). Higher ddf produces earlier, more concentrated melt.

The welfare consequences of varying ddf depend on the interaction between the timing of melt and the timing of agricultural water demand. In regions where spring melt coincides with planting, faster melt can increase water availability during the growing season. In regions where melt occurs before the growing season, faster melt leads to higher runoff and lower stored water when it is needed most.

## A12.2 Deep-Percolation Share

The parameter  $\zeta_i$  controls the allocation of infiltrated water between shallow and deep aquifers. Figure A.8 shows the sensitivity of shallow recharge and surface runoff to  $\zeta_i$ .

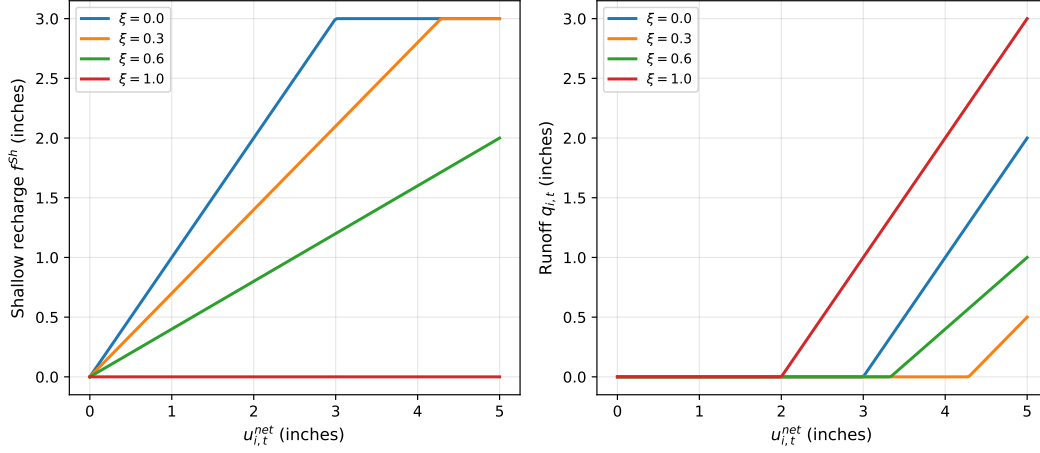


Figure A.8: Left: shallow recharge as a function of post-abstraction input for different values of  $\xi$ . Right: surface runoff. Higher  $\xi$  diverts more water to the deep aquifer, reducing shallow recharge and (when deep capacity fills first) potentially increasing runoff.

The economic implications of  $\xi_i$  depend on extraction technology. Regions that rely primarily on shallow wells are sensitive to reductions in shallow recharge (high  $\xi$ ), while regions with deep-well infrastructure benefit from higher deep-aquifer recharge.

### A12.3 Storage Capacity

Figure A.9 illustrates how storage capacity affects the runoff-infiltration split and the trajectory of groundwater stocks.

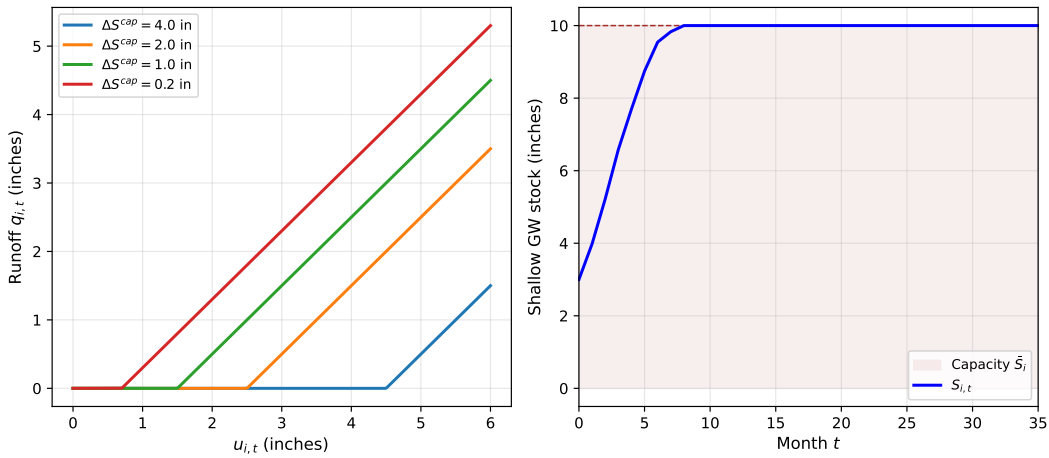


Figure A.9: Left: runoff as a function of liquid input for different available storage capacities. As capacity fills, the marginal propensity to generate runoff increases sharply. Right: illustrative trajectory of shallow GW stocks approaching capacity over a three-year period.

Perturbing  $\bar{S}_i$  and  $\bar{D}_i$  by  $\pm 20\%$  shows that welfare results are most sensitive in semi-arid regions where the infiltration-runoff balance is finely tuned. In humid regions, storage capacity

is rarely binding, so perturbations have little effect on runoff.

## A12.4 Aquifer Depletion Under Alternative Extraction Rates

Figure A.10 shows the projected trajectory of deep groundwater stocks under alternative extraction assumptions for several major aquifer systems.

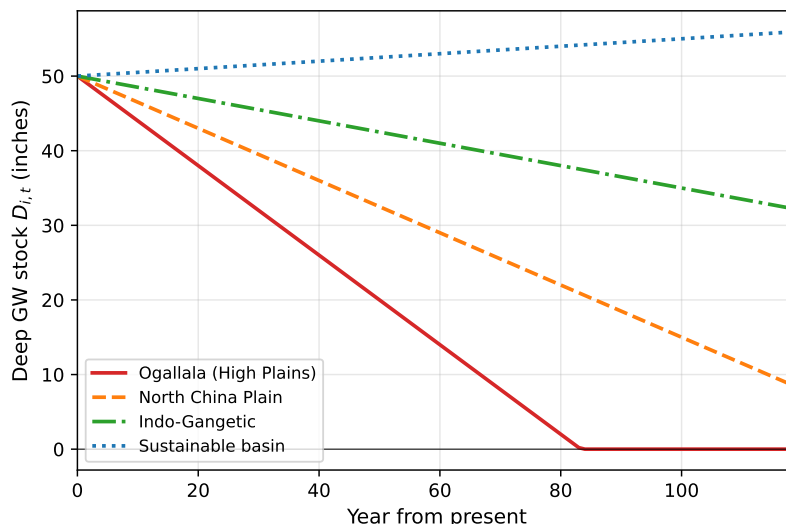


Figure A.10: Deep groundwater stock trajectories under illustrative extraction rates for four basin types. The “sustainable basin” maintains stocks by extracting at or below the recharge rate. Non-sustainable basins deplete at rates depending on the extraction-recharge gap.

## A12.5 Trade Elasticities

The Armington elasticities  $\sigma_j^M$  and  $\sigma_j^D$  govern substitution across foreign sources and between the domestic variety and the import composite, respectively. Sensitivity analysis varies these elasticities around their benchmark sector-specific values and resolves the full equilibrium so that production, land allocation, water demand, and trade adjust jointly.

# A13 Maps and Additional Figures

This section presents global maps of key model inputs and outputs.

## A13.1 Precipitation, Evapotranspiration, and Runoff

Figures A.11–A.13 show global maps of mean annual precipitation, evapotranspiration, and runoff. These patterns drive the spatial distribution of water availability and, through the model, the comparative advantage of each region in water-intensive activities.

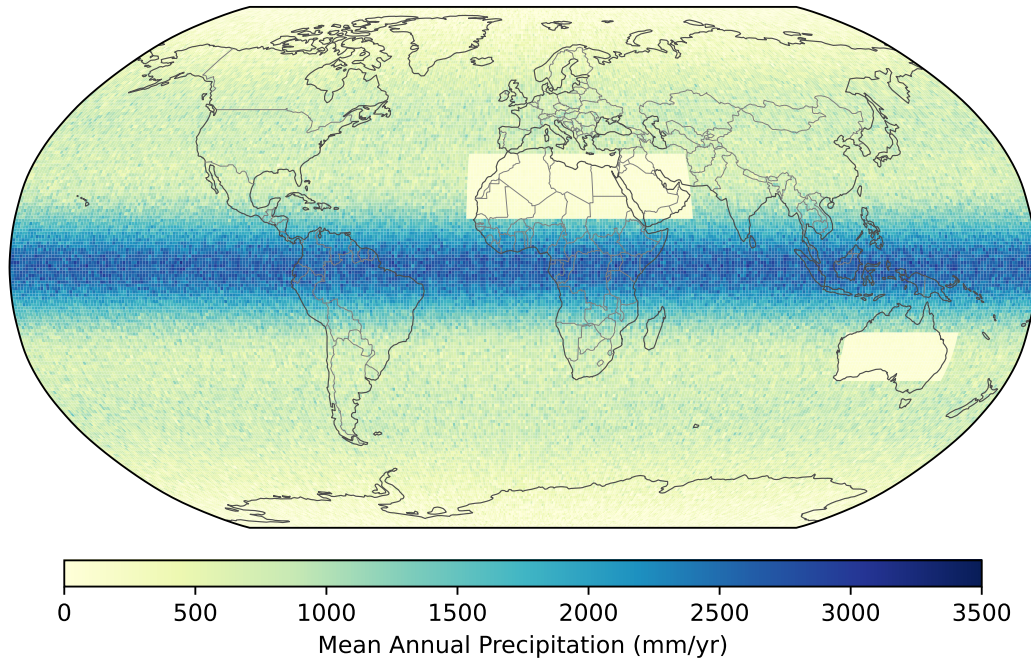


Figure A.11: Mean annual precipitation (mm/yr). The ITCZ and mid-latitude storm tracks are clearly visible. Arid regions (Sahara, Arabian Peninsula, central Australia) receive less than 250 mm/yr.

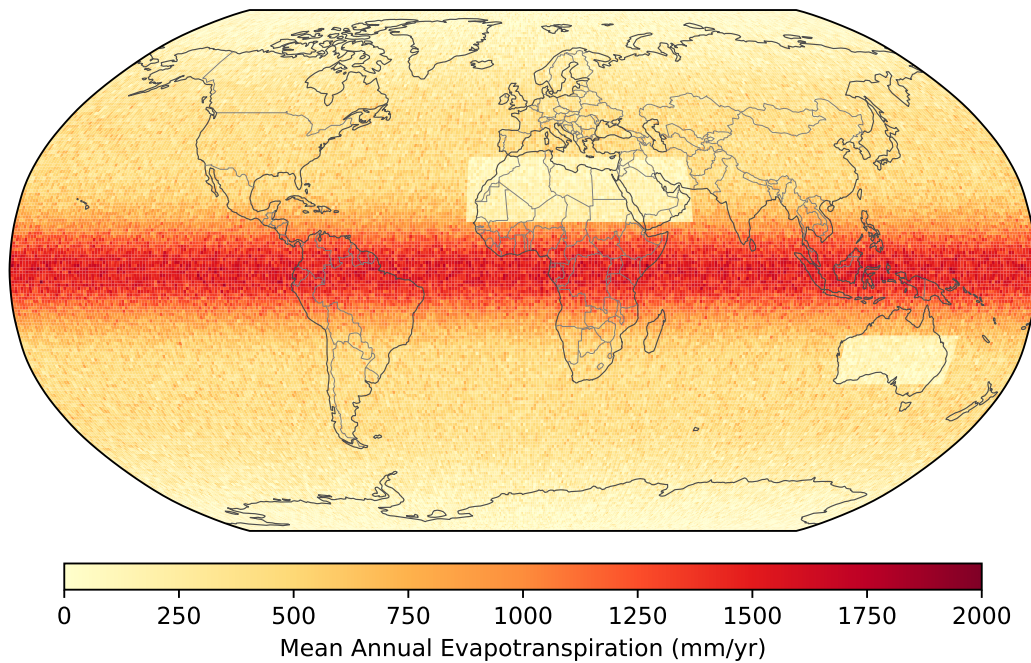


Figure A.12: Mean annual evapotranspiration (mm/yr). ET is highest in the humid tropics (up to 1,500 mm/yr) and lowest in arid and cold regions. ET returns the majority of precipitation to the atmosphere.

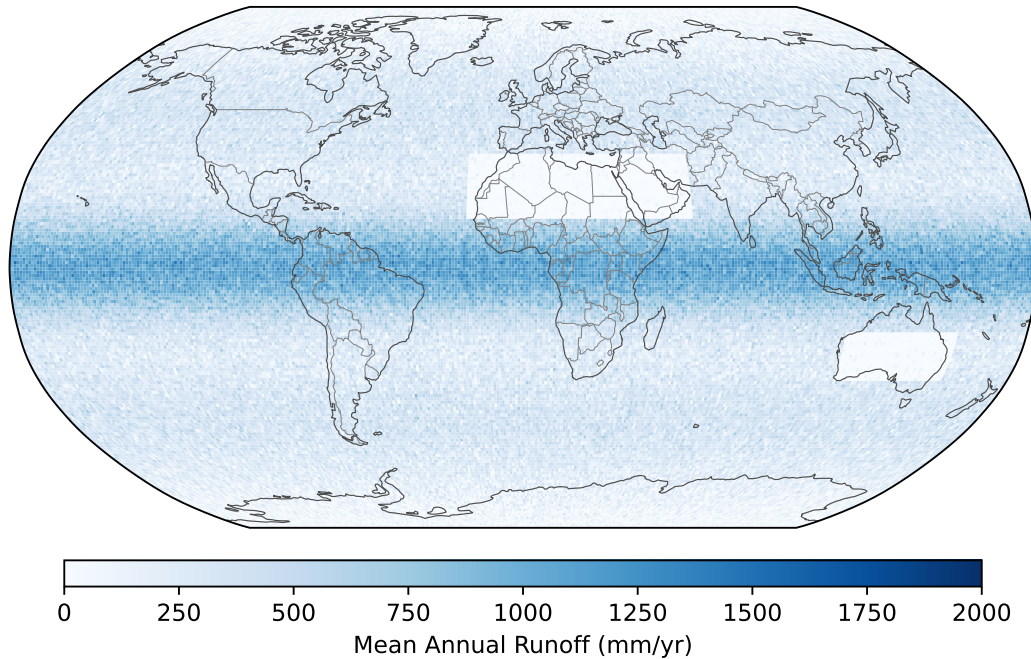


Figure A.13: Mean annual surface runoff (mm/yr). Runoff is the residual of precipitation minus ET minus storage changes. High-runoff regions include the Amazon, Southeast Asia, and high-latitude zones with spring snowmelt.

These three maps together illustrate the fundamental water budget identity at each cell: precipitation equals evapotranspiration plus runoff plus storage change. In tropical regions, ET consumes the majority of precipitation, leaving moderate runoff. In arid regions, ET nearly equals precipitation, leaving negligible runoff and requiring groundwater for economic activity. In high-latitude regions, the seasonal snowpack cycle creates large spring runoff pulses even when annual precipitation is moderate.

### A13.2 Interpretation

The maps show that water availability is distributed very unequally across space, with implications for the spatial distribution of economic activity, the cost of water extraction, and the gains from water-related infrastructure investments. The model transforms these physical patterns into economic outcomes through the hydrological and economic mechanisms described in the main text.

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